

Accept-reject decision-making revealed via a quantitative and ethological study of *C. elegans* foraging

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In this interesting and **important** work, the authors use detailed behavioral analysis and rigorous quantitative modeling to support the idea that *C. elegans* uses an "accept-reject" strategy to make behavioral decisions upon encountering food patches based on learned features of its environment. The work **convincingly** expands our understanding of the behavioral repertoire of this species and provides a strong foundation for future mechanistic studies.

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Abstract

Decision-making is a ubiquitous component of animal behavior that is often studied in the context of foraging. Foragers make a series of decisions while locating food (food search), choosing between food types (diet or patch choice), and allocating time spent within patches of food (patch-leaving). Here, we introduce a framework for investigating foraging decisions using detailed analysis of individual behavior and quantitative modeling in the nematode *Caenorhabditis elegans*. We demonstrate that *C. elegans* make *accept-reject* patch choice decisions upon encounter with food. Specifically, we show that when foraging amongst small, dispersed, and dilute patches of bacteria, *C. elegans* initially *reject* several bacterial patches, opting to prioritize exploration of the environment, before switching to a more exploitative foraging strategy during subsequent encounters. Observed across a range of bacterial patch densities, sizes, and distributions, we use a quantitative model to show that this decision to *explore* or *exploit* is guided by available sensory information, internal satiety signals, and learned environmental statistics related to the bacterial density of recently encountered and *exploited* patches. We behaviorally validated model predictions on animals that had been food-deprived, animals foraging in environments with multiple patch densities, and null mutants with defective sensory modalities. Broadly, we present a framework to study ecologically relevant foraging decisions that could guide future investigations into the cellular and molecular mechanisms underlying decision-making.

Introduction

Decision-making is frequently defined as the selection of a course of action among several alternatives. The ubiquity and importance of decision-making has led to its use in describing a broad range of behaviors from taxes of unicellular organisms to economics and politics in human society [1, 2]. In decision neuroscience, laboratory experiments have investigated the behavioral choices of animals in well-controlled environments or tasks [3]. While these experiments have led to key insights into mechanisms underlying decision-making and related cognitive processes [4], a comprehensive understanding of decision-making remains elusive. Many researchers have advocated for a more neuroethological approach – suggesting that experiments designed to understand problems the brain evolved to solve offer a more rigorous framework for investigation [3, 5, 6]. Thus, one approach to enrich our understanding of decision-making is to look at the decisions made by animals foraging in naturalistic environments.

Foraging animals make a hierarchy of decisions to locate food (food search), choose between different food types (diet or patch choice), and allocate time spent within patches of food items (patch-leaving) [5, 7–10]. These foraging decisions often require that an animal *exploit* an environment for known resources or *explore* it for potentially better opportunities elsewhere [9, 11]. This exploration-exploitation trade-off requires cognitive computations such as learning the spatiotemporal distribution of food, route planning, estimation of food availability, and decision-making [12, 13]. According to optimal foraging theory, foragers may seek to maximize their rate of net energy gained over time [8, 9, 14] by using internal and external information to guide decision-making, especially in environments where resources are sparsely distributed or fluctuating. Although many studies have provided insight into the motivations, behavioral implementations, and genes associated with foraging decisions [9, 10], a more rigorous framework for exploring the neuronal mechanisms underlying foraging decisions is essential to establish a comprehensive understanding of decision-making.

The microscopic nematode *Caenorhabditis elegans* is well suited for investigating the cellular and molecular basis of foraging decisions [15]. A myriad of genetic tools, behavioral assays, and neuronal imaging techniques have been developed to take advantage of the species' quick reproductive cycle, isogeneity, optical transparency, and ease of maintenance [16–18]. While *C. elegans* typically feed upon a diversity of bacterial types in the wild [19], they are commonly maintained in the laboratory on agar plates containing large patches of the bacteria *Escherichia coli* as a food source [16]. However, even in these simplified laboratory conditions and despite having a numerically simple nervous system of only 302 neurons [16, 20], *C. elegans* display complex and robust species-typical behaviors [21] involving learning and memory [22–25], and decision-making [26–31]. Recent studies have demonstrated that ecologically focused environmental enrichment permits identification of novel behaviors and gene functions in *C. elegans* and other animals [32–35].

Foraging animals often must make one of two types of decisions: *stay-switch* or *accept-reject*. *Stay-switch* decisions refer to scenarios where an individual experiences diminishing returns with the current action and must decide when to switch to a new action (e.g., patch-leaving and area-restricted search) [5, 7–10]. In contrast, *accept-reject* decisions refer to situations where an individual must decide between engaging with an option or ignoring it in search of a better one (e.g., diet or patch choice) [5, 8, 9]. *Stay-switch* decisions have been well described in *C. elegans* [25, 26, 36–41] and enable them to successfully explore an environment and exploit the bacteria within [28, 42–44]. However, to our knowledge, *accept-reject* decisions have not yet been demonstrated in *C. elegans*. While *C. elegans* have been shown to alter food preferences in a diet choice assay, this behavior has only been described by a set of *stay-switch* decisions [31, 36, 44]. The ability to make *accept-reject* decisions is likely advantageous for

foraging in fluctuating or variable environments as rejecting an encountered food item of low quality creates the opportunity for a subsequent encounter with preferred food. Thus, while *C. elegans* can successfully forage in patchily distributed environments, it is not known whether *C. elegans* make decisions to *exploit* a patch of food upon encounter and how this decision-making process changes over a series of patch encounters.

In this study, we aimed to identify if *C. elegans* make *accept-reject* decisions upon encounter with bacterial patches. To answer this question, we performed a detailed analysis of the behavior of individual animals foraging in an ecologically inspired environment where bacterial patches were dilute and dispersed. In conditions where the bacterial density was much lower than that of previously visited patches, animals first explored the environment before *accepting* patches for exploitation. This explore-then-exploit strategy persists even when only one bacterial patch is present in the environment. Using a theoretical framework, we found that this initial exploration reflects a series of *accept-reject* decisions guided by signals related to the bacterial density of current and recently *explored* and *exploited* patches as well as the animal's state of satiety.

Results

C. elegans forage in patchy environments with an explore-then-exploit strategy

To investigate whether *C. elegans* make *accept-reject* decisions upon encounter with bacterial patches, we observed the behavior of animals foraging on an agar surface containing an isometric grid of small, low density bacterial patches (**Figure 1A**, Video 1-2). Animals were confined to an arena where their behavior was recorded and tracked for 60 minutes (**Figure 1A**, **Figure 1 – supplement 1**, Video 3). During this time, animals were observed to move about the arena (**Figure 1A-C**, **Figure 1 – supplement 2**, Video 4) with each animal encountering, on average, ~8.4 total patches and ~5.3 unique patches during the one-hour recording (**Figure 1D**). These patch encounters ranged in duration from seconds to tens of minutes and could be classified as either short or long (2+ minutes) using a Gaussian mixture model (**Figure 1E**, **Figure 1 – supplement 3**). Notably, animals were significantly more likely to stay on patch for longer durations at later time points (**Figure 1F**) which corresponded to an overall increase in their probability of residing on patch (**Figure 1G**, **Figure 1 – supplement 4**). Specifically, we found that while animals initially resided on bacterial patches at levels close to those predicted by chance, an animal's probability of residing on a bacterial patch increased significantly after ~10 minutes. These findings, that patch duration and residence increased over time, suggest that animals are more likely to exploit an encountered patch as time goes on.

Given that the bacterial density of the experimental patches was approximately 20 times more dilute than that of conditions animals experienced during development and immediately preceding the assay, it was possible that the initial delay in exploitation could be due to animals' inability to detect the presence of bacteria in these more dilute patches. We therefore set out to determine if animals detected patches at all time points. Previous studies have shown that *C. elegans* display an immediate and marked slowdown upon encounter with the edge of a food patch and sustain these slower speeds on food [28, 45]. Therefore, velocity can be used as a proxy for an animal's ability to sense a bacterial patch. Following this precedent, we quantified the instantaneous velocity of individuals in our assay (**Figure 1H**). We observed significant deceleration of an animal's instantaneous velocity upon encounter with the patch edge (**Figure 1I**), with no obvious difference between short duration, early time point (**Figure 1J**) and long duration, late time point (**Figure 1K**) encounters. Animals consistently displayed marked slowdown when approaching the patch edge (**Figure 1L**), achieving an average deceleration of $-19.5 \mu\text{m/s}^2$ upon encounter with the bacterial patch (**Figure 1M**, **Figure 1 – supplement 5**). This slowdown was statistically significant for both long and short duration encounters (**Figure**

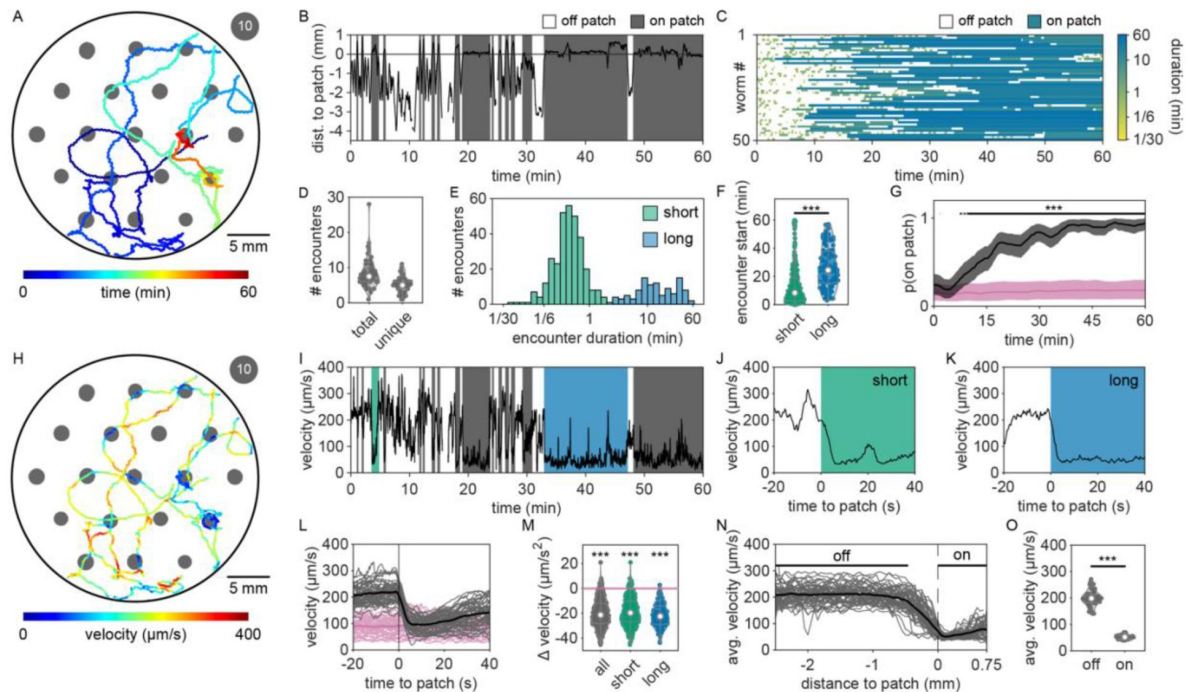


Figure 1

C. elegans forage in a patchy environment with an explore-then-exploit strategy.

(A) An example animal's midbody location (colored to represent time in the experiment) as it forages in an environment bounded by a large (30 mm diameter) arena containing 19 small (~ 1.8 mm diameter) bacterial patches (gray). Each patch was made by pipetting ~ 0.5 μL of OP50 *E. coli* diluted to OD600 ~ 10 and grown at room temperature for ~ 1 hour. (B) Distance between the example animal's midbody position and the nearest patch edge (positive indicates inside patch) is plotted (black) for every time point. Putative encounters with a bacterial patch are indicated (gray). (C) Patch encounters (colored to represent the duration of the encounter) for 50 individuals foraging in these environments are plotted. (D) The number of total and unique patch encounters for each animal are shown. (E) Duration for each patch encounter was computed and classified as either short (0-2 minutes) or long (2-60 minutes) using a Gaussian mixture model. The distributions of all observed short and long encounters are plotted with duration binned logarithmically. (F) The observed start time of each patch encounter is shown for all short and long duration encounters. Long duration encounters occur significantly later (one-tailed Mann-Whitney U-Test). (G) The probability of residing on patch was computed for all worms across time (black) and compared to the probability of residing on patch if patch locations were semi-randomly permuted (pink). Smoothed median values are plotted with bootstrap-derived 2.5% and 97.5% quantiles shown in shaded regions. Time points where observed probabilities of residing on patch significantly exceed permuted probabilities are indicated by a black line (one-tailed Fisher's exact test with Benjamini-Hochberg correction). (H) The track of the example animal in (A) is replotted with color used to represent the animal's instantaneous velocity at each time point. (I) Velocity of the example animal over time is plotted (black) alongside patch encounters (gray) as previously identified in (B). Example encounters – one early, short duration (green) and one late, long duration (blue) – are indicated. (J,K) A 60-second time window surrounding the start of these example encounters is enlarged. (L) Velocity trajectories were aligned to patch entry for every encounter. Mean encounter-aligned (black/gray) and randomly-aligned (pink) trajectories are plotted for each animal (light) and across all animals (dark). (M) Deceleration upon encounter with the patch edge is plotted for every encounter and grouped by duration type. Deceleration was significantly lower for encounter-aligned as compared to randomly-aligned trajectories (pink) for all duration types (one-tailed Mann-Whitney U-Tests with Bonferroni Correction). (N) Mean velocity as a function of the distance from the edge of bacterial patches (computed for 50 μm bins) is shown for every animal (gray) and across all animals (black). (O) Each animal's mean velocity during time spent on and off (midpoint at least ~ 0.46 mm from patch edge) patch is shown. Velocity on patch was significantly slower than off patch (one-tailed paired-sample *t*-test). Sample data for one animal (worm #1) are shown in (A-C,H-K). Summary data for all animals ($N = 50$ worms) and encounters ($N = 419$ total encounters) are shown in (D-G,L-O). Violin plots in (D,F,M,O) give the KDE and quartiles for each measure. Asterisks denote statistical significance ($***p < 0.001$). See also **Figure 1 – supplement 1** and Video 1-4.

1M [↗](#)). Further, consistent with *C. elegans* behavior on larger and more densely seeded patches in other studies [46 [↗](#)], animals in our assay maintained significantly lower average velocities on patch ($\sim 52 \mu\text{m/s}$) compared to off patch ($\sim 198 \mu\text{m/s}$) (Figure 1N,O [↗](#)). These results suggest that, despite detecting the availability of food during early patch encounters, *C. elegans* opt to initially prioritize exploration of the environment before switching to a more exploitative foraging strategy during subsequent patch encounters.

The timing of the switch from exploration to exploitation varies with bacterial density

To determine whether this explore-then-exploit foraging strategy is dependent upon food-related characteristics of the environment, we varied the density of the bacterial patches. By diluting bacterial stocks and controlling growth time, we created 12 bacterial density conditions with relative density ranging from 0 to ~ 200 (Figure 2A [↗](#), Figure 2 – supplement 1 [↗](#)-3 [↗](#)). At the low end of this range (densities less than ~ 0.5), animals do not appear to detect bacterial patches as indicated by on-patch velocities matching those of animals foraging on bacteria-free (density 0) patches (Figure 2B,C [↗](#)). At the high end of the range (density ~ 200), bacterial density is comparable to the environments animals experienced during development and immediately prior to the assay (Figure 2 – supplement 2 [↗](#)-3 [↗](#)).

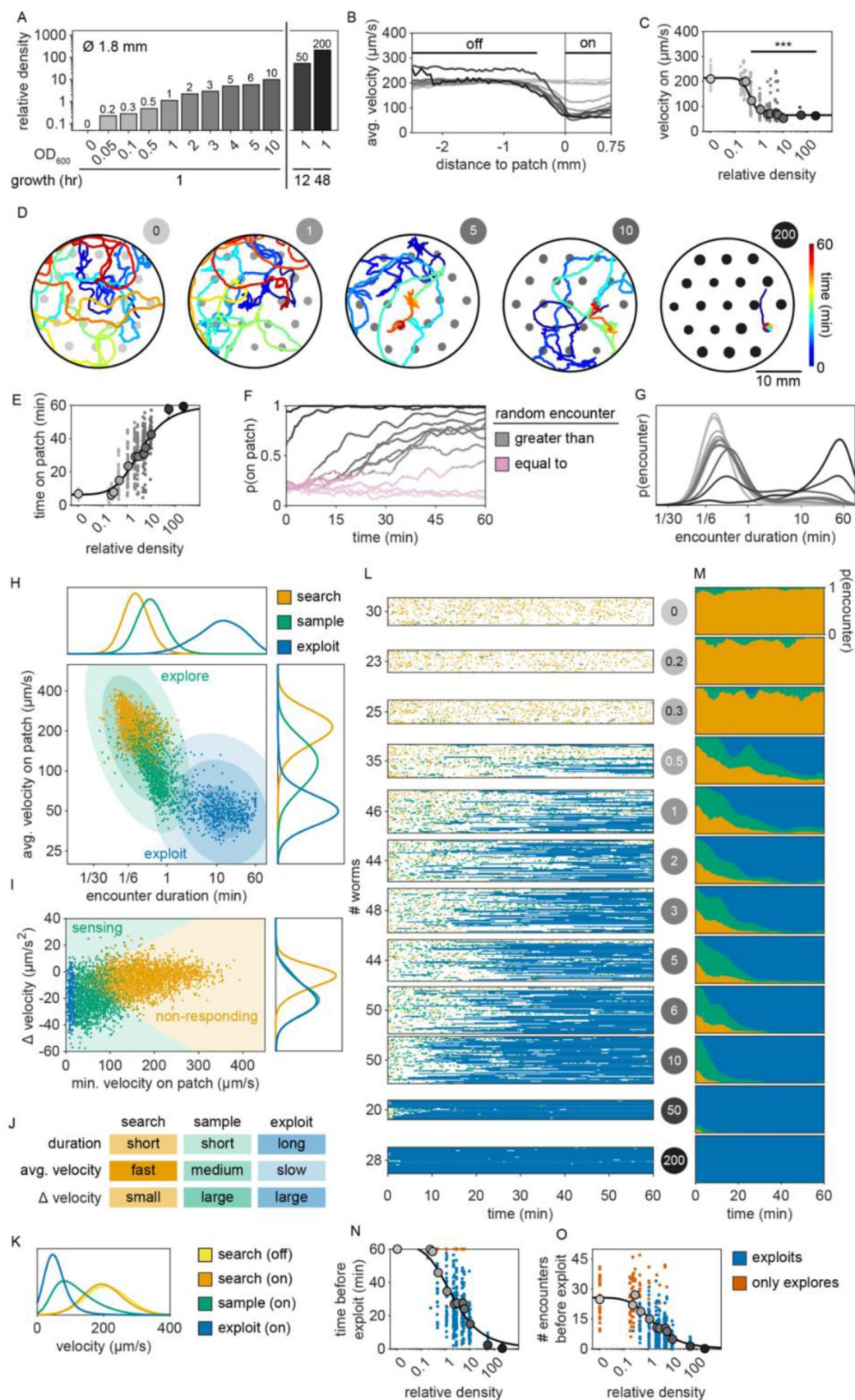


Figure 2

The timing of the switch from explore-to-exploit is density-dependent.

(A) The relative density (as estimated by fluorescently-labeled OP50-GFP) is shown for small (~1.8 mm diameter) bacterial patches made by pipetting ~0.5 μ L droplets of OP50 *E. coli* diluted in LB to a range of optical densities ($OD_{600} = \{0, 0.05, 0.1, 0.5, 1, 2, 3, 4, 5, 10\}$) and controlling growth time at room temperature (hours = $\{1, 12, 48\}$). For (A-G,L-O), gray-scale color saturation is proportional to the relative density of each condition and corresponds to labels in (A). (B) The mean velocities of animals foraging in environments containing patches matching one of the 12 bacterial densities are plotted as a function of the distance from the edge of bacterial patches (computed for 50 μ m bins). (C) Animals' average on-patch velocity is plotted as a function of the relative density of bacteria. Compared to animals foraging amongst bacteria-free patches containing only LB (relative density 0), animals foraging on bacterial patches with relative density of 0.5 or greater display significantly slower on-patch velocities (one-tailed Mann-Whitney U-Tests with Bonferroni Correction). (D) The midbody location (colored to represent time in the experiment) of example animals foraging in environments containing patches (gray) of relative density 0, 1, 5, 10, and 200 are shown. (E) The total time each animal spent on patch is plotted as a function of the relative density of bacteria. Time on patch increased monotonically with increasing bacterial density (Kendall's τ correlation, $p < 0.001$) following a sigmoidal trend. (F) Smoothed median values of the probability of worms residing on patch over time for each density condition are plotted. Time points where observed probabilities of residing on patch either match (pink) or significantly exceed (gray) the probability of residing on patch if patch locations were semi-randomly permuted are indicated (one-tailed Fisher's exact test with Benjamini-Hochberg correction). (G) A KDE of the distribution of encounter durations is plotted for each density condition. (H) For each encounter, the average velocity of the animal during the encounter and the duration of that encounter are plotted on a double-logarithmic plot with color representing the probabilities of clustering classification as *search* (orange), *sample* (green), or *exploit* (blue). Contours showing the first, second, and third standard deviation of the GMM used to classify *explore* and *exploit* encounters are shown as shaded ellipses with saturation corresponding to standard deviation. KDEs for distributions of average on-patch velocity and encounter duration are plotted for each encounter type. (I) For each encounter, the minimum on-patch velocity and maximum change in velocity are plotted. Contours showing the separation of *sensing* and *non-responding* encounters as estimated by semi-supervised QDA are indicated. A KDE for the distribution of the maximum change in velocity is plotted for each encounter type. (J) Features used to classify encounters as *search*, *sample*, or *exploit* are summarized. (K) KDEs of the distributions of animals' velocities are shown for all timepoints during search off and on patch as well as during sample and exploit encounters. (L) Ethograms of patch encounters (colored to represent the probability of classification as *search*, *sample*, and *exploit*) are shown for 443 individuals. (M) The average proportion of each encounter type over time is plotted. (N) Time elapsed and (O) number of encounters occurring prior to the first *exploitation* event are plotted for every animal (blue). In the event that no *exploitation* event occurred, the maximum observed time and encounters are plotted (red-orange). Both time and encounter number before *exploitation* decrease monotonically with increasing patch density (Kendall's τ correlation, $p < 0.001$) following a sigmoidal trend. Summary data for all animals ($N = 443$ total worms; $N = 20$ -50 worms per condition) and encounters ($N = 6,560$ total encounters; $N = 46$ -876 encounters per condition) are shown in (A-C,E-I,K-O). Asterisks denote statistical significance ($***p < 0.001$). See also **Figure 2 – supplement 1** [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) and Video 5-6.

Previous studies have shown that in diet choice assays where animals are given the option between patches of varying quality or density, animals spend more time on preferred patches [31 [1](#), 36 [2](#), 44 [3](#)]. We hypothesized that even when all patches in the environment are of equal density, an efficient forager should still modulate the amount of time spent on the bacterial patches in a density-dependent manner. To test this, behavior was recorded and tracked for one hour as animals foraged in patchy environments matching one of the bacterial density conditions (**Figure 2D** [1](#), **Figure 2** [2](#) – supplement 4). Consistent with our hypothesis, we found that the total time animals spent on bacterial patches increased monotonically with increasing bacterial density following a sigmoidal trend (**Figure 2E** [1](#)). Further, the probability of an animal residing on patch as a function of time was highly dependent on the bacterial density (**Figure 2F** [1](#), **Figure 2 – supplement 5** [1](#)). At the highest densities (50 and 200), animals spent nearly 100% of time in the

assay on bacterial patches, while at the lowest densities (0 to 0.3), animals spent chance levels of time on patch for the duration of the experiment. At intermediate densities (0.5 to 10) animals initially resided on patches at rates predicted by chance, later transitioning to more time spent on patch. Notably, this delayed increase in patch residence is density dependent with animals switching earlier in environments with patches of greater density.

Consistent with our previous observation that animals reside on patches for either short or long durations (**Figure 1E**), we observed two distinct patch encounter types: 1) short (less than 2 minute) patch visits and 2) long (often tens of minutes) patch visits (**Figure 2G**). We validated the existence of two patch types using Silverman's test [47, 48] for bi-modality (**Figure 2 – supplement 6**) and then, using a two-component Gaussian mixture model (GMM), classified all encounters based on the duration of these patch visits and animals' average velocity during the encounter (**Figure 2H**, **Figure 2 – supplement 6**). We identified two behaviorally distinct clusters: 1) short duration and fast velocity (*explore*), and 2) long duration and slow velocity (*exploit*) encounters. The *explore* encounters suggest that an animal has *rejected* a patch, opting to continue exploration of the environment. In contrast, the *exploit* encounters suggest that an animal has *accepted* a patch and is consuming the bacteria within.

While we previously found that animals consistently slowed down upon encounter with the patch edge when foraging in relative density ~ 10 (**Figure 1J–M**), we observed that a large portion of short duration, exploratory encounters were not accompanied by this slow down (**Figure 2 – supplement 7A–B**), especially for the lowest bacterial densities tested. Therefore, to determine which exploratory encounters displayed evidence that the patch was detected by the animal, we further classified encounters by an animal's minimum velocity on patch, deceleration upon patch entry, and maximum change in the animal's velocity upon encounter (i.e., the difference between the peak velocity immediately before the patch encounter and the minimum velocity achieved during the patch encounter) (**Figure 2I**, **Figure 2 – supplement 7**). We again observed two behaviorally distinct clusters: 1) large slow-down and slow on-patch velocity, and 2) small slow-down and fast on-patch velocity. We validated the existence of these two clusters using Silverman's test [47, 48] for bi-modality (**Figure 2 – supplement 7**) and then classified these encounters as *sensing* or *non-responding*, respectively, using a semi-supervised quadratic discriminant analysis (QDA) approach (**Figure 2 – supplement 8**, Video 5). The presence of a slowdown and the subsequent continuation of slower on-patch velocity suggest that an animal sensed the bacteria, while the absence of a slowdown response suggests that the animal either did not perceive the encounter or chose to ignore it. For simplicity, we define: 1) *non-responding* encounters as *searching* (i.e., on-patch behavior that matches off-patch behavior where an animal appears to be searching for food); 2) *sensing*, short duration encounters as *sampling* (i.e., evaluating a patch's suitability as a food source, but ultimately *rejecting* the patch and continuing to explore the environment); and 3) *sensing*, long duration encounters as *exploiting* (i.e., *accepting* a patch and consuming its bacteria) (**Figure 2J**). Consistent with our interpretation of *searching* encounters, the average velocities that animals achieved during *search* on (~ 200 $\mu\text{m/s}$) and off (~ 207 $\mu\text{m/s}$) patch were similar, especially when compared to markedly slower on-patch velocities during *sample* (~ 114 $\mu\text{m/s}$) and *exploit* (~ 57 $\mu\text{m/s}$) encounters (**Figure 2K**). In summary, we have identified three distinct patch encounter types: encounters lacking a slowdown that were either not detected or ignored (*search*) and encounters that were detected with the animal subsequently deciding to *reject* (*sample*) or *accept* (*exploit*) the patch.

We next characterized how the frequency of encounter types varied as a function of time and patch density. The behavior of 443 animals (20–50 worms per condition) was tracked and classified at each time point (**Figure 2L**, Video 6). In low density conditions (0 to 0.3), animals spent nearly all of their time *searching* on or off patch. With increasing bacterial patch density, animals spent less time *searching* and more time *sampling* and *exploiting*. Animals foraging in medium density patches (0.5 to 10) initially spent most of their time *sampling* patches upon encounter. However, animals eventually switched to mostly *exploiting* patches. Animals foraging on high density

patches (50 and 200) almost exclusively *exploited*. Classification of on-patch behavior across all densities tested revealed a consistent switch from *exploratory* behaviors (*searching* and *sampling*) to *exploitation* (Figure 2M) with the timing of the switch occurring earlier for animals foraging in environments with higher bacterial patch density (Figure 2N). The delay in exploitation corresponded to a comparatively large number of initial exploratory encounters with animals on medium density conditions (0.5 to 10) *exploiting* after an average of 4.8 to 18.7 exploratory encounters (Figure 2O). To summarize, for densities similar to those experienced immediately preceding the experiment (50 and 200), animals immediately *exploit* available bacteria. On the other hand, for bacterial densities that appear to be below an animal's sensory threshold (0 to 0.3), animals never *exploit* patches and spend the entire 60-minute assay *searching* for food. When foraging in medium density patches (0.5 to 10), animals initially explore the environment via a combination of *searching* and *sampling* encounters and eventually switch to *exploiting*. Notably, the timing of this switch is density-dependent with animals switching earlier when foraging on higher density patches.

Animals explore before exploiting even when only one patch is available

Given that bacterial patches within an environment were relatively invariable and that bacterial growth during the experiment was negligible (Figure 2 – supplement 3), it was surprising that animals were often willing to *reject* 10+ patches before *exploiting* a patch of the same density as those previously *rejected* (Figure 2O). This *sampling* behavior may be beneficial when food is patchily distributed in the environment as exploiting a dilute patch may result in a lost opportunity for future encounter with a higher quality patch. Thus, we hypothesized that an animal's willingness to *accept* a patch whose density matched that of a previously *rejected* patch could be a specific feature of a patchily distributed environment where numerous observations of individual patches could contribute to learning the features of a changing environment. To address this, we observed whether animals were willing to *reject* a patch numerous times before *accepting* it even when only one patch is available in the environment. We created environments with single large (~8.3 mm) or small (~1.8 mm) diameter patches with relative density ranging from 0 to ~400 (Figure 3A-C). Behavior was tracked for 60 minutes as animals foraged in these environments (Figure 3D-E, Figure 3 – supplement 1-2). We found that even in these single patch environments, animals initially *explore* (*search* and *sample*) before *exploiting* (Figure 3F-G) in a density-dependent manner. Just as when foraging in a patchily distributed environment (Figure 2N), animals *explored* (*search* and *sample*) the small and large single patches numerous times prior to *exploiting* with the timing of this switch occurring significantly earlier with increasing bacterial density (Figure 3H-I). These results suggest that the explore-then-exploit strategy employed by *C. elegans* is not specific to a patchily distributed environment. Rather, animals may *sample* the same patch numerous times before deciding to *exploit*, especially when foraging on dilute patches. The decision of when to stop *exploring* and start *exploiting* is highly dependent upon the density of bacteria within a patch with the switch occurring earlier when animals encounter patches of higher density.

External and internal sensory information guide the decision to explore or exploit

While our behavioral analysis demonstrated that *C. elegans* modulate their decision to *explore* or *exploit* in a density-dependent manner, it remains unclear whether these animals use more complex cognitive processes such as learning and memory to guide their foraging decision. It is plausible that animals are making decisions using simple heuristics given only available sensory information about the current patch. However, previous studies have shown that food-related behaviors in *C. elegans* can be driven by internal states [31] as well as memories of the environment [25, 36, 42]. Thus, to better understand the factors driving the exploitation

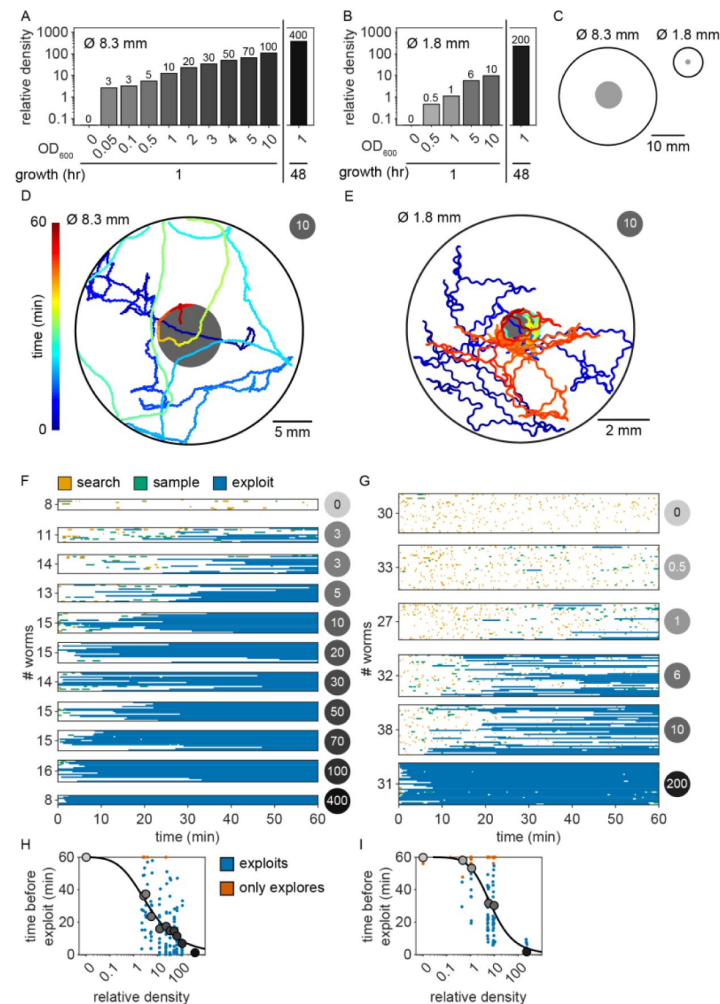


Figure 3

***C. elegans* explore first even when only one patch is available.**

(A) Relative density of large (~8.3 mm diameter) and (B) small (~1.8 mm diameter) bacterial patches was varied by pipetting 20 μ L and 0.5 μ L, respectively, droplets of OP50 *E. coli* diluted in LB to a range of optical densities (OD_{600} = {0, 0.05, 0.1, 0.5, 1, 2, 3, 4, 5, 10}) and controlling growth time at room temperature (hours = {1, 48}). (C) Large patches were formed in the center of 30 mm arenas while small patches were formed in the center of 9 mm arenas. (D-E) The midbody location (colored to represent time in the experiment) of example animals foraging in environments containing a single (D) large or (E) small patch (gray) are shown. (F-G) Ethograms of patch encounters (colored to represent the probability of classification as *search*, *sample*, and *exploit*) for (F) 144 individuals (8-15 per condition) foraging on a single large patch and (G) 191 individuals (27-38 per condition) foraging on a single small patch are shown. (H-I) Time elapsed prior to the first *exploitation* event (blue) for animals foraging on (H) large and (I) small patches is plotted for every animal. When *exploitation* was not observed, time elapsed in the experiment is plotted (red-orange). Animals in large and small patch environments *exploited* higher density patches significantly earlier (Kendall's τ correlation, $p < 0.001$) following a sigmoidal trend. See also **Figure 3 – supplement 1** and **2**.

decision, we implemented a generalized linear model (GLM) to test how foraging decisions are influenced by the density of the current patch as well as additional food-related factors such as an animal's level of satiety and prior experience (**Figure 4A** [↗](#)).

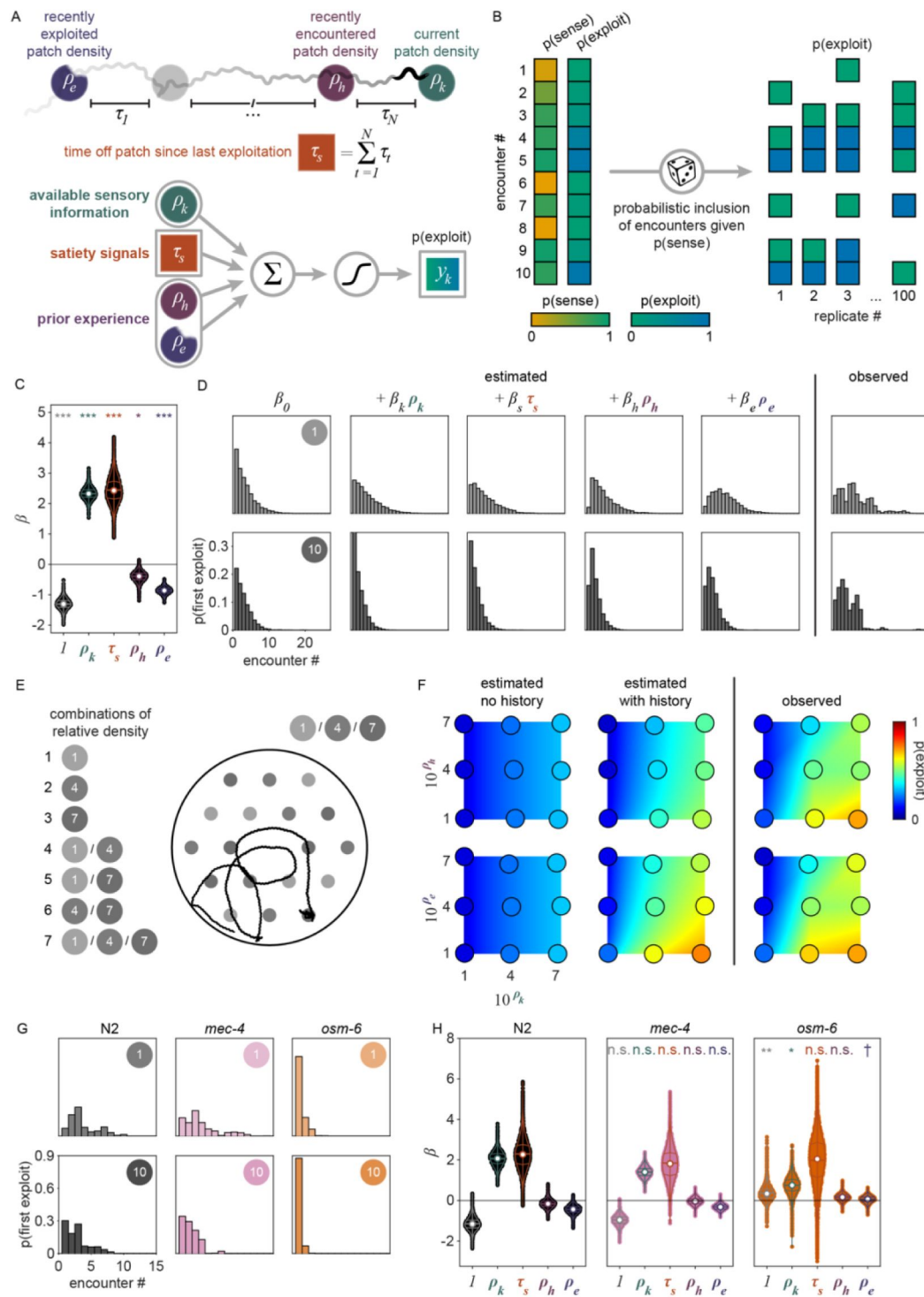


Figure 4

Exploitation decisions are driven by available sensory information, satiety, and prior experience.

(A) Schematic of the covariates $\mathbf{x}_k$ used in our logistic regression model to predict the probability of *exploiting* a patch upon any given encounter $p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k)$. $\mathbf{x}_k$ includes the relative density of the encountered patch k (ρ_k), the duration of time spent off food since departing the last *exploited* patch (τ_s), the relative density of the patch encountered immediately before encounter k (ρ_h), and the relative density of the last *exploited* patch (ρ_e). (B) To account for uncertainty in our classification of sensation, we produced 100 sets of observations wherein we probabilistically included *sensed* encounters ($v_k = 1$) estimated from the distribution $v_k | \mathbf{w}_k \sim \text{Bern}(p(v_k = 1 | \mathbf{w}_k))$ where $p(v_k = 1 | \mathbf{w}_k)$ is the probability that the patch was *sensed* as estimated in [Figure 2I](#). To account for uncertainty in our classification of *exploitation*, we substitute the response variable y_k with our estimate of the probability that the patch was *exploited* $p(y_k = 1 | \mathbf{z}_k)$ as estimated in [Figure 2H](#), a procedure analogous to including *exploitations* ($y_k = 1$) estimated from the distribution $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$. A schematic of these procedures is shown (see Models of exploitation probability). (C) Coefficient values for each covariate in the GLM were estimated across 50000 replicates (500 replicates of hierarchically bootstrapped animals in combination with 100 sets of probabilistically *sensed* encounters). All coefficients are significantly greater than or less than 0 (two-tailed, one-sample bootstrap hypothesis tests with Bonferroni correction). (D) Using observed $p(y_k = 1 | \mathbf{z}_k)$ and estimated $p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k)$ probabilities of *exploitation*, *exploitation* events were simulated from a Bernoulli distribution. Distributions of the probability of first *exploiting* as a function of the number of encounters as estimated by the model with covariates added one-by-one and as observed are shown for animals foraging in single-density, multi-patch environments of relative density 1 or 10. (E) A schematic of multi-density, multi-patch assays where animals foraged in environments containing small (~ 1.8 mm diameter) bacterial patches (gray) of varying combinations of OP50 *E. coli* with relative densities 1, 4, and 7 is shown. (F) The probability of exploitation as estimated with $(p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_h \rho_h + \beta_e \rho_e))$ and without $(p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s))$ the history-dependent terms and as observed $p(y_k = 1 | \mathbf{z}_k)$ is shown for every pairing of current patch density (ρ_k) combined with the density of recently encountered (ρ_h) or *exploited* (ρ_e) patches. Heat maps for observed values were interpolated between the 9 patch density pairings tested. (G) *Exploitation* events were simulated from the distribution $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$ and used to calculate the probability of observing *exploitation* for the first time as a function of the number of encounters for N2, *mec-4*, and *osm-6* animals foraging in single-density, multi-patch environments of relative density 1 or 10. (H) Coefficient values for each covariate were estimated using ridge regression models for each strain across 50000 replicates. A subset of these coefficients significantly varied between wild-type and mutant strains (mean of differences tests with Benjamini-Hochberg correction). Summary data for all animals ($N = 443$ total worms; $N = 20$ -50 worms per condition) and encounters ($\mu = 2659.8$, $\sigma = 15.9$) in the single-density, multi-patch assay are shown in (C,D). Summary data for all animals ($N = 198$ total worms; $N = 20$ -40 worms per condition) and *sensed* encounters ($\mu = 1,724.6$, $\sigma = 9.9$) in the multi-density, multi-patch assay are shown in (F). Summary data for all wild-type and mutant animals ($N = 221$ total worms; $N = 14$ -44 worms per condition) and encounters ($N = 1,352$ total encounters) in a single-density, multi-patch assay are shown in (G,H). Violin plots in (C,H) show the KDE and quartiles for each measure. Asterisks denote statistical significance ($\dagger p_{\text{unadjusted}} < 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). See also [Figure 4 – supplement 1](#).

We considered that the probability of exploiting a patch upon any given encounter can be described by a logistic function:

$$p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k) = \frac{1}{1 + e^{-\boldsymbol{\beta} \cdot \mathbf{x}_k}}$$

where $p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k)$ represents the conditional probability that an animal exploited during patch encounter k ; $\mathbf{x}_k \in R^n$ is a vector of covariates; and $\boldsymbol{\beta} \in R^n$ is a vector of weights that describes how much each covariate influences the animal's choice. In models compared here, $\mathbf{x}_k$ includes a combination of a constant element and covariates that may empirically influence the decision to *exploit*. We considered the simplest model where every encounter is independent with fixed probability of *exploiting* (i.e., $p(y_k = 1 | \beta_0)$) where β_0 represents the average animal's propensity to

exploit. We compared this null model against a set of nested models containing covariates that vary from encounter to encounter and relate to: 1) the relative density of the encountered patch k (ρ_k), 2) the duration of time spent off food since departing the last *exploited* patch (τ_s), 3) the relative density of the patch encountered immediately before encounter k (ρ_h), and 4) the relative density of the last *exploited* patch (ρ_e). Using this model design (i.e., $\beta \cdot \mathbf{x}_k = \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_h \rho_h + \beta_e \rho_e$), we assessed how each of these covariates influences the probability of *exploitation* at each patch encounter k (Figure 4A [↗](#)).

Given that our aim is to understand what factors influence an animal's decision to *explore* or *exploit*, we rationalized that encounters where the animal did not detect the presence of bacteria were unlikely to contribute to decision-making. Therefore, we excluded *searching* encounters from our analysis where animals did *not respond* to the bacterial patch. To account for the uncertainty in our classification of *sensing*, we probabilistically included patch encounters with frequency equal to the probability that the bacterial patch was *sensed* (Figure 4B [↗](#)) as previously estimated by semi-supervised QDA (Figure 2I [↗](#), Figure 2 – supplement 8 [↗](#)). In doing so, we simulated sets of observations of *sensed* (*sample* and *exploit*) encounters. Further, to account for the uncertainty in our classification of patch encounters as *exploration* or *exploitation*, we fit our GLM to the probability of *exploiting* $p(y_k = 1 | \mathbf{z}_k)$ as previously estimated by GMM (Figure 2H [↗](#), Figure 2 – supplement 6 [↗](#)) rather than fitting to direct observations of exploitation y_k (see Models of exploitation probability).

When selecting covariates for our model, we calculated the Bayesian information criterion (BIC) – a model selection test to avoid overfitting by penalizing increases in the number of parameters – and observed the best model performance when all covariates were included (Figure 4 – supplement 1A-B [↗](#)). Further, we found that all covariates (i.e., ρ_k , τ_s , ρ_h , and ρ_e) significantly contribute to the decision to *exploit* $p(y_k = 1 | \beta \cdot \mathbf{x}_k)$ as indicated by coefficient values significantly greater than or less than zero (Figure 4C [↗](#), Figure 4 – supplement 1C [↗](#)). Specifically, we found that β_k and β_s are significantly greater than zero which suggests that animals are more likely to *exploit* patches with increasing patch density ρ_k and duration of food-deprivation τ_s . On the other hand, β_h and β_e are significantly negative, which suggests that the greater the density of a recently encountered or *exploited* patch, ρ_h and ρ_e , the less likely an animal should be willing to *exploit* the current patch.

In order to better understand how each of the covariates in our model affect the *exploitation* decision, we examined the probability of observing *exploitation* for the first time as a function of the cumulative number of patches an animal encountered. Exploitation events were simulated from a Bernoulli distribution with probability estimated by the model (i.e., $y_k | \mathbf{x}_k \sim \text{Bern}(p(y_k = 1 | \beta \cdot \mathbf{x}_k))$) or observed from our classification of *exploitation* (i.e., $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$). Although the probability of *exploiting* for the first time was not expressly fit by our model, this derived metric highlights the model's ability to predict the delay in *exploitation* observed for animals foraging on low to medium density bacterial patches in our experiments (Figure 4D [↗](#), Figure 4 – supplement 2 [↗](#)). For the simplest model where every encounter is independent with fixed probability of *exploiting* $p(y_k = 1 | \beta_0)$, we find that the probability of *exploiting* for the first time is greatest for the first encounter and decreases for every subsequent encounter (Figure 4D [↗](#), Figure 4 – supplement 2 [↗](#)), fitting a geometric distribution.

We next considered whether the observed delay in *exploitation* could be explained by a simple density-dependent model (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k)$). This model predicts that animals are more likely to *exploit* bacterial patches of greater density with probability proportional to the relative density of the current patch ρ_k . While this density-dependent model accurately predicted that the first *exploitation* occurs earlier on average for higher density patches, it also predicted that, regardless of patch density, the first *exploitation* is most frequently observed on the first encounter which did not match the observed distribution (Figure 4D [↗](#), Figure 4 – supplement 2 [↗](#)). We subsequently considered that the onset of food-deprivation during *exploration* of these low-density

environments could drive the observed time-dependent delay in *exploitation*. We assume that animals become satiated during *exploitation* and use the duration of time an animal spent searching off-food since the last *exploitation* event (i.e., τ_s) as a proxy for the animal's likely decrease in satiety. With the addition of this satiety-related signal (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s)$), we observed a slight delay in when animals are predicted to first *exploit* for the lowest density conditions tested (**Figure 4D** [↗](#), **Figure 4 – supplement 2** [↗](#)). To validate our finding that animals use a satiety-related signal to inform decision making, we applied our model to observations of animals foraging in patchily distributed environments following 3 hours of food-deprivation (**Figure 4 – supplement 3A-B** [↗](#)). We found that only with the inclusion of a satiety-related covariate in our model, were we able to reliably predict the immediate *exploitation* that food-deprived animals exhibit (**Figure 4 – supplement 3C** [↗](#)).

We next considered the alternative possibility that this time-dependent signal could actually be related to the physical stress induced by the process of moving animals into the experimental arena rather than the onset of food-deprivation. While we used an agar plug method to mitigate the stress of moving (see Nematode cultures); we sought to examine the impact of satiety as compared to transfer-induced stress on the *exploitation* decision. We quantified both τ_s , the duration of time an animal spent searching off-food since the last *exploitation* event, and τ_t , the duration of time since transfer (i.e., the time elapsed in the experiment). However, as our initial experiments were only one hour, we typically observed only one *exploitation* event. Resultantly, τ_s and τ_t were highly correlated and their individual contributions to the model could not be easily parsed. Therefore, we ran additional experiments with animals foraging in environments with multiple patch densities over two hours (**Figure 4E** [↗](#), **Figure 4 – supplement 4A-C** [↗](#)) which led to observations containing multiple *exploitation* events and a resultant decorrelation of τ_s and τ_t . When the model was re-fit to this two-hour data set using both τ_s and τ_t covariates (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_t \tau_t + \beta_h \rho_h + \beta_e \rho_e)$), only the satiety-related covariate τ_s was statistically significant (**Figure 4 – supplement 4D** [↗](#)). This result suggests that the time-dependent change in *exploitation* is likely due to a satiety-related signal rather than a stress-related signal. Altogether, these results suggest that animals use available external and internal sensory information to guide the decision to *exploit*.

Prior experience guides the decision to *explore* or *exploit*

Although the current patch density and satiety covariates significantly influence the decision to *exploit*, this model (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s)$) does not accurately predict the extent that *exploitation* was delayed. We hypothesized that because animals had experienced very densely seeded bacterial patches immediately prior to the assay, animals may have built an expectation that high density bacterial patches are available in the new environment. We would therefore expect an initial suppression in the probability of *exploiting* while animals updated their expectation using information learned from *sampling* patches in the current environment. To test whether the decision to *exploit* could be influenced by prior experience, we added a history-dependent covariate to our model (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_h \rho_h)$ where ρ_h is the relative density of the patch encountered immediately before encounter k). We find that inclusion of the density of the most recently encountered patch significantly improves our prediction of *exploitation* (**Figure 4D** [↗](#), **Figure 4 – supplement 2** [↗](#)). We then considered whether the animal's expectation of bacterial density in the environment might be a longer lasting memory. Specifically, we considered if the relative density of the most recently *exploited* patch ρ_e could have influenced the animal's decision to *exploit*. The addition of this second history-dependent term to our model (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_h \rho_h + \beta_e \rho_e)$) further improved the fit of our model (**Figure 4D** [↗](#), **Figure 4 – supplement 2** [↗](#)) suggesting that *C. elegans* use recent experiences of encountered and *exploited* patches to guide their decision to *exploit*. Animals may use this learned information to create and update an expectation for the relative density of bacterial patches available in their environment and compare that estimate against the density of a subsequently encountered patch.

Altogether, these results suggest that animals use recent experiences to estimate available resources and modulate their decision to *explore* or *exploit*, a strategy that is highly beneficial for behavioral adaptation in a changing environment.

To validate the role of prior experience in guiding foraging decisions, we observed animals foraging in environments with multiple patch densities (**Figure 4E**, **Figure 4 – supplement 4A–C**). Our model predicts that the lower the density of recently encountered and/or *exploited* patches, the more likely an animal should be to *exploit* a subsequent patch. Without the history-dependent terms ρ_h and ρ_e , our model predicts that no relationship should exist between prior experience and the probability of *exploiting* the current patch. To test which model best fit the animals' behavior, we calculated the conditional probabilities of *exploiting* $p(y_k = 1 | \beta \cdot x_k)$ for this new multi-density, multi-patch data set with more nuanced combinations of current (ρ_k) and recent patch (ρ_h and ρ_e) density using the coefficients β learned previously (**Figure 4C**). We observed significantly better predictions of *exploitation* probability for the model including prior experience as compared to the model without the terms ρ_h and ρ_e (**Figure 4F**, **Figure 4 – supplement 4E**) as confirmed by a likelihood-ratio test. We observed an augmentation in *exploitation* when animals move from low to high density (e.g., 1 to 4 or 1 to 7) as predicted by the model with history-dependence, but not the model without. It is important to note, that this improvement in prediction was achieved without fitting the model to this new multi-density data set. In summary, our modeling results suggest that animals evaluate bacterial density, monitor internal signals of satiety, and leverage recent experiences to drive their decision to *exploit* a bacterial patch upon encounter.

Ciliated sensory neurons are required for the evaluation of patch density

The quantitative modeling applied in this study provides a useful framework for testing the influence of behaviorally relevant features on decision-making. To validate this approach and further elucidate how animals evaluate current and recent patch density, we assessed the foraging behavior of mutants with sensory deficiencies (**Figure 4 – supplement 5**). We found that animals with reduced function of non-ciliated mechanoreceptor neurons due to a null mutation in the *mec-4* gene behave similarly to wild-type N2 animals, displaying increasingly delayed *exploitation* as patch density decreases (**Figure 4G**, **Figure 4 – supplement 5**). Contrastingly, animals with reduced function of ciliated chemoreceptor and mechanoreceptor neurons due to a null mutation in the *osm-6* gene displayed immediate *exploitation* even while foraging on low density patches (**Figure 4G**, **Figure 4 – supplement 5**). These results indicate altered decision-making in the *osm-6* mutants. Specifically, it appears that the influence of patch density on the *exploitation* decision is reduced in *osm-6* mutants and that these animals are more likely to *accept* a patch by default.

To further elaborate on these findings, we fit our GLM to each of the three data sets (i.e., N2, *mec-4*, *osm-6*) and assessed the difference in the estimated distributions of coefficients β between mutant and wild-type models (**Figure 4H**). We hypothesized that the density-related covariates (ρ_k , ρ_h , and ρ_e) would have smaller magnitude (i.e., less influence on decision-making) for animals with reduced sensation of bacterial density. Consistent with this hypothesis, we found a significant reduction in the magnitude of the influence of current patch density (i.e., β_k) on the decision to exploit in *osm-6* mutants. Further, we found a significant increase in the intercept β_0 in *osm-6* mutants which corresponds to the overall increase in *exploitation* observed in these animals (**Figure 4G**, **Figure 4 – supplement 5**). These results suggest that the ciliated sensory neurons play an important role in evaluating the density of the current patch and that when sensation is impaired, animals modify their default behavioral strategy to prioritize *exploitation*. We also observed a marginal increase in our estimate of the influence of recently *exploited* patches (i.e., β_e) in *osm-6* mutants which suggests that reduced efficacy in the evaluation of current patch density also affects learning and memory of patch density. No significant changes were detected in the

mec-4 mutants. Although we cannot rule out a role for non-ciliated mechanoreceptor neurons in contributing to the assessment of patch density, this modality likely provides less salient information. Altogether, these results suggest that ciliated sensory neurons are necessary for the evaluation of patch density and that when the function of these neurons is impaired, animals prioritize *exploitation*. Further, these results demonstrate that this quantitative modeling approach can be used to probe mechanisms underlying decision-making.

Discussion

Like all foraging animals, *C. elegans* must choose between *exploiting* an environment for known resources and *exploring* it for potentially better opportunities elsewhere [5, 7–11]. Previous studies have shown that *C. elegans* use *stay-switch* decisions to efficiently explore a patchily distributed environment and exploit the bacteria within [28, 42–44]. In this study, we show that *C. elegans* also make *accept-reject* decisions while foraging. Specifically, we observe that animals often initially *reject* low density bacterial patches, opting to prioritize exploration of the environment, before switching to a more exploitative foraging strategy during subsequent encounters. This explore-then-exploit strategy was consistent across a range of bacterial patch densities, sizes, and distributions. In order to better understand this phenomenon and identify the factors that contribute to the decision to *exploit*, we leveraged a quantitative modeling approach to hypothesis testing. We found that animals do not employ a default explore-then-exploit strategy; rather, this behavior reflects a series of *accept-reject* decisions guided by multimodal information related to an animal's environment and internal state. Specifically, we show that each decision to *explore* or *exploit* is informed by available sensory information, internal satiety signals, and learned environmental statistics related to the bacterial density of recently encountered and *exploited* patches.

C. elegans make accept-reject foraging decisions

While *C. elegans* have been shown to alter food preferences in a diet choice assay, this behavior has only been described by a set of *stay-switch* decisions [31, 36, 44]. Specifically, *C. elegans* were found to modulate their patch-leaving frequency to *stay* on high quality bacterial patches and *switch* with declining food quality and quantity [26, 36, 40, 41]. The overall effect of leaving low quality patches more frequently is that animals are more likely to reside on high quality patches. Thus, *stay-switch* decision-making enables *C. elegans* to allocate time spent between exploring or exploiting in a manner consistent with optimal foraging theory [5, 7–11]. However, optimal foraging theory also predicts that an animal should never exploit (i.e., always *reject*) a low density patch if higher density patches are sufficiently abundant as inclusion of the low density patch type in the animals' diet decreases the overall rate of energy gained [5, 9]. Thus, while leaving low density patches more frequently drives an overall increase in the rate of energy gained, it can still result in sub-optimal foraging behavior.

Here, we demonstrate that *C. elegans* are capable of making *accept-reject* decisions upon encounter with bacterial patches. We employed quantitative methods – GMM and semi-supervised QDA – to estimate whether animals *sensed* a patch and, if they did, whether they *accepted* or *rejected* it. We found that when transferred from high to low density bacterial patches, animals initially *rejected* several patches, opting to continue exploration of the environment, and that this *reject* decision did not reflect an animal's inability to detect the presence of bacteria in these dilute patches. Further, we found that the decision to *explore* or *exploit* is density-dependent and robust across a range of bacterial densities, distributions, and sizes. The initial preference for *exploration* during foraging on low density patches suggests that animals may be *searching* for denser bacterial patches such as those experienced immediately preceding the assay and throughout development. This explore-then-exploit strategy is advantageous for animals when bacterial patch density is low or variable. On the other hand, we observed immediate *exploitation* of high density

patches. In these environments, an explore-then-exploit strategy is disadvantageous, as delaying *exploitation* when patch density is already high results in spending more energy on food search with no expected increase in energy gained during *exploitation* of a subsequently encountered patch. Thus, consistent with an energy maximizing optimal forager, *C. elegans* initially *explores* when bacterial patch density is low but immediately *exploits* when bacterial patch density is sufficiently high. Altogether, our results demonstrate that *C. elegans* make a distinct decision to either *explore* or *exploit* a patch upon encounter and that the density-dependence of this decision is consistent with predictions from optimal foraging theory.

The concepts of *accept-reject* and *stay-switch* decision-making are well described in foraging theory literature (see Pyke, Pulliam & Charnov 1977 [8] for a review on the subject) and represent the questions “Do you want to eat it?” and, if so, “How long do you want to eat it for?”. These concepts are easily distinguishable for animals that forage using the following framework: 1) search for prey, 2) encounter prey from a distance, 3) identify prey type, 4) decide to pursue (*accept-reject* decision), 5) pursue and capture the prey, 6) exploit prey, and 7) decide to stop exploiting and start searching again (*stay-switch* decision). However, in some scenarios, animals must physically encounter prey prior to identification. In these cases where pursuit and capture are not visualized, it is harder to distinguish between *accept-reject* and *stay-switch* decisions. In our experiments, *C. elegans* do not appear to detect the presence of bacterial patches prior to encounter as slow-down only occurs within one body-length of the patch edge (Figure 2B). Further, we find significant bimodality in encounter duration (Figure 2H) where short duration (*exploratory*) encounters appear to represent a lower bound where animals spend the minimum amount of time possible on a patch (less than 2 minutes), which we interpret as a *rejection* of the patch. On the other hand, *exploitatory* encounters span a large range of durations from 2 to 60+ minutes which we interpret as an initial *acceptance* of the patch followed by a series of *stay-switch* decisions which determine the overall duration of the encounter. While we could certainly model our data using only *stay-switch* decision-making, we ascertain that an encounter of minimal duration is better interpreted ethologically as a *rejection* rather than as an immediate *switch* decision. In accordance with this interpretation, we hypothesize that if *C. elegans* were able to detect the presence of a bacterial patch prior to encountering it, it's possible that animals may be observed to navigate towards (*accept*) or away from (*reject*) a patch. However, it could be the case that, for *C. elegans*, *sampling* the patch is prerequisite to an *accept-reject* decision which would suggest that gustation, mechanosensation, and/or post-ingestive feedback may guide *C. elegans* foraging decisions.

Our finding that *C. elegans* are willing to *reject* numerous encountered patches is somewhat surprising, especially given that *C. elegans* likely possess limited spatial memory [25] and are likely unaware of how many patches are in the environment as evidenced by their willingness to *reject* a patch even when only one patch is available (Figure 3). Taken together, this suggests that animals are “knowingly” passing up an opportunity to *exploit* an immediately available food source for the potential opportunity to *exploit* preferred food later. This type of behavior has been described in humans and other animals in the context of delayed gratification [49, 50], intertemporal choice [49, 51, 52], self-control [53, 54], and optimal stopping [55, 56]. Notably, few animals display the capacity for delayed gratification when given the option between eating an immediately available but less preferred food item and waiting for a preferred food item [57–60]. Even fewer animals are willing to wait if the delayed reward is not visible or increases in quantity rather than quality [61, 62]. Our observation that *C. elegans* can exhibit delayed gratification suggests that this and similar processes require less complex cognition than previously considered or that we may have underestimated the cognitive capacity of *C. elegans*.

External and internal sensory information guide foraging decision-making

Stay-switch decisions in *C. elegans* are strongly modulated by a variety of internal and external sensory cues over a range of behavioral timescales. For example, patch-leaving frequency can be modulated by an animal's prior experience [25, 42], metabolic status, arousal state [31], as well as the presence of pathogens [63, 64], chemorepellents [65], predators [66, 67], and varying levels of environmental O₂ and CO₂ [40, 68].

Here, we used a quantitative framework to show that recent experiences (i.e., the density of current and recently encountered and *exploited* bacterial patches) guide *accept-reject* decision-making. In our study and others [44, 46], *C. elegans* appear to be able to detect the presence of and assess the relative density of bacterial patches. When foraging in environments with low density bacterial patches, we initially observed many *searching* encounters where animals did not appear to slowdown; however, over time, the proportion of *searching* encounters decreased (Figure 2L,M). This increased responsiveness to dilute patches over time could have resulted from small changes in bacterial growth or a modulation of patch detection sensitivity. Contrastingly, animals with reduced function of ciliated chemoreceptor and mechanoreceptor neurons due to a null mutation in the *osm-6* gene almost never *searched*. Rather, they displayed immediate *exploitation* even while foraging on low density patches. This suggests that *osm-6* mutants display an augmented ability to detect bacterial patches but a diminished ability to assess the bacterial density of those patches. In the absence of the ability to assess bacterial density (i.e., all patches appear equally dense), it is reasonable that animals should prioritize *exploitation* as *exploration* would not be expected to result in an encounter with a better patch. Therefore, it is not surprising that *osm-6* mutants default to *exploitation*. Collectively, these results suggest an important role for sensory modalities in the detection and assessment of bacterial patch density and highlight a need to investigate the mechanisms behind these effects in future studies.

The difficulty in assessing bacterial patch density is more than just a sensory challenge. Within-patch spatial inhomogeneity resulting from areas of active proliferation of bacteria [43, 69] complicate an animal's ability to accurately assess the quantity of bacteria within a patch and, consequently, our ability to accurately compute a metric related to our assumptions of what the animal is sensing. In our study, we used the relative density of the patch edge where bacterial density is highest as a proxy for an animal's assessment of bacterial patch density (Figure 2 – supplement 1). This decision was based on a previous finding that the time spent on the edge of a bacterial patch affected the dynamics of subsequent area-restricted search [25]. While within-patch spatial inhomogeneity likely affects an animal's ability to assess patch density, we do not believe that this qualitatively affects the results of our study. Both the patch densities tested (Figure 2 – supplement 3A) as well as our observations of time-dependent changes in *exploitation* (Figure 2E,N-O; Figure 3H-I) maintained a monotonic relationship. Therefore, alternative methods of patch density estimation should yield similar results.

According to early models in optimal foraging theory, foragers behave as if they have *complete information* about the environment (e.g., animals are assumed to know the average density of bacterial patches) [9]. Later theoretical and empirical work expanded models of patch choice to accommodate situations where animals have *incomplete information* (e.g., environments with fluctuating resources) and, therefore, must both acquire information and forage [8, 10, 70]. This work suggests that animals may continuously sample areas in their environment to keep track of resource availability. Here, we found that animals initially choose to *sample* patches and that the duration of this period of *sampling* is correlated with the degree of mismatch between current and recently experienced and *exploited* patches (i.e., animals spend more time *sampling* as the magnitude of difference between the patch density experienced immediately before and during the experiment increases). When there is no difference between current and prior

experience (i.e., animals foraging on patches of relative density 200), animals rarely *sample*. These results suggest that *C. elegans* use *sampling* as a way of re-evaluating the overall quality of patches available in the environment when a change is detected. Further, the results of our model suggest that animals likely use the information learned during *sampling* encounters to guide their decision to *explore* or *exploit* on subsequent patch encounters. We behaviorally validated this hypothesis by applying our model predictions to animals foraging in environments with multiple patch densities. Consistent with our predictions, we observed that the lower the density of recently encountered or *exploited* patches and the higher the density of the current patch, the more likely an animal would be to *exploit* (**Figure 4F**). These results suggest that *C. elegans* can learn and remember features of recent experiences and use that learned information to guide efficient decision-making. Incorporating prior experience into the decision to *explore* or *exploit* currently available food is highly beneficial when foraging in a fluctuating environment. *C. elegans* thus may have evolved this ability to maximize foraging in the wild where animals experience a boom-and-bust environment [71]. Future studies investigating the effects of varying patch density during animal development could provide additional insights into the effects of more distant experiences on the *explore-exploit* decision. However, testing this is non-trivial as maintaining stable bacterial density conditions over long timescales requires matching the rate of bacterial growth with the rate of bacterial consumption.

In addition to external sensory information, internal state signals serve a critical role in modifying decision-making in *C. elegans* [31, 72]. For example, food-deprived animals are more likely to cross an aversive barrier (e.g., copper, primaquine, glycerol) in search of food as compared to well-fed controls [72, 73]. Here, we find that a satiety-related signal likely drives the decision to *exploit*. Specifically, our model suggests that as animals' satiety decreases, their willingness to accept a lower quality patch increases. We confirmed this prediction by analyzing the behavior of food-deprived animals, which immediately *exploit* even when bacterial patches are very dilute (**Figure 4 – supplement 3**). Further, we showed that this temporal signal was not likely to have been caused by transfer-induced stress (**Figure 4 – supplement 4D**). Combined with the effect of prior experience, these results suggest that the decision to *explore* or *exploit* likely requires integration of food-related internal and external cues. Future studies should probe if additional factors related to the animal's biotic and abiotic environment [35, 74–77] as well as alternative motivations (e.g., reproduction and predatory avoidance) [78–81] further modify these foraging decisions.

Quantitative and ethological approach to investigating decision-making

Naturalistic observations of *C. elegans* behavior are necessary to achieve a more complete cellular, molecular, and genetic understanding of how the nervous system has evolved to drive animal behavior. For example, large patches of densely seeded bacteria have primarily been used in experiments assessing *C. elegans* foraging [20, 21, 82–84]. However, wild nematodes experience a boom-and-bust environment [71] that may be more consistent with sparse, patchily distributed bacteria [28]. Thus, just as quiescence is only observed on high quality bacterial patches [85], additional behavioral states may be discovered when assessing foraging in dilute patchy environments. By conducting detailed analyses of ecologically inspired foraging behaviors in individual animals, we discovered that *C. elegans* make *accept-reject* decisions upon encounter with bacterial patches. By studying foraging in these environments containing small, dilute bacterial patches, we increased the frequency of opportunities for *accept-reject* decision-making (i.e., patch encounters and patch-leaving events are more frequent) and decreased the condition-specific bias towards *accept* decisions (i.e., animals are less likely to *accept* patches when moved between environments of differing bacterial density). We expect that further investigation into naturalistic behaviors could reveal additional insights into more complex behaviors and decision-making in *C. elegans* and other animals.

In this study, we developed an assay that enables observation of ecologically relevant decision-making in freely moving animals. We posit that future studies can leverage this assay to investigate the neuronal mechanisms underlying decision-making. The frequency of decision-making opportunities (i.e., numerous *accept-reject* decisions could be observed in the span of minutes) and consistency of patch choice decisions across hundreds of animals and a range of environmental conditions tested supports this claim. Further, we anticipate that the *accept-reject* decision occurs during a narrow time window – possibly at the time of encounter with the patch edge or shortly thereafter. Therefore, we expect that this assay could be adapted for experiments utilizing calcium imaging in freely-moving animals [17 [↗](#), 30 [↗](#), 86 [↗](#)–90 [↗](#)] to monitor neuronal activity as animals make these decisions. Additionally, the foraging task used here could be leveraged as an accumulation of evidence paradigm, which has been predominantly studied using mammalian animal models [91 [↗](#)].

Finally, we used simple quantitative models to dissect and validate the components underlying decision-making. Specifically, we employed a logistic regression GLM which we adapted to accommodate our uncertainty in classification of *sensing* and *exploitation* for each encounter. While alternative models may provide more mechanistic predictions, the GLM enabled us to conduct a sequence of hypothesis tests probing the influence of food-related signals on the decision to *exploit*. Indeed, we conducted a great number of analyses involving model selection and optimization. Ultimately, we focused our study on asking whether a set of simple covariates could be used to predict foraging decisions. The GLM enabled us to identify the simplest model that could explain the observed delay in *exploitation*. While even the null model predicted that, on average, numerous *exploratory* encounters were likely to occur before the first *exploitation*, models including satiety and prior experience significantly better predicted the extent of this delay. Further, we demonstrated that we could validate these model predictions in food-deprived animals, animals exploring multiple patch densities, and animals with sensory deficiencies. We regard our study as an initial foray into demonstrating *accept-reject* decision-making in nematodes. The exact mechanisms and, consequently, the best model design require further investigation. For example, it is possible that the density of recently *exploited* patches is predictive of subsequent *exploitation* because post-ingestive feedback is required for the memory and/or because more distant experiences are remembered (i.e., long-term memory is involved). Future studies should seek to differentiate between these possibilities and characterize the time window of the memory. Altogether, we suggest that this quantitative and ecologically inspired approach to investigating behavior and decision-making is incredibly powerful for refining hypotheses and enables subsequent investigation into the underlying cellular, molecular and circuit pathways in *C. elegans*, a strategy that could be adopted across species.

Materials and Methods

Key Resources Table

Reagent type (species) or resource	Designation	Source or reference	Identifier
bacterial strain (<i>Escherichia coli</i>)	OP50	Caenorhabditis Genetics Center	WBStrain00041969
bacterial strain (<i>Escherichia coli</i>)	OP50-GFP	Caenorhabditis Genetics Center	WBStrain00041972
organism/strain (<i>Caenorhabditis elegans</i>)	N2: wild isolate	Caenorhabditis Genetics Center	WBStrain00000001
organism/strain (<i>Caenorhabditis elegans</i>)	PR811: [osm-6(p811) V]	Caenorhabditis Genetics Center	WBStrain00030796
organism/strain (<i>Caenorhabditis elegans</i>)	TU253: [mec-4(u253) X]	Caenorhabditis Genetics Center	WBStrain00035037
software, algorithm	StreamPix 8	NorPix	RRID:SCR_015773
software, algorithm	WormLab	MBF Bioscience	RRID:SCR_017669
software, algorithm	MATLAB 2024a	Mathworks	RRID:SCR_001622
software, algorithm	Photoshop 2024	Adobe	RRID:SCR_014199
software, algorithm	Illustrator 2024	Adobe	RRID:SCR_010279
software, algorithm	Code for analysis	https://github.com/shreklab/Haley-et-al-2024	

Data and Code Availability

All data reported in this paper will be shared by the lead contact upon request. All original code has been uploaded to GitHub (<https://github.com/shreklab/Haley-et-al-2024>) and are publicly available as of the date of publication. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request. Source data files contain the summarized data for all plots.

Bacterial cultures

Stock liquid cultures of the OP50 strain of *Escherichia coli* were prepared via inoculation of a single colony in sterile lysogeny broth (LB) grown overnight at room temperature. Stock liquid cultures were subsequently stored at 4°C for up to 6 weeks.

Solutions of OP50 for each experiment were prepared via a series of dilutions in LB (**Figure 1 – supplement 1A**). 50 mL of OP50 stock solution was centrifuged at 3000 rpm for 5 minutes. After removal of supernatant, approximately 500 to 1000 µL of saturated liquid culture remained. The bacterial density of this saturated solution was estimated by measuring the optical density at 600 nm (OD₆₀₀) of a 1:50 dilution of the homogenized solution on a spectrophotometer (Molecular Devices SpectraMax Plus 384). The saturated solution was subsequently diluted to achieve an OD₆₀₀ of ~10 ($\mu = 10.22$, $\sigma = 0.31$). Measurements of the number of colony forming units in the “10” solution estimated 13.1×10^9 cells per mL on average. Additional densities (OD₆₀₀ = {0.05, 0.1, 0.5, 1, 2, 3, 4, 5}) were prepared via dilution of the “10” solution with LB and kept on ice to prevent bacterial growth. A “0” density solution was prepared with just LB.

For all experiments, these density solutions were seeded onto cold, low-moisture nematode growth medium (NGM) plates (3% agar) to facilitate pipetting of small, circular, quick-drying patches. Unless otherwise noted, seeded plates were immediately returned to 4°C after patches dried to prevent bacterial growth. Plates were stored at 4°C for an average of 6 days before experimentation.

For experiments using an OP50 strain expressing green fluorescent protein (OP50-GFP) [92], cultures were prepared as above, but with the addition of 100 µg/mL of carbenicillin to the liquid LB. All bacterial strains used in this study are listed in the Key Resources table.

Nematode cultures

Caenorhabditis elegans strains were maintained under standard conditions at 20°C on NGM plates (1.7% agar) seeded with stock liquid culture of OP50 [16]. All experiments were performed on young adult hermaphrodites, picked as L4 larvae the day before the experiment ($\mu = 24.6$ hours, $\sigma = 3.4$). Unless otherwise indicated, well-fed animals of the standard *C. elegans* strain N2 Bristol were used for experiments. Transgenic strains were always compared to matched controls tested in parallel on the same days. All *C. elegans* strains used in this study are listed in the Key Resources table.

Assay preparation and recording

Single-density, multi-patch assay

Several days before the experiment, condition and acclimation plates were prepared (**Figure 1 – supplement 1A**). For the condition plates, a large circular arena (30 mm in diameter) made of clear transparency film (6 mil, PET) was cut using a computer-controlled cutting machine (Cricut Maker 3) and placed in the center of a 10 cm petri dish filled with 25 mL of NGM (3% agar). These arenas function to corral animals with a low probability of escape [93]. A pipetting template made of the same material was designed and cut with an isometric grid consisting of 19 circular holes spaced 6 mm apart from center-to-center. The pipetting template was overlaid on top of the arena and 0.5 µL of OP50 solution ($OD_{600} = \{0, 0.05, 0.1, 0.5, 1, 2, 3, 4, 5, 10\}$) was pipetted into each hole. After drying, the template was immediately removed, and the petri dish lid replaced to prevent contamination and dehydration. For two conditions (“1 (12H)” and “1 (48H)”), plates were seeded with $OD_{600} = 1$ and then left at room temperature for 12 or 24 hours, respectively, prior to storage at 4°C. For all other conditions, plates were immediately stored at 4°C after drying. Acclimation plates were prepared by seeding NGM (3% agar) plates with one large patch of 200 µL of $OD_{600} = 1$ grown for 24 hours at room temperature.

Approximately 24 hours before the experiment, acclimation plates and “1 (48H)” condition plates were transferred from 4°C to room temperature. After plates warmed to room temperature (~1 hour), 30–60 L4 animals were picked onto the acclimation plates. Plates were then stored at 20°C until the experiment for a combined bacterial growth time of approximately 48 hours (24 hours from freshly seeded to 4°C; 24 hours from 4°C to experiment). The combined bacterial growth time for all condition plates was approximately 1 hour ($\mu = 1.16$ hours, $\sigma = 0.23$) unless otherwise noted as “12” ($\mu = 13.63$ hours, $\sigma = 0.50$) or “48” hours ($\mu = 49.28$ hours, $\sigma = 2.94$).

On the day of the experiment, each set of condition plates was transferred from 4°C to room temperature. After one hour, 4 young adult animals were gently transferred to an empty NGM plate to limit the spread of bacteria from acclimation to condition plates. Animals were then quickly transferred to a condition plate. Animals were transferred using the flat surface of cylindrical plugs excised from clean 3% agar [66, 93]. Importantly, the use of agar as a medium to transfer animals provides minimal disruption to their environment as all physical properties (e.g. temperature, humidity, surface tension) are maintained. Qualitatively, we observe no marked change in behavior from before to after transfer with the agar plug method, especially as compared to the often drastic changes observed when using a metal or eyelash pick.

Condition plates were subsequently placed face down on a piece of glass suspended above an edge-lit backlight (Advanced Illumination). Recordings were acquired using Pixelink cameras (PL-B741F) combined with Navitar lenses (1-60135 and 1-6044) and Streampix 8 software. Behavior

was recorded for one hour at 1024x1024 pixels and 3 frames per second (fps) with spatial resolution of ~33 pixels per mm ($\mu = 32.68$ pixels/mm, $\sigma = 1.70$).

The assay was repeated over numerous days with every condition assayed on each day when possible. The order of seeding assay plates and recording behavior was randomized for each day to compensate for the potential effects of time on animal age and bacterial density. Further, the average temperature ($\mu = 21.99^\circ\text{C}$, $\sigma = 0.95$) and humidity ($\mu = 51.39\%$, $\sigma = 11.67$) during each experiment were recorded. As a result of randomization, animal age, bacterial growth time, temperature, and humidity did not significantly vary between density conditions. Replication of the assay was constrained by the set-up (~4 hours per day), recording (1-2 hours per animal), and analysis (0.5-1 hours per animal) time required. A maximum of 24 animals (2 cameras run for 12 hours) could be assayed per day. Given these resource constraints, we collected replicates until saturation of qualitative observations (i.e., new animals merely replicated earlier observations without adding new information) [94] resulting in sample sizes comparable to similar studies [28, 42].

Large, single-patch assay

Plates were prepared as described for the single-density, multi-patch assay with the exception that only one large patch was created by pipetting 20 μL of OP50 solution ($\text{OD}_{600} = \{0, 0.05, 0.1, 0.5, 1, 2, 3, 4, 5, 10\}$) into the center of the 30 mm diameter arena.

Small, single-patch assay

Plates were prepared as described for the single-density, multi-patch assay with the exception that only one small patch was created by pipetting 0.5 μL of OP50 solution ($\text{OD}_{600} = \{0, 0.5, 1, 5, 10\}$) into the center of a 9 mm diameter arena. Given the smaller arena size, only one young adult animal was transferred into the arena. Behavior was recorded for one hour at higher spatial ($\mu = 104.76$ pixels/mm, $\sigma = 17.6$) and temporal resolution (8 fps). The acquisition set-up was otherwise unaltered except for the removal of a 0.25x Navitar lens (1-6044) from the light path.

Multi-density, multi-patch assay

Plates were prepared as described for the single-density, multi-patch assay with the exception that varying combinations of $\text{OD}_{600} = \{1, 5, 10\}$ (i.e., $\text{OD}_{600} = \{1, 5, 10, 1+5, 1+10, 1+5+10\}$) were pipetted onto each assay plate in a patterned isometric grid of 18 0.5 μL droplets. Behavior was recorded for two hours. To compensate for the longer duration recordings, NGM agar plates were poured without BactoTM peptone (BD 211677) to prevent bacterial growth. The result of this change was that the relative density of bacterial patches was ~30% lower as compared to comparable 0.5 μL patches in the single-density, multi-patch assay and the small, single-patch assay (**Figure 2 – supplement 2-3**). Further, to prevent censorship of our data at the beginning of the recording, we formed a small droplet of S-Complete solution [95] in the middle of the arena where no bacterial patch was placed. Animals were transferred to this droplet using an eyelash pick immediately prior to the assay plate being placed on the imaging set-up. The recording was started once the droplet evaporated and animals were free to crawl about the arena. This process ensured that recordings began prior to an animal's first patch encounter.

Bacterial patch location detection

Most bacterial densities tested in these experiments were too dilute to be visible under normal imaging conditions. Therefore, several strategies were employed to accurately detect the location of bacterial patches. First, a small reference dot was cut into the arena and pipetting templates enabling consistent and traceable patch placement. Further, prior to each behavior recording, a “contrast” video was acquired wherein a piece of dark cardstock was passed between the light source and the condition plate (**Figure 1 – supplement 1B-D**, Video 2). This produced an effect

where diffraction of light through the patches provided enough contrast to view the patches. Binary masks of the circular arena and patches were extracted from these videos in MATLAB using the Image Processing Toolbox and further refined manually in Photoshop (Adobe, 2024). Arena masks were used to calculate the scale (pixels/mm) of each image. In early experiments, the “contrast” video was acquired prior to worms being added to the condition plate, which resulted in displacement of the arena within the camera’s field-of-view. Image registration using MATLAB’s Computer Vision Toolbox was performed to accurately map the patches detected in the “contrast” video onto the behavioral recording.

Bacterial patch density estimation

As described above (see Assay preparation and recording), we varied the relative density of bacterial solutions by diluting OP50 in LB and growing patches for different lengths of time. As a result of these procedures, the relative bacterial density throughout the assay was not known and needed to be estimated. To determine the relative density of these bacterial patches, we seeded plates with fluorescently labeled OP50-GFP under identical conditions as in our assays (**Figure 2 – supplement 1A** [↗](#)). We imaged these bacterial patches for several hours using a Zeiss Axio Zoom.V16. Images were analyzed using the Image Processing Toolbox in MATLAB. After correcting for inconsistent illumination across the field-of-view and normalizing images using matched controls (**Figure 2 – supplement 1B-C** [↗](#)), we detected the location of each bacterial patch and extracted a fluorescence intensity profile along each patch’s radius (**Figure 2 – supplement 1D** [↗](#)). Consistent with previous studies, we found that even in extremely dilute conditions, bacterial density was greater at the patch edge where actively proliferating bacteria are concentrated [43 [↗](#), 69 [↗](#)]. Therefore, we detected the patch edge and computed the peak amplitude of each bacterial patch at all time points assayed (**Figure 2 – supplement 1E** [↗](#)). We then performed linear regression on the values for each condition to create models of peak amplitude as a function of time. For low density conditions (i.e., $OD_{600} = \{0.05, 0.1\}$) of small patches, fluorescent signals could not be detected. We therefore estimated these functions from a multinomial regression of values taken from $OD_{600} = \{0.5, 1, 2\}$. Through this process, we used the total time that bacteria were allowed to grow at room temperature for each condition plate to estimate the peak amplitude at each time point (**Figure 2 – supplement 3** [↗](#)). Finally, to minimize confusion about density conditions between experiments, we computed values for relative density by linearly scaling estimated peak amplitude values so that 0.5 μ L patches seeded with $OD_{600} = 10$ and grown for 1 hour at room temperature would have relative density equal to 10. We labeled each condition using the relative density rounded to one significant digit.

Behavioral tracking

WormLab[®] (MBF Bioscience) was used for tracking animal behavior (**Figure 1 – supplement 1E-G** [↗](#), Video 3). For each video, we adjusted thresholds for pixel intensity and worm dimensions to enable automatic tracking. WormLab then fit the worm’s body at every frame and stitched together a track of each worm across frames. Subsequent manual corrections were required for instances where the software created new worm tracks. This occurred most frequently when animals: 1) encountered the arena border, 2) collided with each other, or 3) overlapped with bubbles or dust particles embedded in the agar. To correct for these discontinuities, we manually “stitched” worm tracks together enabling us to keep track of each animal’s location throughout the duration of the experiment. We exported the location of each animal’s midpoint for further analysis in MATLAB (Mathworks, 2024a). For small, single-patch assays where higher spatial resolution enabled reliable distinction between the head and tail of the animal, we manually confirmed the head-tail and exported 25 points along the animal’s midline (**Figure 1 – supplement 2A-C** [↗](#)).

Animals were excluded from analyses when 1) the animal’s location could only be tracked for less than 75% of the video due to the animal escaping the arena and when 2) the location of bacterial patches could not be reliably assessed due to missing or low-quality “contrast” videos (see

Bacterial patch location detection).

Patch encounter detection

As the head position of animals in our experiments was difficult to track due to low spatial resolution and limited automated detection of head position using WormLab, we instead tracked the midpoint position of the worm's body and estimated patch encounters using a set of measured criteria. To do this, we tracked the behavior of worms in higher resolution (~ 105 pixels/mm compared to ~ 33 pixels/mm and 8 fps compared to 3 fps), single patch assays (**Figure 1 – supplement 2A-C** [↗](#)). We found that when the head of the animal was in contact (i.e., within 1 pixel) with the patch edge (see Bacterial patch location detection), the animal's midpoint was on average 0.46024 mm away (**Figure 1 – supplement 2D** [↗](#)). We therefore defined a patch encounter as the time when an animal's midpoint came within 0.46024 mm of the patch edge. However, this definition led to two types of errors: 1) putative entry events where the animal merely passed by the patch and 2) putative leaving events where the animal did not fully leave the patch.

To remove putative entry events where the animal nearly passed by the patch, we defined an additional distance threshold based on the distance between the animal's midpoint and the patch edge for all time points when the head was within the patch (**Figure 1 – supplement 2E** [↗](#)). The midpoint was almost always (99% of time) within 0.28758 mm of the patch edge. Therefore, we removed patch encounters where the midpoint never got closer than 0.28758 mm from the patch edge. Although this criterion removes most “near miss” events, some events remain where the animal approached the patch, but its head never entered. Further decreasing this threshold (e.g., midpoint must be on patch) would result in real events being excluded due to the animal taking on specific postures where the midpoint remains outside the patch while feeding within the patch. Therefore, we chose to be more liberal (i.e., more false positive patch encounters than false negatives) in our detection of patch encounters. Subsequent analysis of these putative patch encounters removed the majority of remaining false positives (see Patch encounter classification as sensing or non-responding).

To address putative leaving events and avoid splitting a single patch encounter into multiple encounters, we considered that a lawn leaving event is most frequently defined as an event where all body parts have left the patch [\[31\]](#) [↗](#). When tracking only the midpoint, it is not possible to ensure that the worm's entire body has exited the patch. Using the criterion that an animal exits a patch when its midpoint exceeds 0.46024 mm from the patch edge accurately predicts exit events most of the time, but under certain scenarios (e.g., animals maintain an outstretched feeding posture [\[66\]](#) [↗](#)) false leaving events were detected. To exclude these leaving events, we analyzed the variability in the distance from the midpoint to the patch edge during on- and off-patch events (**Figure 1 – supplement 2F** [↗](#)). We found that on-patch events had low distance variability while off-patch events displayed a bimodal distribution, with some events having high variability as expected, and some events having low variability. We fit a two-component Gaussian mixture model to the off-patch distance variability and found that low variability leaving events could be reliably excluded when the standard deviation of off patch distances was less than 0.22221. We subsequently combined putative patch encounters where variability was below this threshold, resulting in fewer overall encounters.

Patch encounter classification as exploration or exploitation

To classify patch encounters as *exploration* or *exploitation* events, patch encounters were detected (see Patch encounter detection) and two features were computed: 1) duration of patch encounter and 2) average on-patch velocity. Both the duration and velocity features displayed a bimodal distribution with two visible clusters: one with short duration encounters and high on-patch velocity and one with long duration encounters and low on-patch velocity (**Figure 2 – supplement 6A** [↗](#)). To confirm the existence of two clusters, we conducted a modified version of Silverman's test for bimodality [\[47\]](#) [↗](#), [\[48\]](#) [↗](#). Broadly, Silverman's test uses kernel density estimation (KDE) to

investigate the number of modes in a distribution. By solving for a critical bandwidth at which the KDE switches modality from j to $j + 1$ modes, one is able to test the null hypothesis that a distribution has j modes, versus the alternative that it has more than j modes. Following a previously described protocol [47, 48], we tested the null hypothesis that our data set was unimodal (i.e. $j = 1$). We first mean-centered the matrix $\mathbf{z}_k$ containing the log transforms of the duration and average velocity of each patch encounter k . We then calculated the projection of each data point onto the first principal component (PC1) of the data to get a one-dimensional distribution representing the data in $\mathbf{z}_k$ (Figure 2 – supplement 6B). The use of principal components instead of principle curves as described previously was justified as both methods are indistinguishable when $j = 1$ [47]. Two modes are visible in both the histogram and KDE of the distribution of PC1 projected data when the KDE is computed using a Gaussian kernel and Silverman's rule of thumb for bandwidth [96] defined as $\hat{h} = \sigma \left(\frac{4}{(d+2)n} \right)^{\frac{1}{d+4}}$, where, for this data set, σ is the standard deviation, $d = 1$ is the dimensionality, and $n = 6560$ is the number of observations. To test the significance of this distribution being bimodal rather than unimodal, we computed the critical bandwidth h^* defined as the minimum bandwidth for which the KDE has only one mode. Following Silverman's test procedure, we computed a smoothed bootstrap across 2000 replicates by sampling the data with replacement and adding noise such that for any data point z and bootstrap resample b , we make noisy $\tilde{z}_{i,b} = \frac{z_i + h^* \epsilon_{i,b}}{\sqrt{1 + \left(\frac{h^*}{\sigma}\right)^2}}$ where $\epsilon_{i,b}$ where $\epsilon_{i,b}$ is a standard normal

random variable independently drawn for each i and b . We found that the critical bandwidth h^* of our data set was significantly ($p < 0.001$) greater than that of the bootstrapped data sets which allows us to reject the null hypothesis that the distribution of $\mathbf{z}_k$ is unimodal and justifies the use of a two-cluster model in classifying these data (Figure 2 – supplement 6C).

After confirming the bimodality of the data set, we classified patch encounters using a two-component Gaussian mixture model (GMM) to estimate $p(y_k = 1 | \mathbf{z}_k)$ where for each encounter k , $\mathbf{z}_k$ are the log transforms of the duration and average velocity of the patch encounter, $y_k = 1$ indicates an *exploitation*, and $y_k = 0$ indicates an *exploration* (Figure 2 – supplement 6D). We optimized the GMM across cross-validated replicates with respect to the regularization value α to minimize the posterior variance given by

$$\sum_k p(y_k = 1 | \mathbf{z}_k) p(y_k = 0 | \mathbf{z}_k)$$

We found that $\alpha = 0.025$ best separated the *exploration* and *exploitation* clusters (Figure 2 – supplement 6E). The posterior probability of *exploitation* $p(y_k = 1 | \mathbf{z}_k)$ was subsequently estimated for all encounters (Figure 2 – supplement 6F-G).

Patch encounter classification as sensing or non-responding

To classify patch encounters as *sensing* or *non-responding* events, we computed animals' 1) deceleration upon encounter with the patch edge, 2) minimum velocity during the patch encounter, and 3) maximum change in velocity between the peak velocity immediately prior to the start of the patch encounter and the minimum velocity during the encounter (Figure 2 – supplement 7A-B). Deceleration was defined as the slope of the line fit on an animal's velocity between 1.5 seconds before and 6.5 seconds after the start of an encounter. Minimum velocity was defined as the absolute minimum velocity observed during the duration of the patch encounter. The maximum change in velocity was computed by subtracting the minimum velocity from the peak velocity within 10 seconds of the patch encounter. The combination of these three metrics reveals two non-Gaussian clusters confirmed by Silverman's test ($p = 0.003$) as described in the previous section (Figure 2 – supplement 7C-F). These clusters represent encounters where animals sensed the patch (i.e., slow minimum velocity, large changes in velocity) or did not respond to it (i.e., maintained fast minimum velocity, with little to no change).

We estimated the probability of *sensing* $p(v_k = 1 | \mathbf{w}_k)$ where for each encounter k , $\mathbf{w}_k = (s_k, t_k, u_k)$ are the three velocity-related features, $v_k = 1$ indicates an encounter that was *sensed*, and $v_k = 0$ indicates an encounter where *no response* was detected. Using a semi-supervised approach to quadratic discriminant analysis (QDA), we labeled a subset of all encounters and iteratively estimated labels on the remaining unlabeled data (**Figure 2 – supplement 8A–C** [↗](#), Video 5). To label the data, we made two simple assumptions: 1) animals must have sensed the patch if they exploited it and 2) animals must not have sensed the patch if there was no bacteria to sense. Therefore, all encounters with bacteria-free patches (i.e., LB only, relative density 0) were labeled as true negatives $v_k = 0$ and, across 1000 replicates, we probabilistically included *exploitation* encounters $y_k = 1$ estimated from the distribution $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$ as true positives $v_k = 1$. The semi-supervised QDA method then used these initial labels to iteratively fit a paraboloid that best separated the labeled data, by minimizing the posterior variance of classification. Using this approach, the conditional probability of *sensing* $p(v_k = 1 | \mathbf{w}_k)$ was estimated for all encounters using QDA and averaged across replicates (**Figure 2 – supplement 8D–E** [↗](#)).

Although we found that the two clusters were successfully discriminated by this semi-supervised QDA approach, a small subset of encounters (252 of 20109 encounters) could not be reliably classified in this manner as the onset of the patch encounter was not observed. This type of data censoring occurred when the behavioral recording was started after the animal had already entered the patch. Although we could compute deceleration-related metrics for these encounters with the assumption that the animal entered the patch at the first frame, these measurements are censored with a bias towards lower magnitudes of deceleration. Therefore, to better predict the conditional probabilities of *sensing* for these 252 encounters, we assumed that the minimum velocity on-patch s_k was a reliable metric and marginalized the conditional probabilities over the other two metrics (t_k and u_k) as defined by

$$p(v_k = 1 | s_k) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(v_k = 1 | s_k, t_k, u_k) p(t_k, u_k | s_k) dt du$$

We numerically integrated using an adaptive quadrature method over the product of the QDA estimated conditional probability distribution and the kernel density estimate of the joint probability distribution. This procedure resulted in a slight increase ($\mu = +0.0615$, $\sigma = 0.1832$) in our estimation of the probability of *sensing* (**Figure 2 – supplement 8F** [↗](#)).

Altogether, this classification approach resulted in low false positive and false negative rates. Specifically, 3.35% of the time encounters with bacteria-free patches were incorrectly identified as *sensing* while 2.92% of the time *exploitatory* encounters were incorrectly identified as *non-responding*. Further, this classification removed many of the remaining near miss patch encounters in which animals came close to but did not truly enter the patch (see Patch encounter detection). For all analyses, we excluded near miss encounters where the animal's midpoint never entered the patch and the probability of *sensing* was less than 5% (i.e., $p(v_k = 1 | \mathbf{w}_k) < 0.05$).

Models of exploitation probability

We consider that the probability of *exploiting* a patch upon any given encounter can be described by a logistic function:

$$p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k) = \frac{1}{1 + e^{-\boldsymbol{\beta} \cdot \mathbf{x}_k}}$$

where $p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k)$ represents the conditional probability that an animal exploited during patch encounter k ; $\mathbf{x}_k \in R^n$ is a vector of covariates; and $\boldsymbol{\beta} \in R^n$ is a vector of weights that describes how much each covariate influences the animal's choice. In models compared here, $\mathbf{x}_k$ includes a combination of a constant element as well as covariates that may empirically influence the decision to exploit: 1) the log-transformed relative density of the encountered patch k (ρ_k), 2) the

duration of time spent off food since departing the last *exploited* patch (τ_s), 3) the log-transformed relative density of the patch encountered immediately before encounter k (ρ_h), and 4) the log-transformed relative density of the last *exploited* patch (ρ_e).

The observations of *exploitation* and *sensation* for our data are derived predictions from classification models rather than direct measurements (see Patch encounter classification as exploration or exploitation and Patch encounter classification as sensing or non-responding). As such, the labels for both whether a patch encounter was *sensed* by the animal $p(v_k = 1 | \mathbf{w}_k)$ and whether the animal *exploited* the patch $p(y_k = 1 | \mathbf{z}_k)$, are both reported as probabilities rather than binary, deterministic variables.

To account for uncertainty in *sensation*, we produced 100 sets of “encounter sampled” observations wherein we probabilistically included *sensed* encounters $v_k = 1$ estimated from the distribution $v_k | \mathbf{w}_k \sim \text{Bern}(p(v_k = 1 | \mathbf{w}_k))$. In doing so, we assume that only patch encounters recognized by the animal guide the decision to *exploit* (Figure 4B). After removing *non-responding* encounters where $v_k = 0$, we defined observations of all covariates (i.e., ρ_k , τ_s , ρ_h , and ρ_e) using only *sensed* encounters.

To account for uncertainty in *exploitation* we modified the standard maximum likelihood. To understand our approach, consider that standard logistic regression models binary observations as realizations of the Bernoulli distribution $y_k | \mathbf{x}_k \sim \text{Bern}(p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k))$ and that the likelihood for this model is given by

$$\prod_k p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k)^{y_k} (1 - p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k))^{1-y_k}$$

However, we do not have direct observations of y_k and cannot evaluate the logistic likelihood directly. Rather, we have posterior probabilities of *exploitation* determined by a classifier, denoted by $p(y_k = 1 | \mathbf{z}_k)$, where $\mathbf{z}_k$ are a set of velocity-related features that are distinct from $\mathbf{x}_k$ (see Patch encounter classification as exploration or exploitation). Thus, rather than maximizing the Bernoulli likelihood, we accommodate our uncertainty about y_k by learning the regression parameters that minimize the Kullback-Leibler (KL) divergence between $p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)$ and the reference probability distribution $p(y_k | \mathbf{z}_k)$ as given by

$$\begin{aligned} \mathcal{D}_{KL}[p(y_k | \mathbf{z}_k) || p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)] &\equiv \sum_{y_k} p(y_k | \mathbf{z}_k) \log \left[\frac{p(y_k | \mathbf{z}_k)}{p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)} \right] \\ &= C_{\boldsymbol{\beta}} - \sum_{y_k} p(y_k | \mathbf{z}_k) \log[p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)] \\ &= C_{\boldsymbol{\beta}} - \sum_{y_k} p(y_k | \mathbf{z}_k) \log \left[p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)^{y_k} (1 - p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k))^{1-y_k} \right] \\ &= C_{\boldsymbol{\beta}} - p(y_k = 1 | \mathbf{z}_k) \log[p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)] \\ &\quad - (1 - p(y_k = 1 | \mathbf{z}_k)) \log[1 - p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)] \\ &= C_{\boldsymbol{\beta}} \\ &\quad - \log \left[p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)^{p(y_k = 1 | \mathbf{z}_k)} (1 - p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k))^{1-p(y_k = 1 | \mathbf{z}_k)} \right] \end{aligned}$$

where $C_{\boldsymbol{\beta}}$ is a term that is constant with respect to $\boldsymbol{\beta}$, and the second term is equivalent to the logarithm of the logistic likelihood with $p(y_k = 1 | \mathbf{z}_k)$ replacing y_k . Put plainly, minimizing the sum of the KL divergence between our classifier probabilities and logistic regression probabilities over

all encounters k is equivalent to maximizing the logistic likelihood with our classifier-based *exploitation* probabilities $p(y_k = 1 | \mathbf{z}_k)$ as observations instead of direct measurements of *exploitation* y_k as given by

$$\sum_k \log \left[p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k)^{p(y_k = 1 | \mathbf{z}_k)} (1 - p(y_k | \boldsymbol{\beta} \cdot \mathbf{x}_k))^{1 - p(y_k = 1 | \mathbf{z}_k)} \right]$$

Notably, when there is no observation uncertainty $p(y_k = 1 | \mathbf{z}_k)$ collapses to a delta function on $y_k = 1$ and we recover the standard likelihood function. Thus, our method is a generalization of maximum likelihood. To train our model, we therefore fit a standard logistic regression GLM where we provided sets of observations of covariates $\mathbf{x}_k$ and response variables $p(y_k = 1 | \mathbf{z}_k)$ for *sensed* encounters.

The covariates $\mathbf{x}_k$ included the relative density of the encountered patch k (ρ_k), the duration of time spent off food since departing the last *exploited* patch (τ_s), the relative density of the patch encountered immediately before encounter k (ρ_h), and the relative density of the last *exploited* patch (ρ_e). For every encounter k , ρ_k is the log-transformed relative density of the encounter patch as estimated from experiments using fluorescent bacteria OP50-GFP (as described in Bacterial patch density estimation). τ_s is the duration of time spent off food since the beginning of the recorded experiment (i.e., total time elapsed minus duration of time on patch). For the first patch encounter (i.e., $k = 1$), this is equivalent to the total time elapsed. ρ_h is the log-transformed relative density of the patch encountered immediately before encounter k . For the first patch encounter, we initialized this parameter using the approximated relative density of the bacterial patch on the acclimation plates (see Assay preparation and recording in Methods). Acclimation plates contained one large 200 μL patch seeded with $\text{OD}_{600} = 1$ and grown for a total of ~ 48 hours. The relative density of these acclimation plates was estimated using the same procedure used to estimate values of ρ_k (see Bacterial patch density estimation). ρ_e is the log-transformed relative density of the most recently *exploited* patch. For the first patch encounter as well as every encounter prior to the first observed *exploitation*, we assume that animals must have *exploited* within the last 24 hours while on the acclimation plates and thus initialized ρ_e using the approximated relative density of the bacterial patch on the acclimation plates.

Exploitation of single-density, multi-patch environments

As described above, we used an “encounter sampling” protocol to generate 100 replicate observations of only *sensed* encounters by removing all encounters where *sensing* could not be determined (i.e., $v_k = 0$) as randomly estimated from the distribution $v_k | \mathbf{w}_k \sim \text{Bern}(p(v_k = 1 | \mathbf{w}_k))$. For example, we can consider an animal foraging in the single-density, multi-patch environment (Figure 4B). Although we detected 10 total encounters $K = k_1, k_2, \dots, k_{10}$, several of these encounters had low probabilities of *sensing* $p(v_k = 1 | \mathbf{w}_k)$ (Figure 4B). We use our “encounter sampling” protocol to simulate *sensation* of the patch during each of these 10 encounters and find that $v_k = 0$ for k_1, k_3, k_6 and k_8 . By removing these 4 no response encounters, we generate a new observation comprised of the 6 *sensed* encounters $K_1^* = k_2, k_4, k_5, k_7, k_9, k_{10} = k_{1*}, k_{2*}, \dots, k_{6*}$. Using only the data related to this new sequence of encounters K^* , we generated a vector of covariates (i.e., ρ_k, τ_s, ρ_h , and ρ_e). Importantly, this procedure considers time spent during the *no response* encounters k_1, k_3, k_6 and k_7 the same as time spent searching off patch. We repeated this process 100 times to generate new observations of encounter sequences (i.e., $K_1^*, K_2^*, \dots, K_{100}^*$). In doing so, we reduced the number of encounters from 6,560 total encounters (as defined in Patch encounter detection) to 2,604–2,698 ($\mu = 2659.8, \sigma = 15.9$) *sensed* encounters for each of the 100 replicates. Additionally, we used a “worm sampling” protocol to generate 500 hierarchically bootstrapped replicates [97]. Specifically, we took our vector of 443 animals $A = a_1, a_2, \dots, a_{443}$ and resampled with replacement (e.g., $A_1^* = a_{317}, a_{88}, \dots, a_{130} = a_{1*}, a_{2*}, \dots, a_{443*}$) to generate 500 new samples $A_1^*, A_2^*, \dots, A_{500}^*$. When combined, the “encounter sampling” and “worm sampling” protocols created 50,000 unique replicates of observations of the covariates $\mathbf{x}_k$ and response variables $p(y_k = 1 | \mathbf{z}_k)$ which were used to estimate the coefficients $\boldsymbol{\beta}$ (Figure 4C). A null distribution of

coefficients β^* was estimated by shuffling the response variable vector (**Figure 4 – supplement 1C**). A two-tailed, one-sample bootstrap hypothesis test was used to assess whether our covariate estimates were significantly greater than or less than 0 (i.e., $p = 2 \times \min[P(\beta \leq 0), P(\beta \geq 0)]$).

Exploitation of food-deprived animals in single-density, multi-patch environments

To validate model predictions related to the satiety signal covariate τ_s , a novel data set of well-fed and 3-hour food-deprived animals foraging in single-density, multi-patch environments of relative density 5 was generated (**Figure 4 – supplement 3A-B**). The “encounter sampling” protocol described above was used to generate 100 replicates of observations of the covariates $\mathbf{x}_k$ and response variables $p(y_k = 1 | \mathbf{z}_k)$ for this data set. As a result of “encounter sampling”, we reduced the number of encounters from 42 total encounters to 30-37 ($\mu = 34.6$, $\sigma = 1.3$) *sensed* encounters for food-deprived animals and 235 total encounters to 94-107 ($\mu = 99.8$, $\sigma = 2.1$) *sensed* encounters for well-fed animals. Coefficient values β were re-estimated in the absence of the satiety signal τ_s (**Figure 4 – supplement 3C**) using the original single-density, multi-patch data set. Using the mean coefficient values β for both models (with and without satiety), we predicted the conditional probabilities of exploiting $p(y_k = 1 | \beta \cdot \mathbf{x}_k)$ for the new data set for well-fed and food-deprived animals. We subsequently simulated a series of *exploitation* events from a Bernoulli distribution using the estimated (i.e., $y_k | \mathbf{x}_k \sim \text{Bern}(p(y_k = 1 | \beta \cdot \mathbf{x}_k))$) and observed (i.e., $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$) *exploitation* probabilities. These simulated *exploitations* were used to generate distributions of the probability of an *exploitation* occurring for the first time as a function of the number of encounters (**Figure 4 – supplement 3D**).

Exploitation of multi-density, multi-patch environments

To investigate the individual contributions of a satiety-related signal and transfer-induced stress, a novel data set of animals foraging on multi-density, multi-patch environments was generated (**Figure 4E**, **Figure 4 – supplement 4A-C**). The “encounter sampling” protocol described above was used to generate 100 replicates of observations of the covariates $\mathbf{x}_k$ and response variables $p(y_k = 1 | \mathbf{z}_k)$ for this data set. As a result of “encounter sampling”, we reduced the number of encounters from 4,296 total encounters (350-984 total encounters per condition) to 1,702-1,745 ($\mu = 1,724.6$, $\sigma = 9.9$) *sensed* encounters. Coefficient values β were estimated for models including combinations of τ_s , the duration of time an animal spent searching off-food since the last *exploitation* event, and τ_t , the duration of time since transfer (i.e., the time elapsed in the experiment) (**Figure 4 – supplement 4D**) using the new multi-density, multi-patch data set. While all coefficients were significant in models containing only τ_s or τ_t , only τ_s was significant in a model containing both terms (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_t \tau_t + \beta_h \rho_h + \beta_e \rho_e)$).

Further, to validate model predictions related to the prior experience covariates ρ_h and ρ_e , coefficient values β were re-estimated in the absence of the recently encountered and *exploited* patch density terms ρ_h and ρ_e (**Figure 4 – supplement 4E**) using the original single-density, multi-patch data set. Using the mean coefficient values β for both models (with and without prior experience), we predicted the probabilities of *exploiting* $p(y_k = 1 | \beta \cdot \mathbf{x}_k)$ for the new data set. To generate the heat maps of predicted behavior (**Figure 4F**), we varied ρ_k and ρ_h as well as ρ_k and ρ_e across a range of values and set the remaining covariate terms to the average values observed in the data set. To generate the heat maps of observed behavior (**Figure 4F**), we averaged the observed probabilities of *exploiting* $p(y_k = 1 | \mathbf{z}_k)$ for each pairing of ρ_k and ρ_h as well as ρ_k and ρ_e and linearly interpolated values between these 9 points.

Exploitation of sensory mutants in single-density, multi-patch environments

To test the utility of our model in identifying covariate-specific phenotypes in animals with varied genotypes, a novel data set of wild-type animals and sensory mutants foraging on single-density, multi-patch environments was generated (**Figure 4 - supplement 5A-B**). The “encounter sampling” and “worm sampling” protocols described above were used to generate 50,000 replicates of observations of the covariates $\mathbf{x}_k$ and response variables $p(y_k = 1 | \mathbf{z}_k)$ for this data set. As a result of “encounter sampling”, we reduced the number of encounters from 1,352 total encounters (27-193 total encounters per condition) to 862-894 ($\mu = 879.4$, $\sigma = 7.4$) *sensed* encounters. As the number of encounters were significantly less in these data sets as compared to the original single-density, multi-patch data set, we used ridge regression to regularize the magnitude of the covariates. We optimized each model across cross-validated replicates by varying the regularization value λ . We found that $\lambda_{N2} = 0.0175$, $\lambda_{mec-4} = 0.0223$, and $\lambda_{osm-6} = 0.0614$ maximized the mean log-likelihood of each model (**Figure 4 - supplement 5E**). Coefficient values β were estimated for each of the three strains tested using these values (**Figure 4 - supplement 5D**). To assess whether the magnitude of coefficients significantly differed between wild-type and mutant models, we conducted a two-sample test of the mean of differences as given by

$$Z = \frac{\mu_{\text{mutant}} - \mu_{N2}}{\sqrt{\sigma_{\text{mutant}}^2 + \sigma_{N2}^2}}$$

where μ and σ are the mean and standard deviation, respectively, of the distribution of a covariate across replicates. Statistical significance was calculated as a left-tailed test $\Phi(Z)$ for the density coefficients (β_k , β_h , and β_e) and a two-tailed test $2(1 - \Phi(|Z|))$ for the other coefficients (β_0 and β_s) where Φ is the standard normal cumulative distribution function.

Figure Supplements

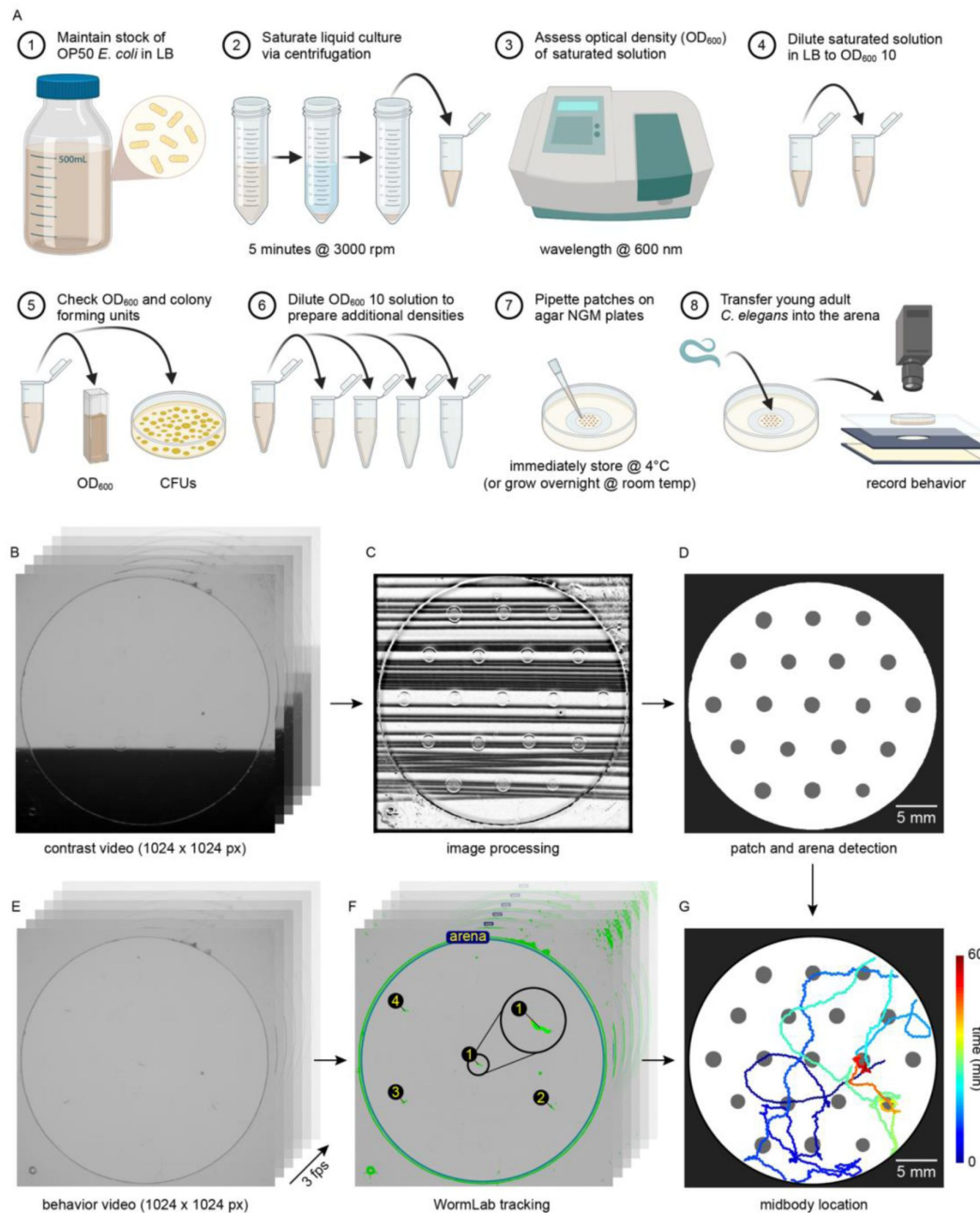


Figure 1 – supplement 1

Assay preparation, arena and patch detection, and behavioral tracking.

(A) Graphic describing the steps to prepare assay plates. This panel was created with [BioRender.com](https://www.biorender.com). (B) Example frames from a “contrast video” showing how a piece of dark cardstock passed between the light source and the assay plate enables visualization of dilute bacterial patches. (C) An image of the bacterial patches is generated from the “contrast video”. (D) Masks of the arena (black) and bacterial patches (gray) were constructed using image processing techniques and, when applicable, subsequent manual corrections (see Bacterial patch location detection). Arena masks were used to calculate the image scale in pixels/mm. (E) Example frame from a behavior video containing four adult *C. elegans* is shown. (F) WormLab was used to identify the midbody location of animals in each frame (see Behavioral tracking). An inset shows detection of one of the four worms. (G) An example animal’s tracked location as it forages in this environment is plotted with color used to represent time (dark blue = 0 mins; dark red = 60 mins).

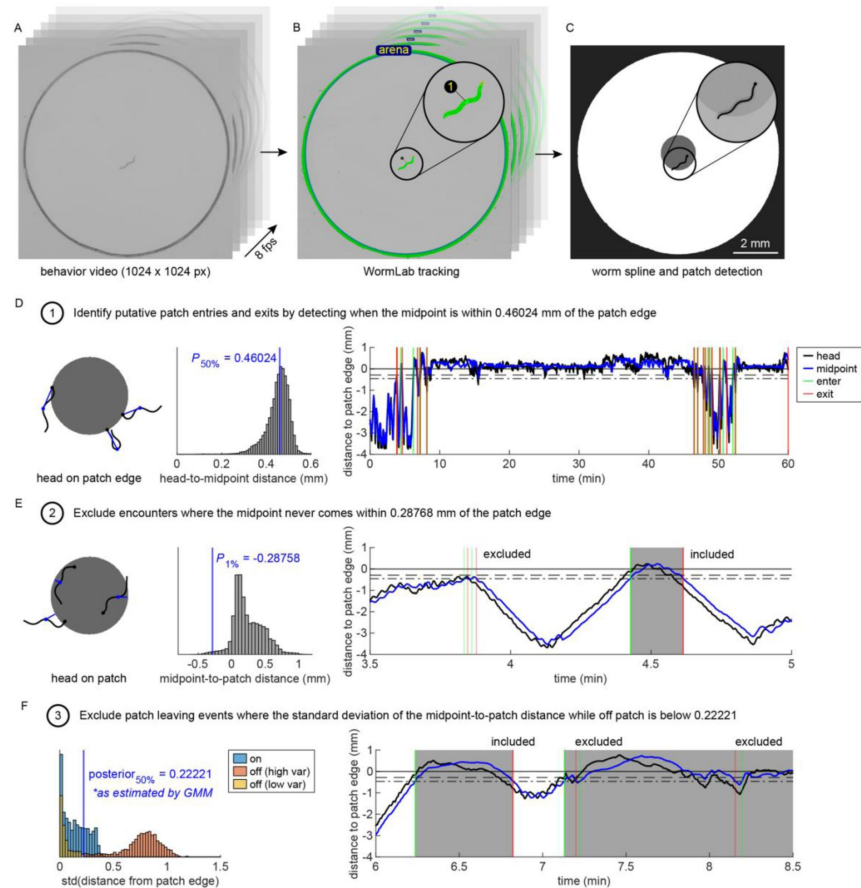


Figure 1 – supplement 2

Defining a patch encounter using high resolution behavioral recordings.

(A) Example frame from a behavior video containing one adult *C. elegans* is shown. (B) WormLab was used to track the body of the animal in each frame (see Behavioral tracking). (C) An example image showing the center spline of the example animal as well as the location of the bacterial patch (gray) and arena (black). The inset shows the animal's center spline and bacterial patch superimposed on top of the raw image. (D) Example body postures/orientations (black) of an animal during frames when its head is on the patch edge. The distance (blue lines) between the head (black dot) and midpoint (blue dot) of the animal was calculated. A histogram of this head-to-midpoint distance is plotted, and the median value (blue) is indicated (sample size: 220 animals; 74839 frames). The median head-to-midpoint distance was subsequently used as a threshold for detection of patch encounter and leaving events. For one example animal, the distance between the head (black) or midbody (blue) position and the nearest patch edge (positive = inside patch; negative = outside patch) is plotted for every time point. The edge of the patch (solid black line) and threshold for encounter detection (dashed -.- black line) are indicated. Putative patch encounter (green) and leaving (red) events are shown. (E) Example body postures/orientations of an animal when its head is on the patch. The distance between the midpoint of the animal and the patch edge (blue lines) was calculated. A histogram of this midpoint-to-patch distance is plotted, and the 1st percentile (blue) is indicated (sample size: 220 animals; 1571163 frames). This value is used as a threshold for excluding putative encounters where the animal's midpoint never comes within 0.28768 mm of the patch edge. A 1.5 minute time window of the example animal's position in (D) is shown. The edge of the patch (solid line), threshold for encounter detection (dashed -.- line), and threshold for encounter exclusion (dashed -.- line) are indicated. Example putative encounters are shown. The first two sets of enter and exit events were excluded. The third set shown was included. The included encounter is shaded in gray. (F) The standard deviation of the distance between the location of the patch edge and the midpoint of the animal was calculated for every putative on-patch encounter and off-patch event (sample size: 220 animals; 2945 putative on-patch events; 2725 putative off-patch events). A subset of putative off-patch events had low standard deviations matching those of on-patch events. A one-dimensional Gaussian mixture model was fit to the off-patch standard deviation values and a threshold was set where the posterior probability of events falling in the high or low variance cluster was 0.5. A 1.5 minute time window of the example animal's position in (D) is shown. Examples of two excluded off-patch events and one included event are shown.

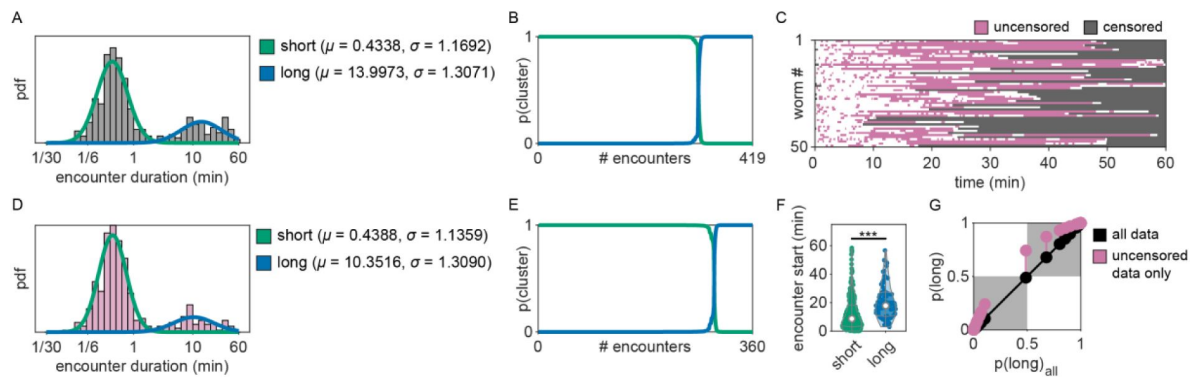


Figure 1 – supplement 3

Classifying encounters based on duration using a Gaussian mixture model.

(A) As in [Figure 1E](#), duration for each patch encounter ($N = 419$ events; 50 worms) was computed and classified as either short or long using a Gaussian mixture model (GMM). The distributions of all observed short and long encounters are plotted with duration binned logarithmically. The probability density functions for the two component Gaussians are shown. (B) The posterior probabilities for classification as short or long are sorted for all encounters, highlighting the low posterior variance of the GMM. (C) Given that some patch encounters were not fully observed (i.e., ~22% of animals were already on patch at the start of the recording and ~96% of animals were on patch when the recording ended), we identified that ~14% of observed patch durations were left- censored (i.e., having actual duration $\geq$ observed duration). Thus, to confirm findings in [Figure 1E-F](#), we re-classified only non-censored observations where both the entry and exit of a patch encounter were observed. As in [Figure 1C](#), patch encounters for 50 individuals foraging in these patchy environments are plotted across time. Encounters that were uncensored (pink) and censored (gray) are indicated. (D) The distribution of the duration of only fully observed (uncensored) encounters ($N = 360$ events; 50 worms) is plotted. The component Gaussians for a GMM fitting this subset of data are shown. (E) Sorted posterior probabilities for the uncensored model are shown. (F) The observed start time of each uncensored patch encounter is shown for all short and long duration encounters. Long duration encounters occur significantly later (Mann-Whitney U-Test, $***p < 0.001$). Violin plots show the KDE and quartiles for each duration type. (G) Posterior probabilities for classification in the long duration cluster are shown for the two models. The uncensored data model results in a slight increase in posterior probabilities of long duration classification but ultimately affected classification of only one of the 419 encounters.

Figure 1 – supplement 4

Permuting patches to test for significance of the observed time-dependent increase in patch residence.

(A) Locations of bacterial patches (dark gray) are shown with the threshold for putative encounters (light gray) set at 0.46024 mm (as described in [Figure 1 – supplement 2](#) and Patch encounter detection). (B) Patches were permuted via randomized rotation and translation with the restriction that patches could not overlap with other patches nor with the arena. (C,D) Midbody location of the animal over time was overlaid on the original and semi-randomly permuted patches. (E,F) On patch events were subsequently identified. This permutation procedure was repeated for 1000 replicates for every worm tested. Smoothed median values of the probability of worms residing on patch over time was compared for every time point in [Figure 1G](#), [2F](#) and [Figure 2 – supplement 5](#).

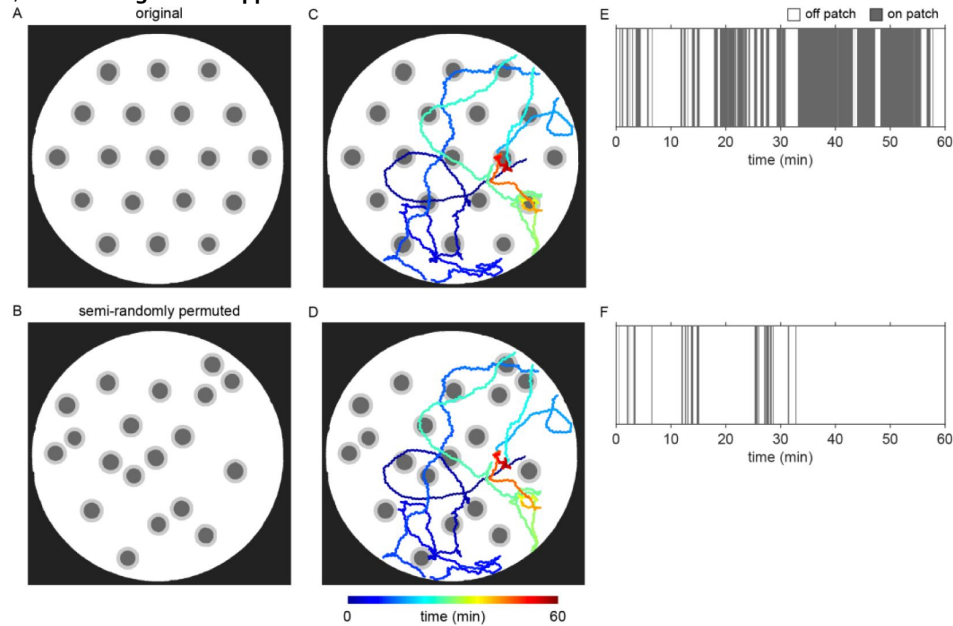
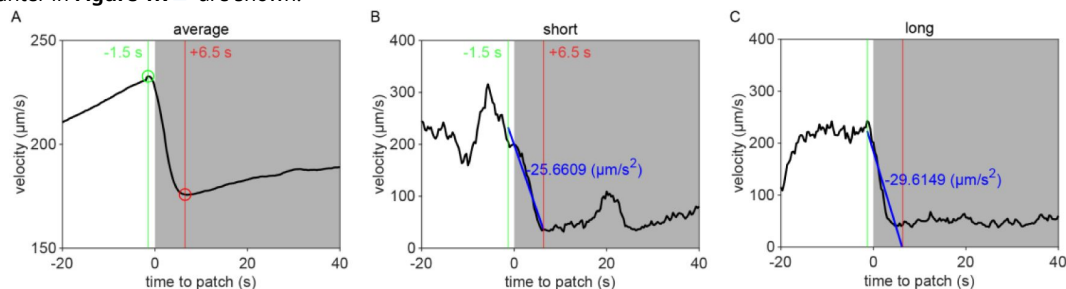


Figure 1 – supplement 5

Analyzing deceleration upon encounter with a patch.

(A) To quantify the magnitude of slow-down upon encounter with the patch edge, we averaged the velocity trajectories across encounters (black) and found the velocity reached a maximum at 1.5 seconds before (green) and a minimum at 6 seconds after (red) the start of the encounter (gray). (B,C) Subsequently, for every encounter, a line (blue) was fit to the animal's velocity (black) for that 8 second time window surrounding the start of the encounter. Deceleration was defined as the slope of this line. Examples matching (B) the short duration encounter in [Figure 1J](#) and (C) the long duration encounter in [Figure 1K](#) are shown.



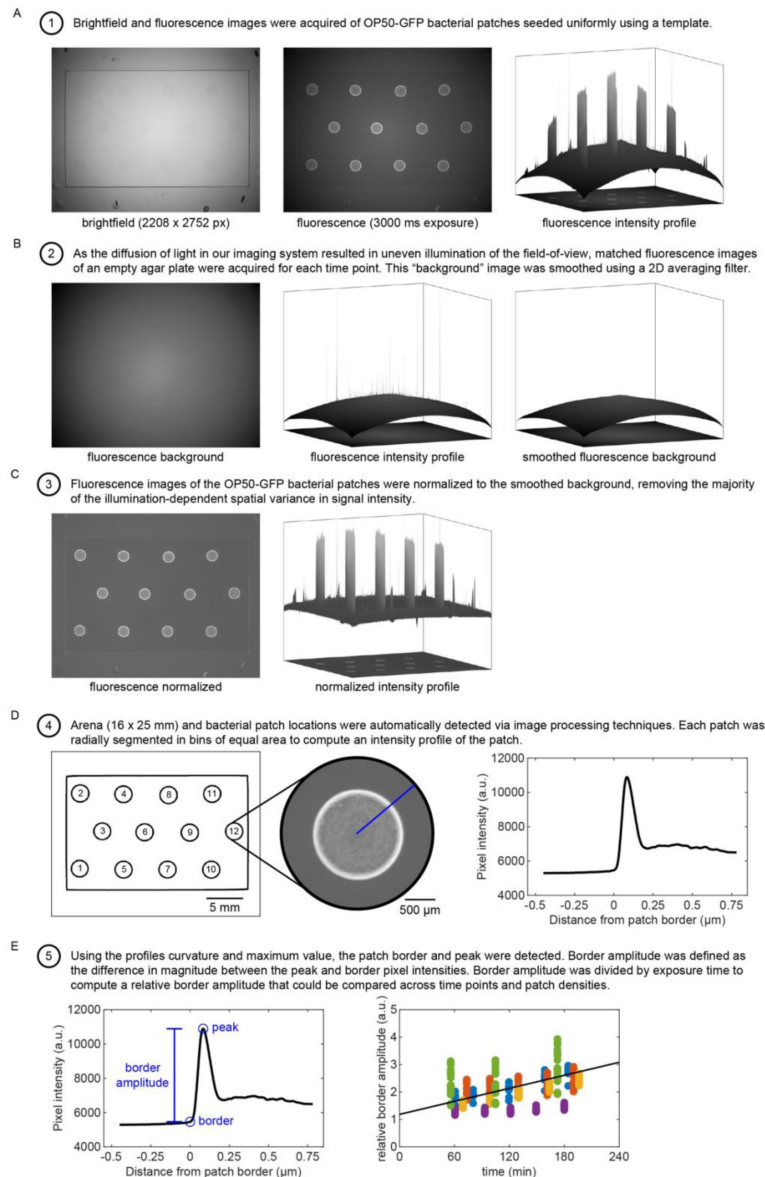


Figure 2 – supplement 1

Obtaining intensity profiles for fluorescent bacterial patches.

(A) Fluorescently labeled OP50-GFP bacterial patches were seeded onto agar plates under conditions matching those in the experimental assay. Small patches (0.5 μL) were pipetted in an isometric grid (6 mm center-to-center spacing) on a 16 x 25 mm rectangular template comparable to the one used in behavioral assays. Large patches (20 or 200 μL) were seeded directly onto agar plates. Brightfield and fluorescence images were acquired for every condition at multiple time points of bacterial growth. The fluorescence intensity profile of these images was obtained. (B) Due to uneven illumination within the field-of-view of our imaging system, matched fluorescence images of an empty agar plate were acquired at each time point. These "background" images were smoothed using a two-dimensional averaging filter to remove noise. (C) Fluorescence images of the bacterial patches were normalized to these background images. (D) Locations of bacterial patches within the template were automatically detected using image processing techniques. Each patch was then radially segmented in bins of equal area. The mean pixel intensity value for each bin was computed and used to create an intensity profile of the patch. (E) The patch border and peak were detected using signal processing techniques. Specifically, the patch border was identified by finding a peak in the curvature κ of the edge profile. Border amplitude was defined as the magnitude of the difference between the patch border and peak. Relative border amplitude was defined as the border amplitude divided by the exposure time. A line was fit to the values of relative border amplitude across numerous time points and experiment days (5 replicates shown as varying color points).

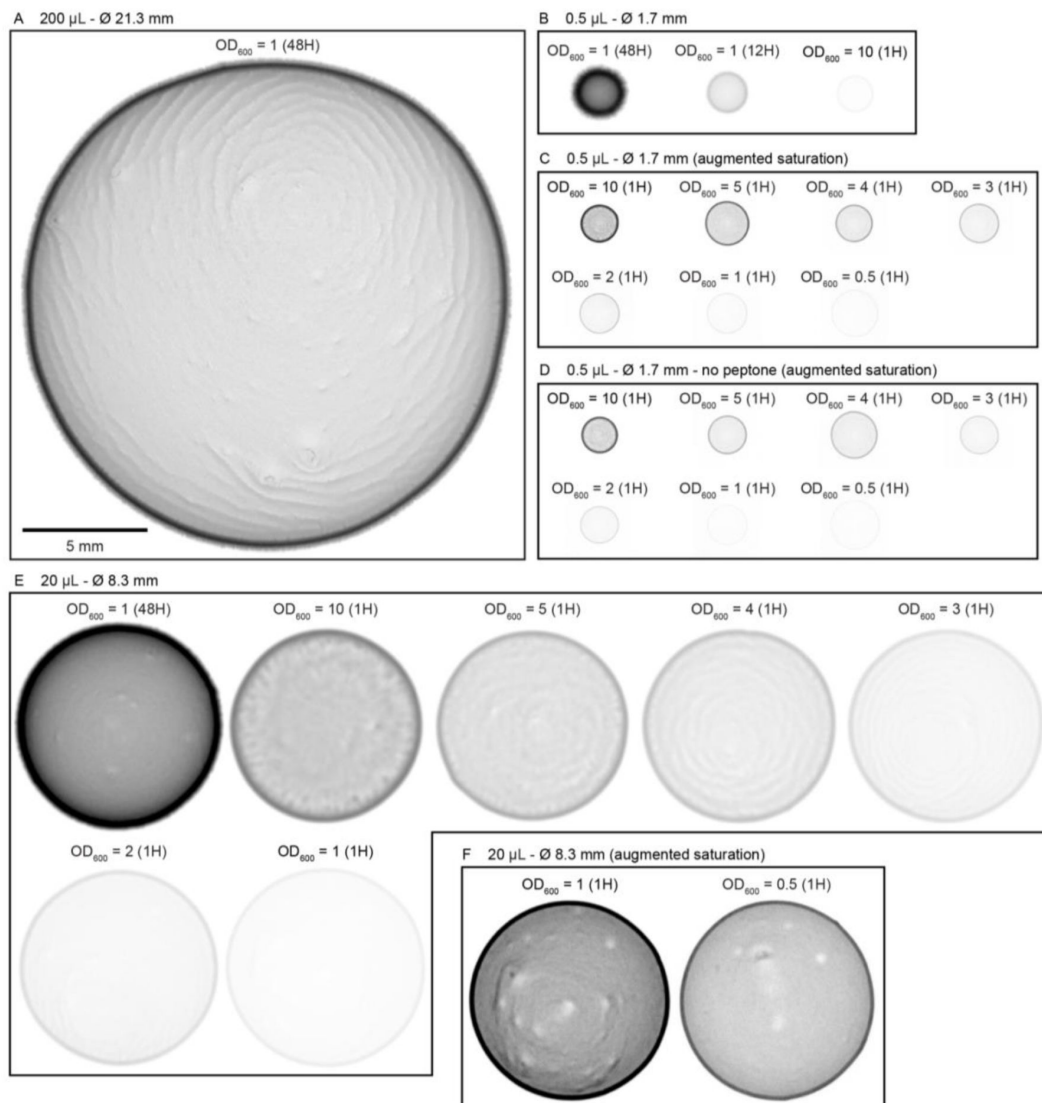


Figure 2 – supplement 2

Example fluorescence profiles of bacterial patches under varied growth conditions.

Fluorescently-labeled OP50-GFP bacterial patches were imaged (as described in [Figure 2 – supplement 1](#) and Bacterial patch density estimation). Inverted images (darker saturation = more bacteria) are shown here to highlight the range of bacterial densities tested in these experiments. All patches are shown on the same spatial scale and using one of the two linear saturation scales: 1) regular – normalized to show variance in high density patches; 2) augmented (~22x) – normalized to show variance in low density patches). Each patch shown is labeled with the optical density (OD_{600}) at time of seeding and the approximate amount of time bacteria was grown for at room temperature (i.e., 1, 12, or 48 hours). (A) An example large (200 μ L) bacterial patch of the type that animals experienced immediately prior to the assays in this paper. (B) Example small (0.5 μ L) bacterial patches of greatest density. (C) Example small (0.5 μ L) bacterial patches of lower density. The saturation of each image has been uniformly augmented to ~22x darker to the extent that the OD_{600} 10 (1H) bacterial patch is the same patch as the one in B. (D) Example small (0.5 μ L) bacterial patches of lower density seeded on NGM agar plates lacking peptone. (E) Example medium (20 μ L) bacterial patches of greater density. (F) Example medium (20 μ L) bacterial patches of lowest density. The OD_{600} 1 (1H) bacterial patch is the same patch as the one shown in E, but ~22x darker.

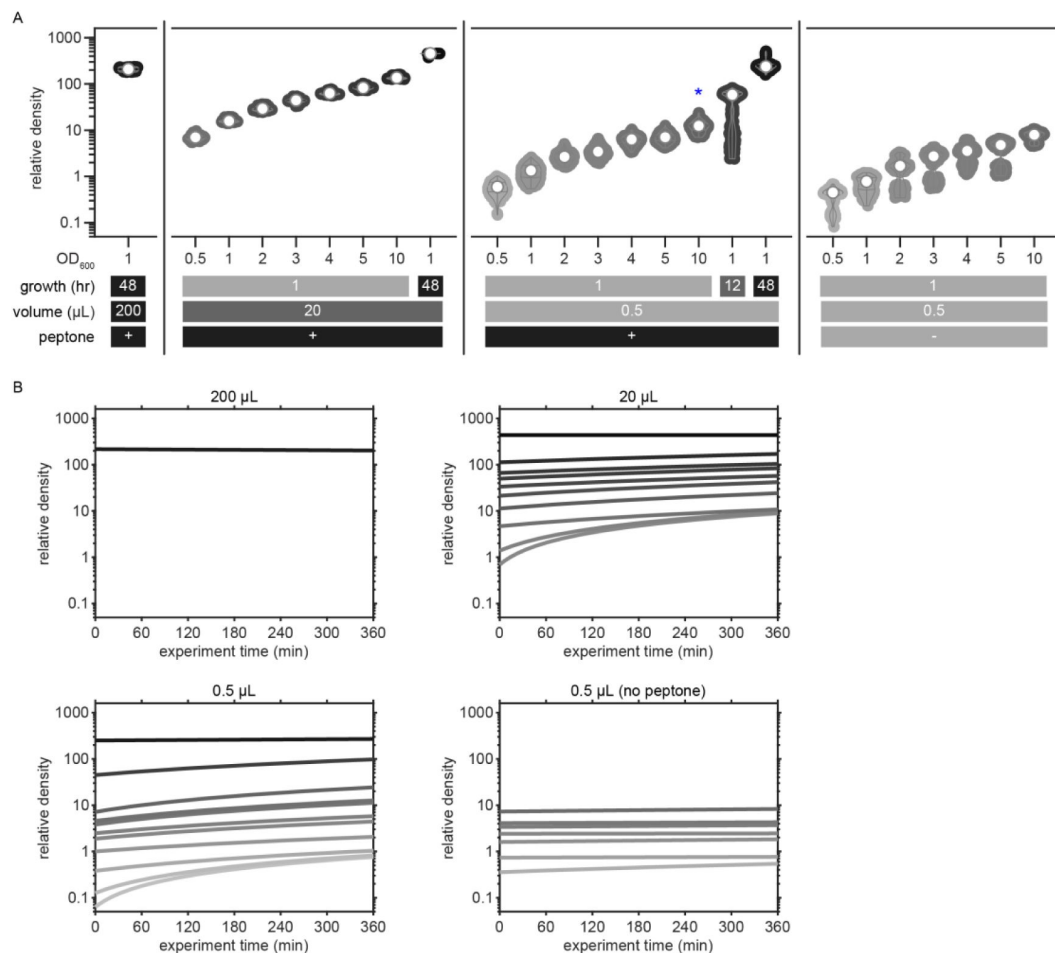


Figure 2 – supplement 3

Quantifying the relative density of bacterial patches across conditions and time points.

(A) Relative density of OP50-GFP bacterial patches matched to all assay conditions in this paper are shown. Relative density represents the relative border amplitude (as defined in [Figure 2 – supplement 1](#) and Bacterial patch density estimation) normalized to the mean density of patches seeded with 0.5 μL of OD₆₀₀ = 10 grown for 1 hour on plates with peptone (blue asterisk). In this manner, the bacterial patches experienced by animals in [Figure 1](#) have been set to an average relative density of 10. Violin plots show the KDE and quartiles for each condition. (B) Linear fits for the relative density of bacterial patches in each condition as a function of time are plotted. Saturation of each line corresponds to the relative density as depicted in (A). For each experimental plate, the relative density of bacterial patches was estimated by applying the coefficients from these linear regressions to the total amount of time the bacteria was grown at room temperature.

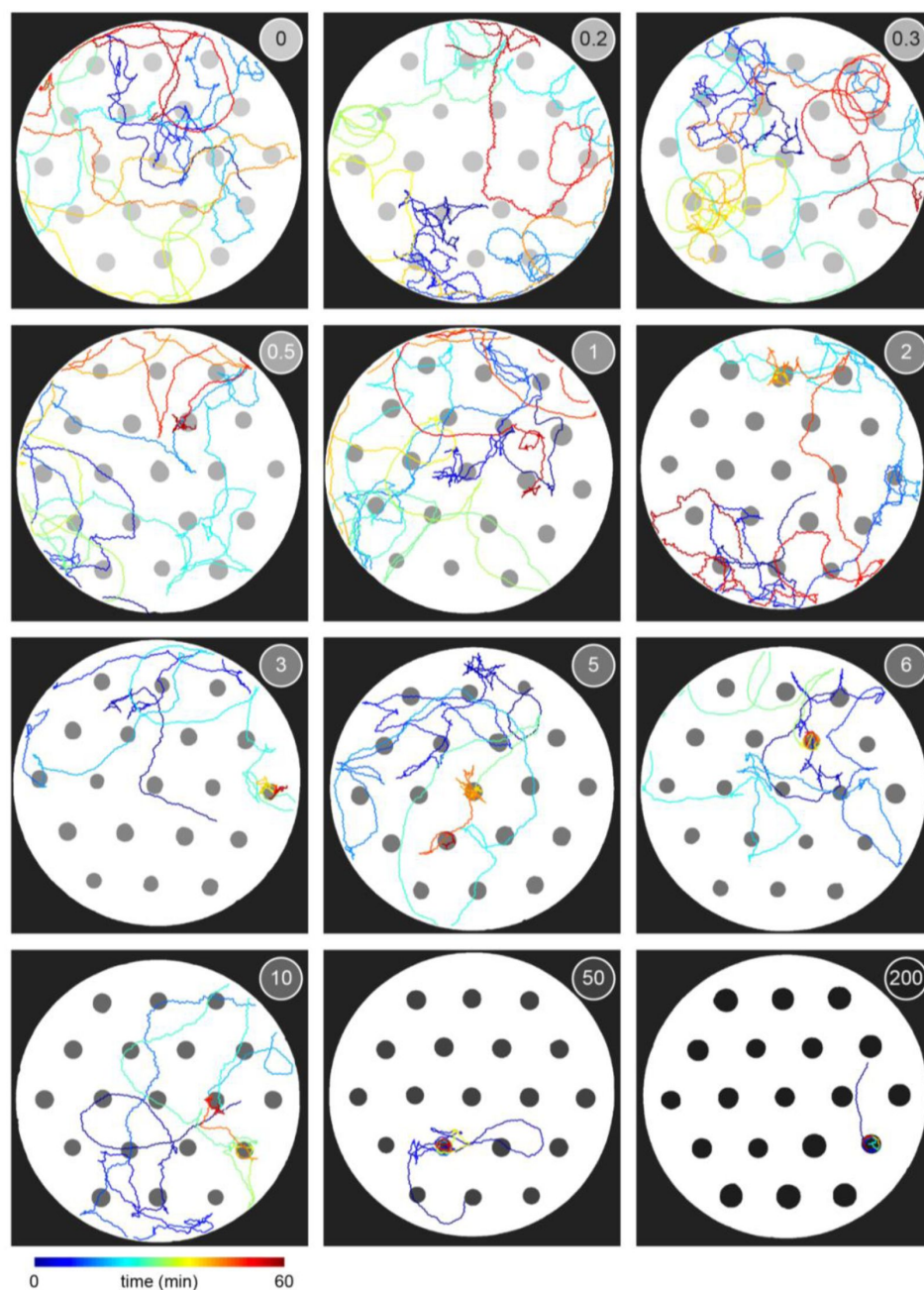


Figure 2 - supplement 4

Example traces of animals foraging in environments with varying bacterial density.

Traces showing the midbody location of 12 example animals as they forage within a 30 mm arena are plotted with color used to represent time (dark blue = 0 mins; dark red = 60 mins). Locations of bacterial patches are indicated in gray. The relative density of bacterial patches in each environment is noted. Animals shown foraging on relative density 0, 1, 5, 10, and 200 are the same as those in [Figure 2D](#).

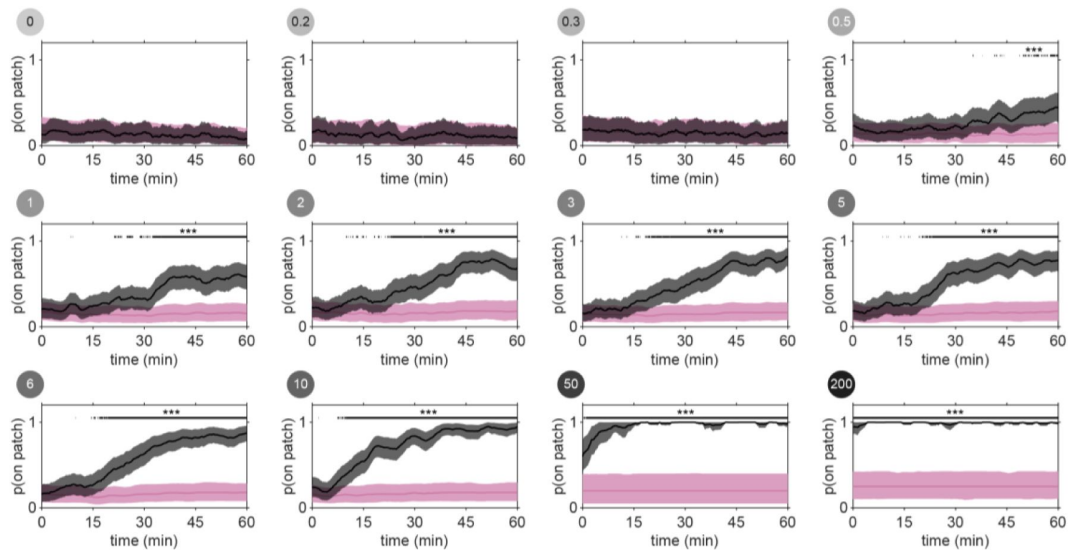


Figure 2 – supplement 5

Time- and density-dependent increase in patch residence.

Probability of residing on patch was computed for all worms across time (black) and compared to the probability of residing on patch for semi-randomly permuted patch locations (pink) (as described in [Figure 1 – supplement 4](#)). Smoothed median values are plotted with bootstrap-derived 2.5% and 97.5% quantiles shown in shaded regions. Time points where observed probabilities of residing on patch significantly exceed permuted probabilities are indicated by a black line (one-tailed Fisher's Exact Tests with Benjamini-Hochberg correction, $***p < 0.001$). The smoothed median values for observed data are shown in [Figure 2F](#). Data for relative density 10 match those in [Figure 1G](#).

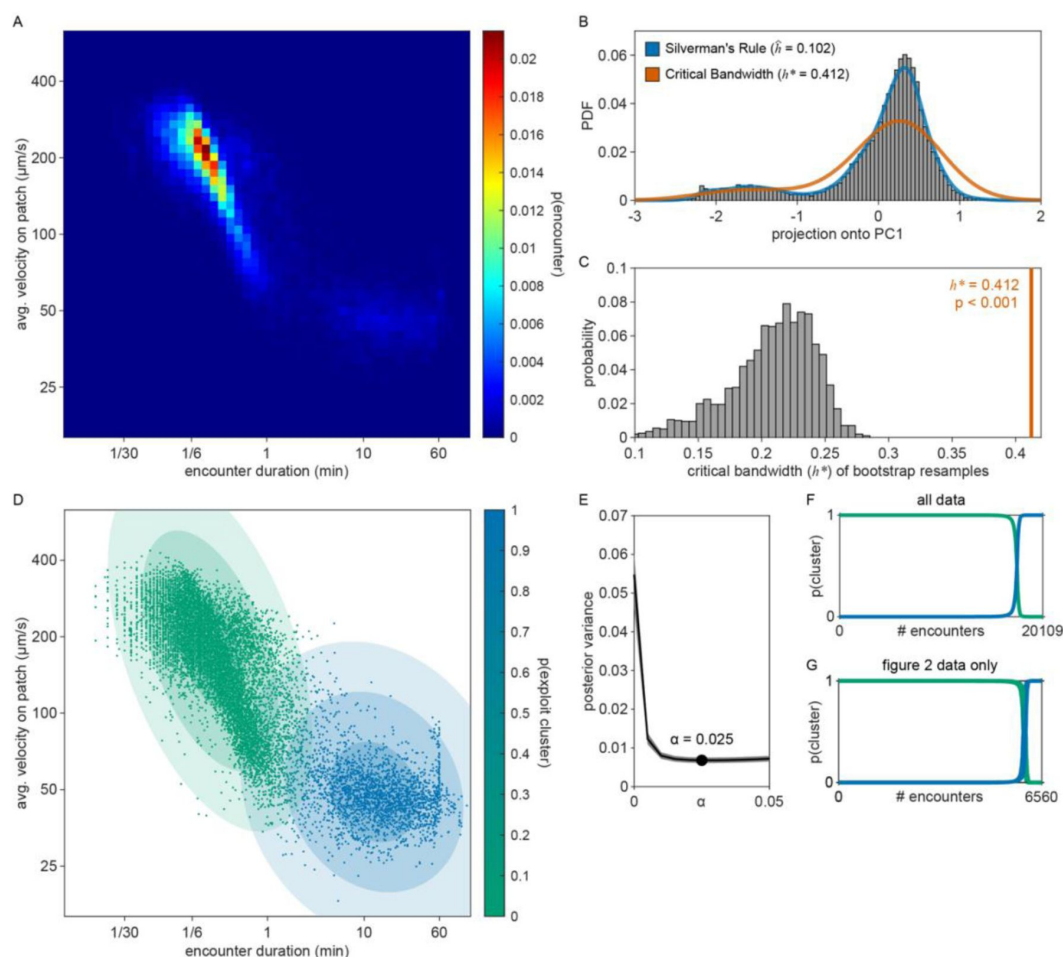


Figure 2 – supplement 6

Classifying encounters as exploration or exploitation.

(A) A two-dimensional histogram shows the frequency of events for given pairs of values for the average velocity of the animal during an encounter and the duration of that encounter for all encounters in the paper. Probabilities are represented as a heat map of color. (B) The first principal component (PC1) for these two metrics was computed. The distribution of projections of the data onto PC1 is given by a histogram (gray). Kernel density estimates (KDE) were computed using the bandwidth $\hat{h}$ defined by Silverman's Rule of Thumb (blue) and using the critical bandwidth $\hat{h}^*$ (red-orange) at which the KDE is at the threshold between uni- and bi-modal (see Patch encounter classification as exploration or exploitation). (C) Critical bandwidth values were computed across 2000 smoothed bootstrap resamples. The distribution of these bootstrapped critical values is given as a histogram and compared against the critical bandwidth of the original data set (red-orange). (D) The average velocity of the animal during an encounter and the duration of that encounter are plotted for each encounter as visualized on a double-logarithmic plot. Contours showing the first, second, and third standard deviation of the two-dimensional Gaussian mixture model (GMM) are shown as shaded ellipses with saturation corresponding to standard deviation. Encounters for all data in the paper are individually plotted as in [Figure 2H](#) but with color representing the posterior probability of clustering classification as *explore* (green) or *exploit* (blue). (E) Posterior variance of the data as defined in Patch encounter classification as *exploration* or *exploitation* was calculated for 1000 replicates of GMMs with varied regularization value α . The classifier with minimum posterior variance (i.e., $\alpha = 0.025$) was used. (F) The posterior probabilities for classification as *explore* (green) or *exploit* (blue) are sorted for all encounters, highlighting the low posterior variance of the GMM. (G) Posterior probabilities for only the data in [Figure 2](#) are shown.

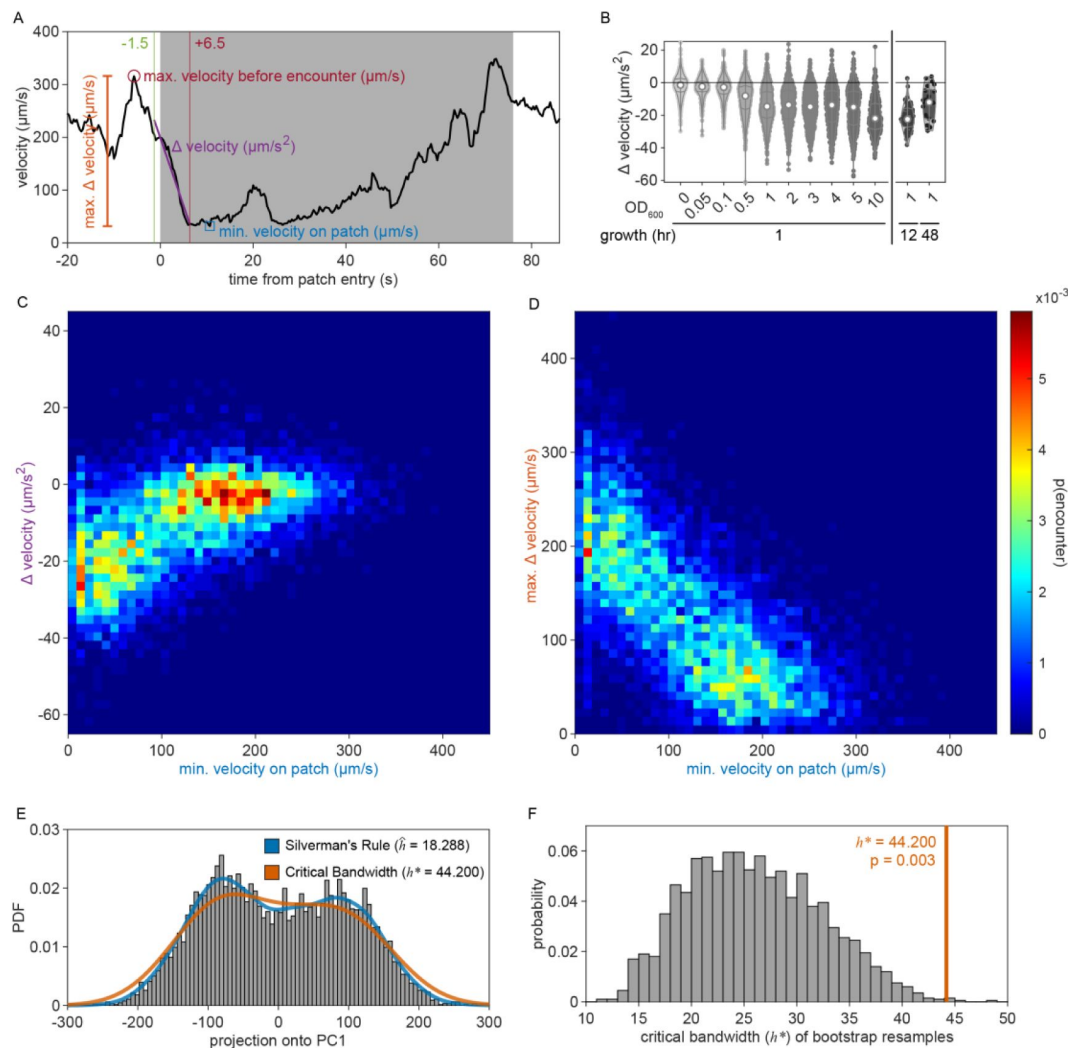


Figure 2 – supplement 7

Classifying encounters as sensing or non-responding.

(A) Minimum velocity (min. velocity on patch), maximum change in velocity (max. Δ velocity), and deceleration (Δ velocity) were calculated (as described in Patch encounter classification as sensing or non-responding). (B) Maximum change in velocity for every density condition is shown. Violin plots show the KDE and quartiles for each condition. Two-dimensional histograms show the frequency of events for given pairs of values for (C) deceleration and minimum on-patch velocity and for (D) maximum change in velocity and minimum on-patch velocity for all encounters in the paper. Probabilities are represented as a heat map of color. (E) The first principal component (PC1) for these three metrics was computed. The distribution of projections of the data onto PC1 is given by a histogram (gray). Kernel density estimates (KDE) were computed using the bandwidth $\hat{h}$ defined by Silverman's Rule of Thumb (blue) and using the critical bandwidth h^* (red-orange) at which the KDE is at the threshold between uni- and bi-modal (see Patch encounter classification as sensing or non-responding). (F) Critical bandwidth values were computed across 2000 smoothed bootstrap resamples. The distribution of these bootstrapped critical values is given as a histogram and compared against the critical bandwidth of the original data set (red-orange).

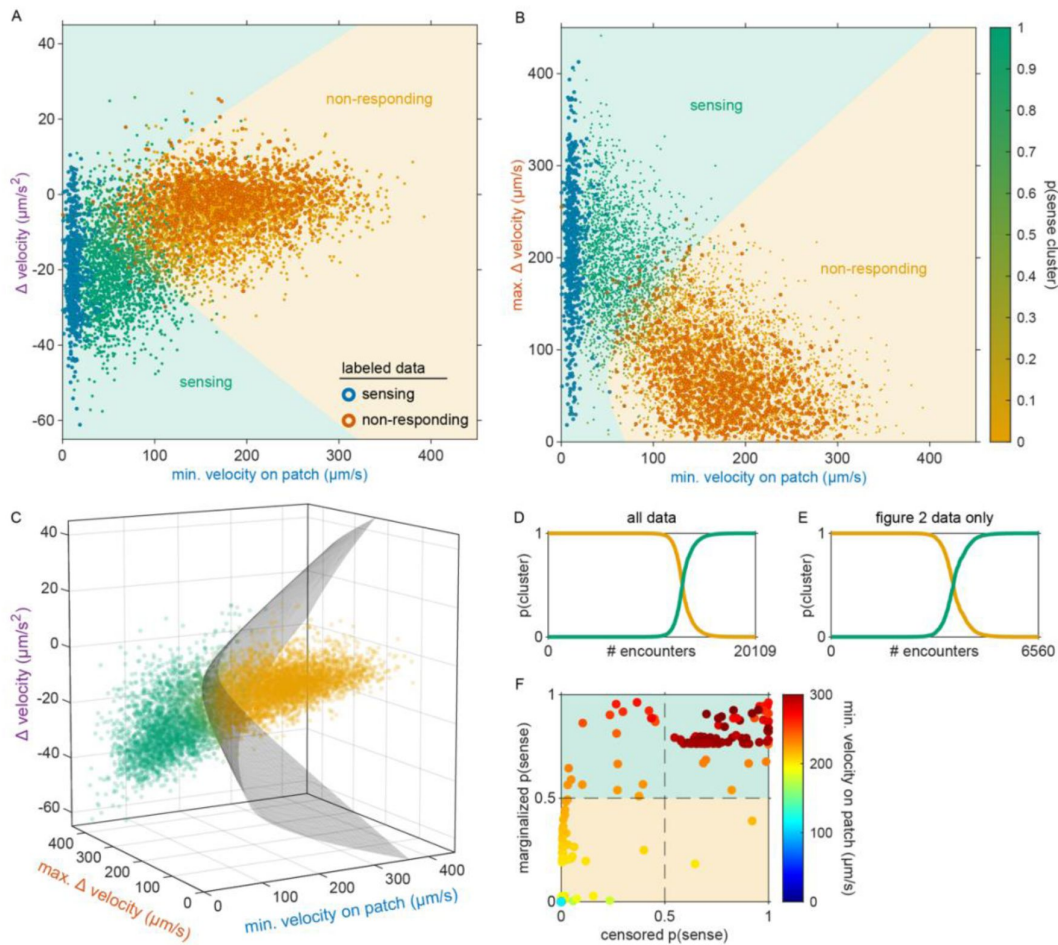


Figure 2 – supplement 8

Classifying encounters as sensing or non-responding.

(A) Deceleration upon encounter and minimum on-patch velocity during the encounter are plotted for each encounter. A three-dimensional parabolic boundary between *sensing* and *non-responding* clusters was fit using semi-supervised QDA and a subset of labeled data (blue circles = *sensing*, dark orange circles = *non-responding*). Regions identified by the classifier as *non-responding* or *sensing* are indicated by colored contours (orange = *non-responding*, green = *sensing*). Encounters for all data in the paper are individually plotted as in **Figure 2I** but with color representing the conditional probability of clustering classification as *non-responding* (orange) or *sensing* (green). (B) Maximum change in velocity and minimum on-patch velocity during the encounter are plotted for each encounter in the paper as in (A). (C) Minimum velocity on patch, maximum change in velocity, and deceleration are shown for every encounter in the paper. See Video 5 for three-dimensional rotation of this plot. (D) The probabilities for classification as *sensing* (green) or *non-responding* (orange) are sorted for all encounters. (E) Probabilities for only the data in **Figure 2** are shown. (F) For a subset of encounters that were censored (i.e., recording did not observe animals entering the patch), probabilities of *sensing* were estimated using the semi-supervised QDA approach as well as via marginalization over the conditional probabilities using only the observed minimum velocity on patch. Marginalized probabilities were used for these values (see Patch encounter classification as sensing or non-responding).

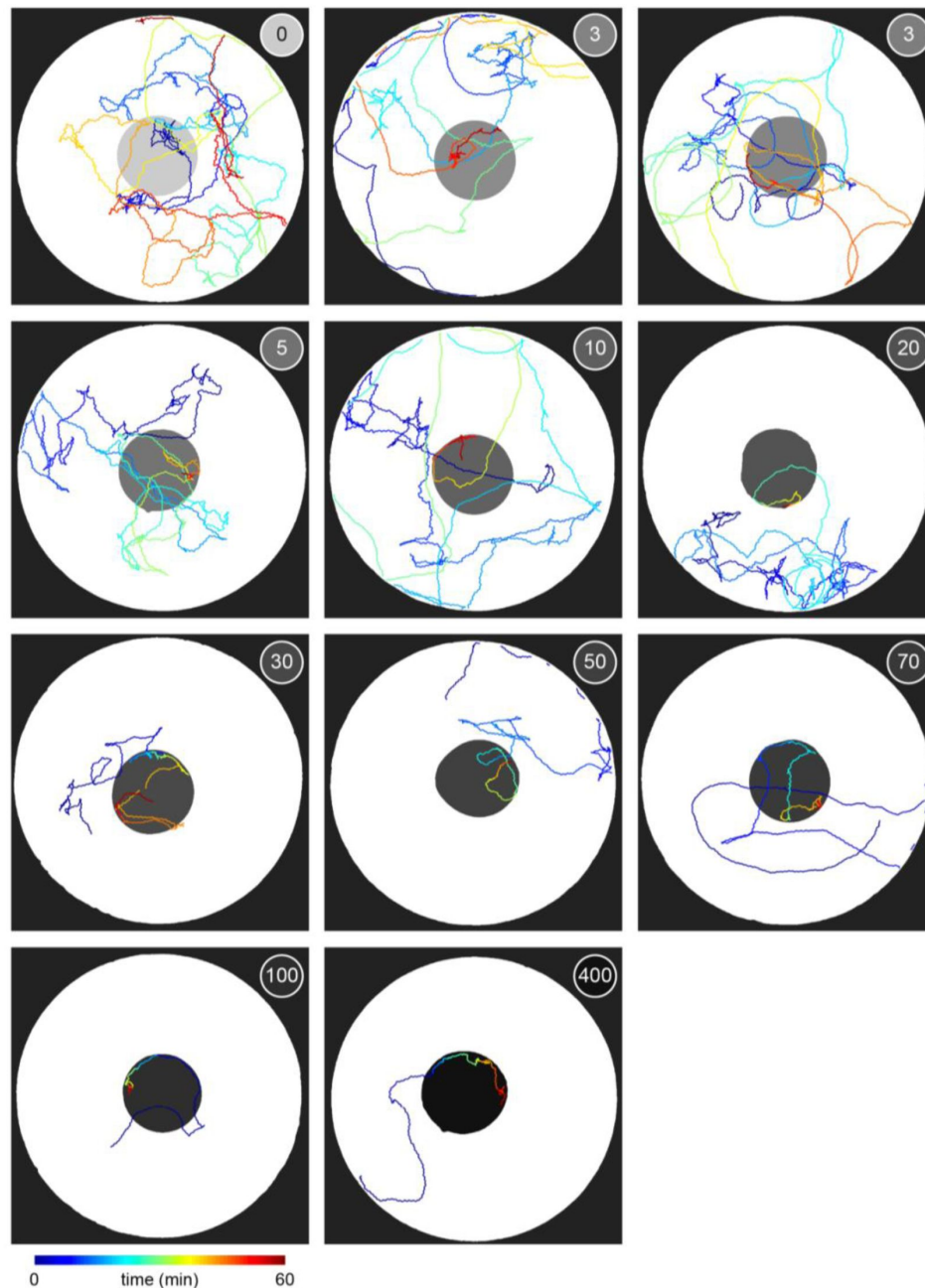


Figure 3 - supplement 1

Example traces of animals foraging in environments with one large bacterial patch.

Example traces showing the midbody location of 11 animals as they forage within a 30 mm arena are plotted with color used to represent time (dark blue = 0 mins; dark red = 60 mins). Location of bacterial patches are indicated in gray. The relative density of bacterial patches in each environment is noted. The animal shown foraging on relative density 10 is the same as that in [Figure 3D](#).

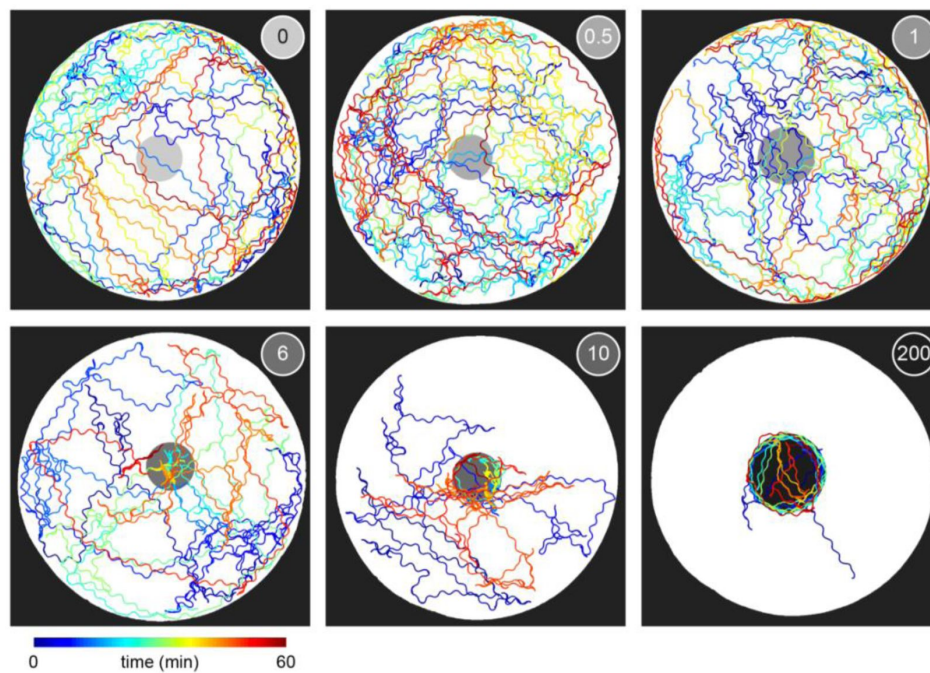


Figure 3 – supplement 2

Example traces of animals foraging in environments with one small bacterial patch.

Example traces showing the midbody location of 6 animals as they forage within a 9 mm arena are plotted with color used to represent time (dark blue = 0 mins; dark red = 60 mins). Location of bacterial patches are indicated in gray. The relative density of bacterial patches in each environment is noted. The animal shown foraging on relative density 10 is the same as that in [Figure 2E](#).

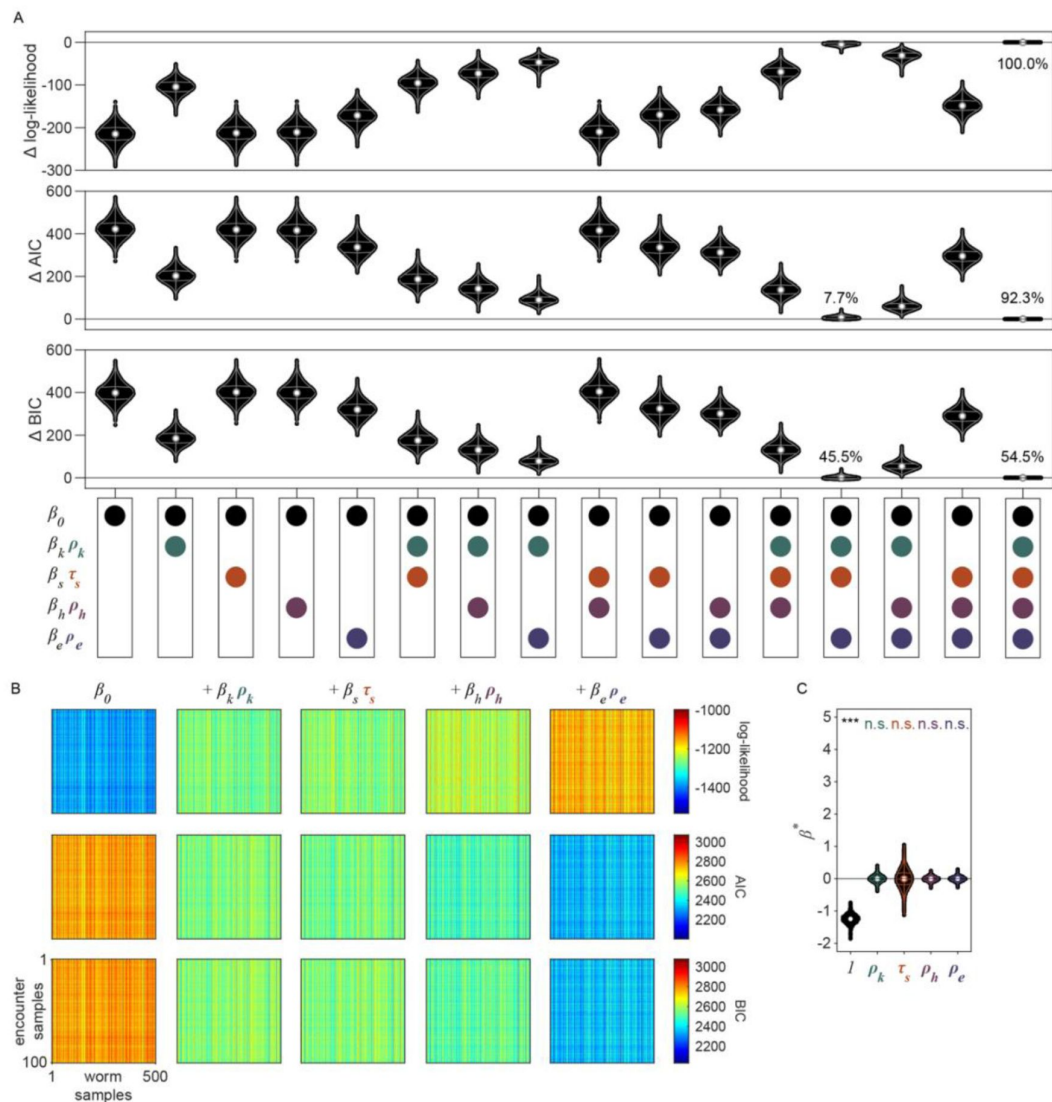


Figure 4 - supplement 1

Model selection for nested GLMs.

(A) Values of log-likelihood, Aikake Information Criterion (AIC), and Bayesian Information Criterion (BIC) were computed for 50,000 replicates of the linear regression model described in Models of exploitation probability. The replicate-by-replicate difference in value of these metrics between the full model (i.e., $\beta \cdot \mathbf{x}_k = \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_h \rho_h + \beta_e \rho_e$) and every combination of covariate subsets is plotted here. Log-likelihood values less than zero and AIC or BIC values greater than zero indicate worse model performance of the partial term models as compared to the full model. Percentages indicate the proportion of replicates where the model performed best according to that criterion. Even using BIC, the strictest model criterion, the full model performs better than any other model on the majority of replicates. (B) To illustrate the replicate-by-replicate variance, log-likelihood, AIC, and BIC are plotted across all 100 replicates of “encounter samples” (i.e., we removed encounters where the animal likely did not *sense* the encounter $v_k = 0$ as estimated from the distribution $v_k | \mathbf{w}_k \sim \text{Bern}(p(v_k = 1 | \mathbf{w}_k))$) and 500 replicates of hierarchically bootstrapped “worm samples” (i.e., we resampled animals with replacement and included all encounters of the resampled animals). Values for each metric at each replicate are colored to match their respective color bar. Metrics are shown for this subset of models to illustrate the addition of each covariate to the model. Lower AIC and BIC values for the full model (i.e., $\beta \cdot \mathbf{x}_k = \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_h \rho_h + \beta_e \rho_e$), represent better model performance even with penalization for increasing the number of parameters. (C) Coefficients β^* for the null distribution (i.e., observations of the response variable $p(y_k = 1 | \mathbf{z}_k)$ shuffled relative to the covariates $\mathbf{x}_k$) are plotted. Only β_0 was significantly greater than or less than 0 (two-tailed, one-sample bootstrap hypothesis tests with Bonferroni correction). These β^* represent the null distribution for the β in Figure 4C. Violin plots in (A,C) show the KDE and quartiles for each covariate across the 50,000 replicates. Asterisks denote statistical significance (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

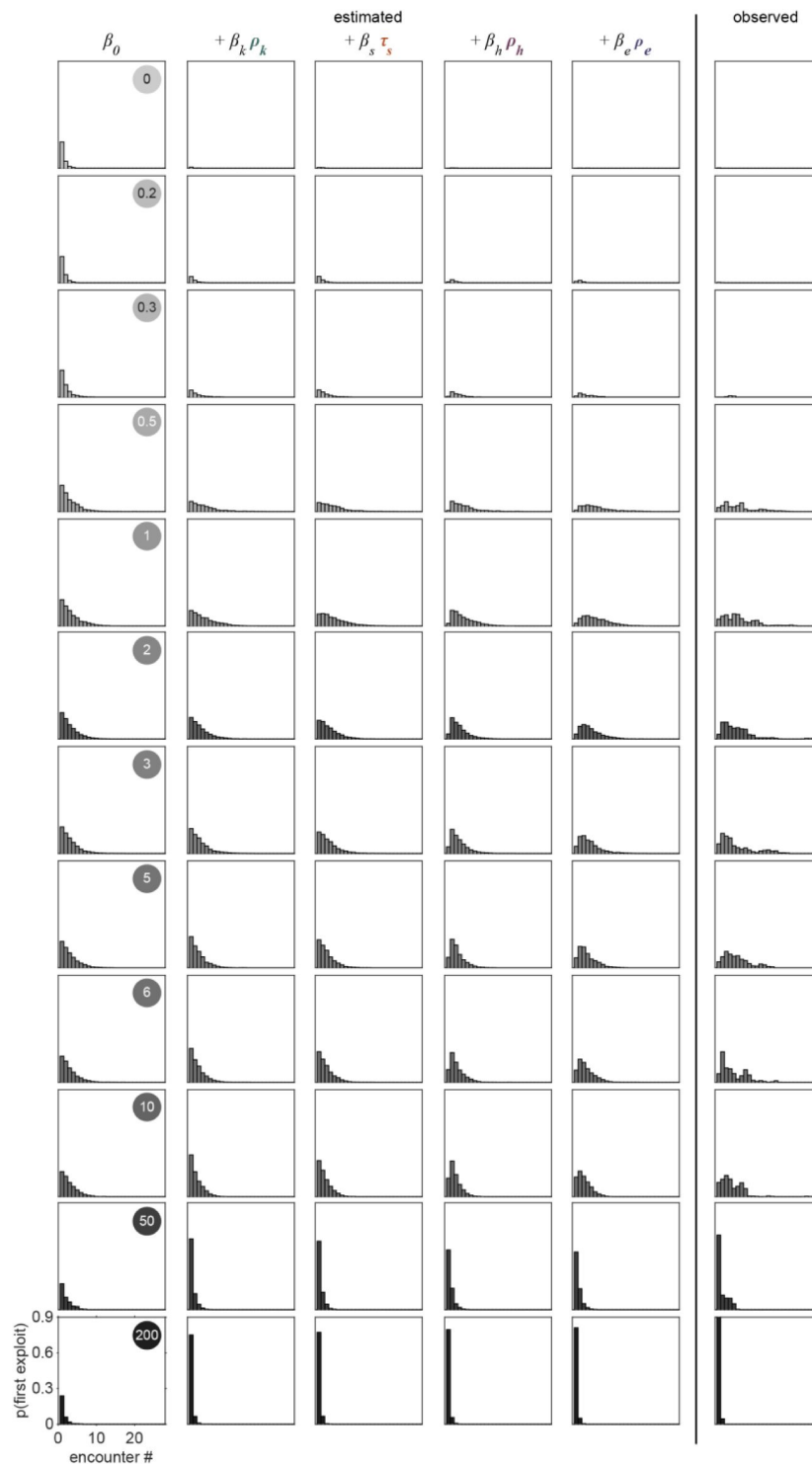


Figure 4 – supplement 2

Probability of first exploitation as a function of the number of encounters.

Histograms of the probability of first *exploiting* as a function of the number of encounters as estimated by the models and as observed are shown for animals foraging in single-density, multi-patch environments of all 12 density conditions. *Exploitation* events were simulated from a Bernoulli distribution with probability estimated by the model $y_k | \mathbf{x}_k \sim \text{Bern}(p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k))$ where covariates were added one at time as well as from the observed probabilities estimated by the Gaussian mixture model classification $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$ in [Figure 2H](#). Histograms for relative density 1 and 10 are the same as those in [Figure 4D](#).

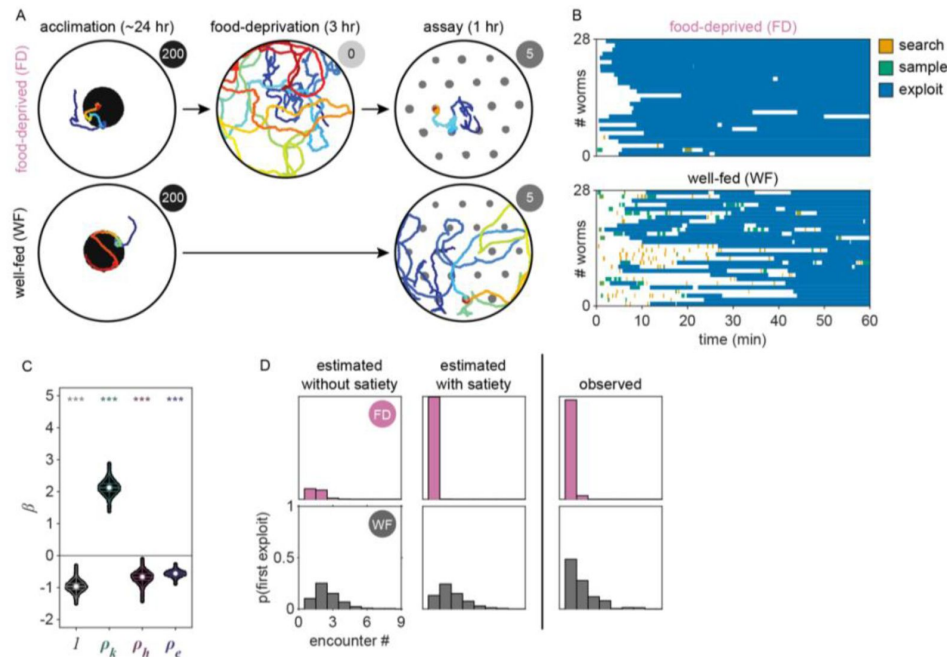


Figure 4 – supplement 3

Exploitation decision of food-deprived animals is well-predicted by the model.

(A) Schematic of experiments used to test satiety's influence on the decision to *exploit*. As in all experiments, animals were acclimated to high density (relative density ~200) patches for ~24 hours. Well-fed animals were then removed from these acclimation plates, cleaned of bacteria, and then immediately transferred to the assay plate with relative density 5. Food-deprived animals moved about a bacteria-free arena for 3 hours prior to the assay. Examples of animal behavior while foraging in acclimation-like, bacteria-free, and single-density, multi-patch environments of relative density 5 are shown. (B) Patch encounters for 28 food-deprived and 28 well-fed individuals are plotted across time. Each encounter is colored to match its probability of classification as *search* (orange), *sample* (green), and *exploit* (blue). (C) Coefficient values were re-estimated across 50000 replicates as in [Figure 4C](#) for a model with the satiety term removed (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_h \rho_h + \beta_e \rho_e)$). All coefficients are significantly greater than or less than 0 (two-tailed, one-sample bootstrap hypothesis tests with Bonferroni correction; ***p < 0.001). Violin plots show the KDE and quartiles for each measure. (D) Histograms of the probability of first *exploiting* as a function of the number of encounters are shown for food-deprived and well-fed animals. Model predictions of *exploitation* events were simulated from a Bernoulli distribution $y_k | \mathbf{x}_k \sim \text{Bern}(p(y_k = 1 | \boldsymbol{\beta} \cdot \mathbf{x}_k))$ where the coefficient values $\boldsymbol{\beta}$ corresponded to those previously estimated for our linear regression model with ([Figure 4C](#)) and without ([Figure 4 – supplement 3C](#)) the satiety term using the data set shown in [Figure 2](#) (i.e., the 443 well-fed animals foraging in one of 12 density conditions). Observed *exploitations* were simulated from a Bernoulli distribution using the probability of *exploitation* estimated by our GMM classifier $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$ in [Figure 2H](#). Summary data for all animals (N = 56 total worms; N = 28 worms per condition) and encounters (N = 277 total encounters; N = 42-235 encounters per condition) are shown in (B,D).

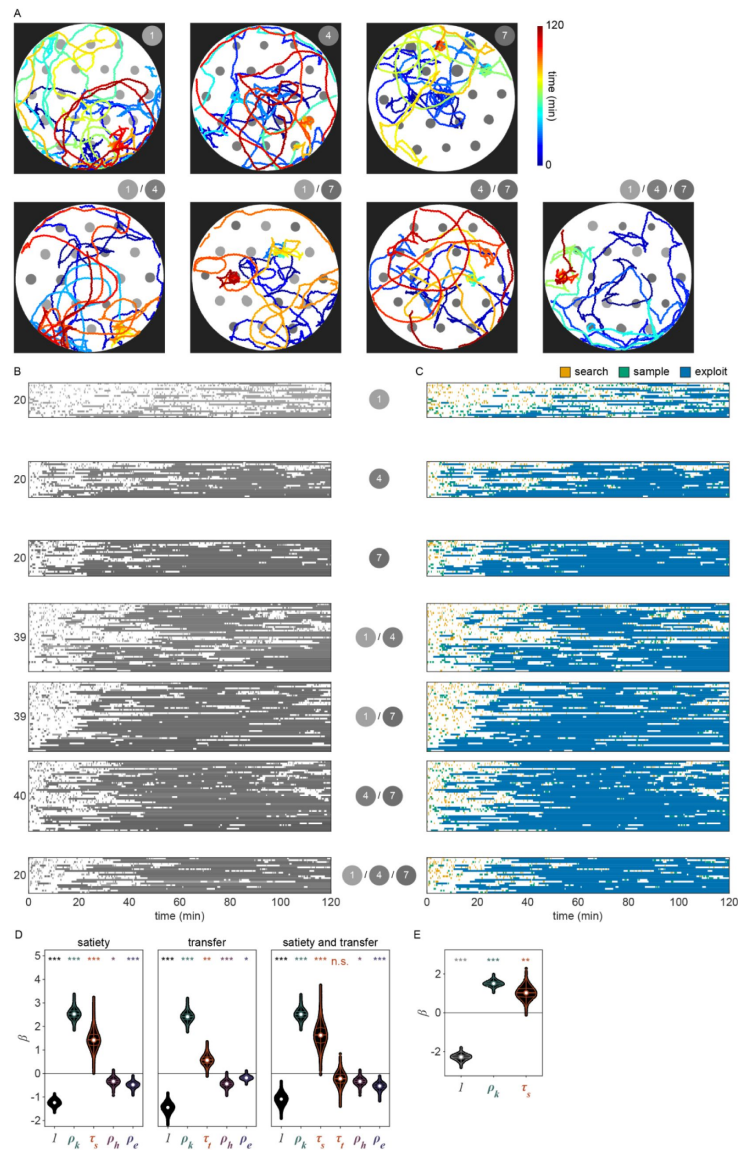


Figure 4 - supplement 4

Example traces and classified behavior of animals foraging in environments with multiple patch densities.

(A) Example traces showing the midbody location of 7 animals as they forage within a 30 mm arena are plotted with color used to represent time (dark blue = 0 mins; dark red = 60 mins). Locations of bacterial patches are indicated in gray with saturation indicating the relative density. Environments contained combinations of patches with relative density 1, 4, and 7 as noted. (B) Patch encounters for 200 individuals (20-40 per condition) are plotted across time. Each encounter is colored to match its relative density. (C) Patch encounters are colored to match their probability of classification as *search* (orange), *sample* (green), and *exploit* (blue). (D) Coefficient values were estimated across 50000 replicates for the multiple density data set for models containing only the satiety-related covariate (i.e., $\beta \cdot \mathbf{x}_k = \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_h \rho_h + \beta_e \rho_e$), only the transfer-related covariate (i.e., $\beta \cdot \mathbf{x}_k = \beta_0 + \beta_k \rho_k + \beta_t \tau_t + \beta_h \rho_h + \beta_e \rho_e$), and for both the satiety and transfer covariates (i.e., $\beta \cdot \mathbf{x}_k = \beta_0 + \beta_k \rho_k + \beta_s \tau_s + \beta_t \tau_t + \beta_h \rho_h + \beta_e \rho_e$). All coefficients are significantly greater than or less than 0 (two-tailed, one-sample bootstrap hypothesis tests with Bonferroni correction) with the exception of β_t in the satiety and transfer model. (E) Coefficient values were re-estimated across 50000 replicates for the original single density data set as in [Figure 4C](#) for a model with the history-dependent terms removed (i.e., $p(y_k = 1 | \beta_0 + \beta_k \rho_k + \beta_s \tau_s)$). All coefficients are significantly greater than or less than 0 (two-tailed, one-sample bootstrap hypothesis tests with Bonferroni correction; ** $p < 0.01$; *** $p < 0.001$). Violin plots show the KDE and quartiles for each measure. Summary data for all animals ($N = 198$ total worms; $N = 20-40$ worms per condition) and encounters ($N = 3,493$ total encounters; $N = 282-795$ encounters per condition) are shown in (B,C).

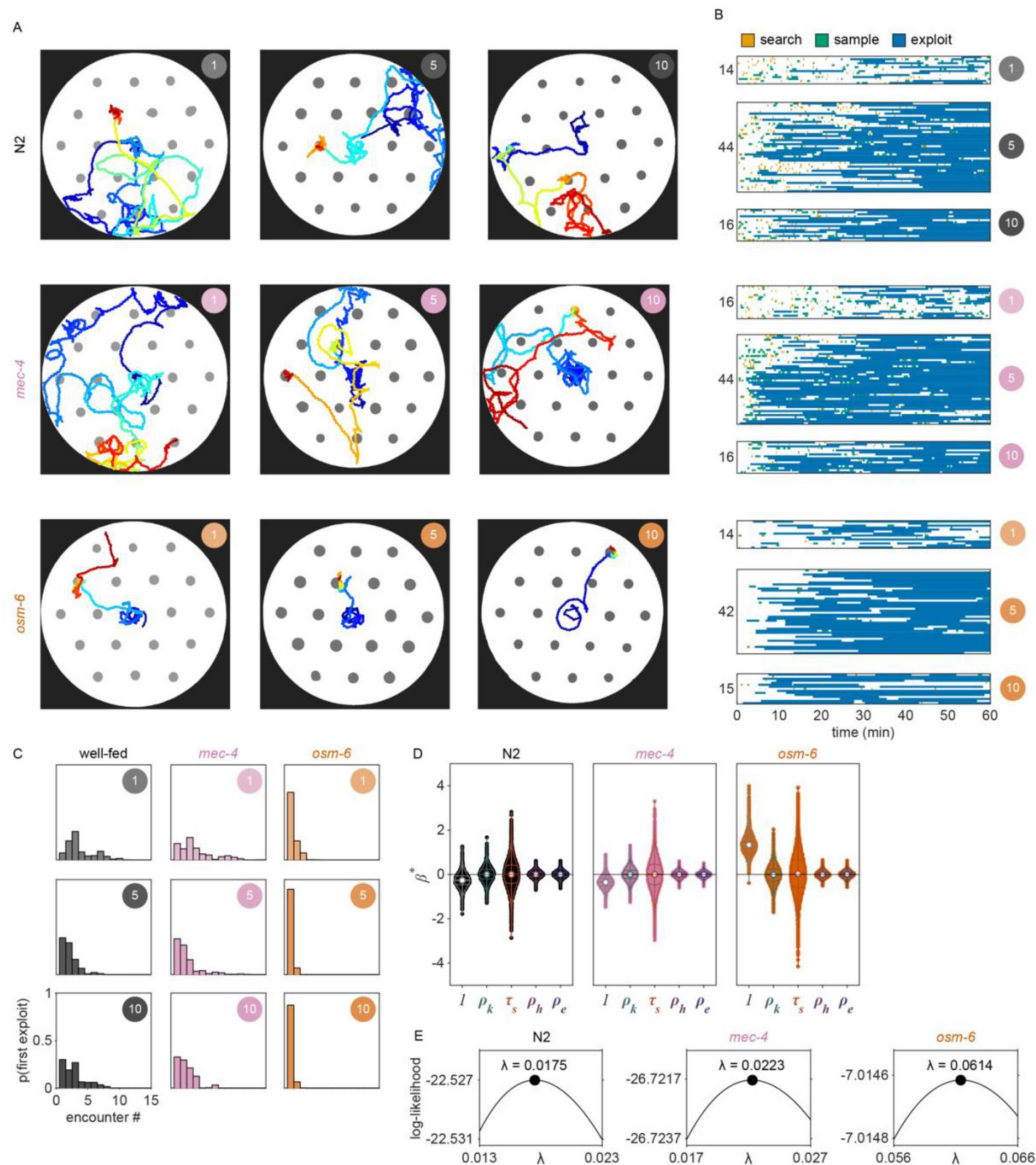


Figure 4 – supplement 5

Example traces and classified behavior of animals with chemosensory or mechanosensory deficiencies.

(A) Example traces showing the midbody location of 9 animals as they forage within a 30 mm arena are plotted with color used to represent time (dark blue = 0 mins; dark red = 60 mins). Location of bacterial patches are indicated in gray. The relative density of bacterial patches in each environment is noted. 3 strains of *C. elegans* were tested (grey = N2, pink = *mec-4*, orange = *osm-6*) across 3 conditions of relative density (1, 5, and 10). (B) Patch encounters for 76 individuals (16-44 per density condition) of each strain are plotted across time. Each encounter is colored to match its probability of classification as search (orange), sample (green), and exploit (blue). (C) Histograms of the probability of first exploiting as a function of the number of encounters are shown for each strain and density condition. Observed exploitations were simulated from a Bernoulli distribution using the posterior probability of exploitation defined by our GMM classifier $y_k | \mathbf{z}_k \sim \text{Bern}(p(y_k = 1 | \mathbf{z}_k))$ in Figure 2H. Histograms of each strain for relative density conditions 1 and 10 are the same as those in Figure 4G. (D) Coefficients β^* for the null distribution (i.e., observations of the response variable $p(y_k = 1 | \mathbf{z}_k)$) were shuffled relative to the covariates $\mathbf{x}_k$ are plotted. Violin plots show the KDE and quartiles for each covariate across the 50,000 replicates. These β^* represent the null distribution for the β in Figure 4H. (E) The ridge regression parameter λ was optimized for each strain to maximize the mean log-likelihood of the model. Summary data for all animals (N = 221 total worms; N = 14-44 worms per condition) and encounters (N = 1,352 total encounters; N = 27-193 encounters per condition) are shown in (B,C).

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Additional information

Author Contributions

Jessica Haley: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Reviewing & Editing, Visualization, Supervision, Project administration, Funding acquisition. **Tianyi Chen:** Investigation, Data Curation, Writing – Reviewing & Editing. **Mikio Aoi:** Methodology, Formal analysis, Writing – Reviewing & Editing, Visualization, Supervision. **Sreekanth Chalasani:** Conceptualization, Resources, Writing – Reviewing & Editing, Supervision, Funding acquisition.

Abbreviations

- BIC: Bayesian information criterion
- GMM: Gaussian mixture model
- GLM: generalized linear model
- KDE: kernel density estimate
- KL divergence: Kullback-Leibler
- LB: lysogeny broth
- NGM: nematode growth medium
- OD₆₀₀: optical density at 600 nm
- PC1: first principal component
- QDA: quadratic discriminant analysis

Additional files

Video 1. *C. elegans* foraging in a patchily distributed environment. Four adult *C. elegans* forage for one hour in a 30 mm arena containing an isometric grid of small, low density bacterial patches. [View video](#)

Video 2. Visualizing dilute bacterial patches using diffraction of light. A piece of dark cardstock passed between the light source and the assay plate enables visualization of dilute bacterial patches. [View video](#)

Video 3. Tracking the midbody location of an animal foraging in a patchily distributed environment. The midbody location of one of the four animals shown in Video 1 is shown as the animal forages for one hour in a 30 mm arena containing an isometric grid of small, low density bacterial patches. The animal shown corresponds to the example animal observed in Figure 1A-B,H-K. [View video](#)

Video 4. Tracking the entire body of a foraging animal. The position of the entire body of an animal is shown as the animal forages for one hour in a 9 mm arena containing one small, low density bacterial patch. The head is indicated by a small dot. The animal shown corresponds to the example animal observed in Figure 3D. [View video](#)

Video 5. *Classifying encounters as sensing based on quantification of slow-down. Three metrics quantifying the magnitude of slow-down upon patch encounter were computed. A semi-self-supervised quadratic discriminant analysis was trained on a subset of labeled data and used to classify all encounters as sensing (green) or non-responding (orange). Data correspond with Figure 2I and Figure 2 – supplement 8A-C.* [🔗](#)

Video 6. *Foraging behavior of an animal with encounters labeled as search, sample, or exploit. The midbody location of the animal in Video 3 is shown and labeled by the probability of classification as searching (i.e., off patch exploration and patch encounters lacking a slow-down response), sampling (i.e., patch encounters that were sensed, but not exploited), and exploiting at every time point. The animal shown corresponds to the example animal observed in Figure 1A-B,H-K.* [🔗](#)

Figure 1 – source data 1. *Number of total and unique patch encounters in Figure 1D.* [🔗](#)

Figure 1 – source data 2. *Duration of patch encounters in Figure 1E.* [🔗](#)

Figure 1 – source data 3. *Start time of patch encounters in Figure 1F.* [🔗](#)

Figure 1 – source data 4. *On patch residence in Figure 1G.* [🔗](#)

Figure 1 – source data 5. *Velocity of animals upon encounter with patch edge in Figure 1L.* [🔗](#)

Figure 1 – source data 6. *Deceleration of animals upon encounter with patch edge in Figure 1M.* [🔗](#)

Figure 1 – source data 7. *Velocity of animals upon encounter with patch edge in Figure 1N.* [🔗](#)

Figure 1 – source data 8. *Velocity on and off patch in Figure 1O.* [🔗](#)

Figure 2 – source data 1. *Relative density of bacterial patches in each condition in Figure 2A.* [🔗](#)

Figure 2 – source data 2. *Velocity of animals upon encounter with patch edge in Figure 2B.* [🔗](#)

Figure 2 – source data 3. *Velocity on patch in Figure 2C.* [🔗](#)

Figure 2 – source data 4. *Time on patch in Figure 2E.* [🔗](#)

Figure 2 – source data 5. *On patch residence in Figure 2F.* [🔗](#)

Figure 2 – source data 6. *Duration of patch encounters in Figure 2G.* [🔗](#)

Figure 2 – source data 7. *Encounter classification as explore or exploit using a Gaussian Mixture Model (GMM) in Figure 2H.* [🔗](#)

Figure 2 – source data 8. *Encounter classification as sense or no response using semi-supervised Quadratic Discriminant Analysis (QDA) in Figure 2I.* [🔗](#)

Figure 2 – source data 9. *Velocity during search, sample, and exploit behaviors in Figure 2K.* [🔗](#)

Figure 2 – source data 10. *Encounter classification as explore or exploit and sense or no response in Figure 2L.* [🔗](#)

Figure 2 – source data 11. *Probability of an encounter type over time in Figure 2M.* [🔗](#)

Figure 2 – source data 12. *Time before first exploitation in Figure 2N.* [🔗](#)

Figure 2 – source data 13. Number of encounters before first exploitation in Figure 2O. [↗](#)

Figure 3 – source data 1. Relative density of large (20 μ L) bacterial patches in Figure 3A. [↗](#)

Figure 3 – source data 2. Relative density of small (0.5 μ L) bacterial patches in Figure 3B. [↗](#)

Figure 3 – source data 3. Encounter classification as explore or exploit and sense or no response in Figure 3F. [↗](#)

Figure 3 – source data 4. Encounter classification as explore or exploit and sense or no response in Figure 3G. [↗](#)

Figure 3 – source data 5. Time before first exploitation in Figure 3H. [↗](#)

Figure 3 – source data 6. Time before first exploitation in Figure 3I. [↗](#)

Figure 4 – source data 1. Coefficients of linear regression model in Figure 4C. [↗](#)

Figure 4 – source data 2. Probability of exploitation occurring for the first time as a function of the number of encounters in Figure 4D. [↗](#)

Figure 4 – source data 3. Observed and predicted (with and without history-dependence) posterior probabilities of exploitation in Figure 4F. [↗](#)

Figure 4 – source data 4. Probability of exploitation occurring for the first time as a function of the number of encounters for animals with sensory mutations in Figure 4G. [↗](#)

Figure 4 – source data 5. Coefficients of linear regression model in Figure 4H. [↗](#)

References

1. Lee D (2013) **Decision making: From neuroscience to psychiatry** *Neuron* **78**:233–248 <https://doi.org/10.1016/j.neuron.2013.04.008>
2. Budaev S, Jørgensen C, Mangel M, Sigrunn E, Giske J (2019) **Decision-making from the animal perspective: Bridging ecology and subjective cognition** *Frontiers in Ecology and Evolution* **7**:164 <https://doi.org/10.3389/fevo.2019.00164>
3. Mobbs D, Trimmer PC, Blumstein DT, Dayan P (2018) **Foraging for foundations in decision neuroscience: Insights from ethology** *Nature Reviews Neuroscience* **19**:419–427 <https://doi.org/10.1038/s41583-018-0010-7>
4. Gold JI, Shadlen MN (2007) **The neural basis of decision making** *Annual Review of Neuroscience* **30**:535–574 <https://doi.org/10.1146/annurev.neuro.29.051605.113038>
5. Stephens DW (2008) **Decision ecology: Foraging and the ecology of animal decision making. Cognitive, Affective & Behavioral Neuroscience** **8**:475–484 <https://doi.org/10.3758/CABN.8.4.475>
6. Hall-McMaster S, Luyckx F (2019) **Revisiting foraging approaches in neuroscience. Cognitive, Affective & Behavioral Neuroscience** **19**:225–230 <https://doi.org/10.3758/s13415-018-00682-z>
7. Schoener TW (1971) **Theory of feeding strategies** *Annual Review of Ecology and Systematics* **2**:369–404 <https://doi.org/10.1146/annurev.es.02.110171.002101>
8. Pyke GH, Pulliam HR, Charnov EL (1977) **Optimal foraging: A selective review of theory and tests** *The Quarterly Review of Biology* **52**:137–154 <https://doi.org/10.1086/409852>
9. Stephens DW, Krebs JR. (1986) **Foraging Theory** Princeton University Press
10. Stephens DW, Brown JS, Ydenberg RC (2007) **Foraging: Behavior and Ecology** University of Chicago Press <https://doi.org/10.7208/9780226772653>
11. Nonacs P (2000) **State dependent behavior and the Marginal Value Theorem** *Behavioral Ecology* **12**:71–83 <https://doi.org/10.1093/oxfordjournals.beheco.a000381>
12. Hills TT (2010) **Animal foraging and the evolution of goal-directed cognition** *Cognitive Science* **30**:3–41 https://doi.org/10.1207/s15516709cog0000_50
13. Calhoun AJ, Hayden BY (2015) **The foraging brain** *Current Opinion in Behavioral Sciences* **5**:24–31 <https://doi.org/10.1016/j.cobeha.2015.07.003>
14. Charnov EL (1976) **Optimal foraging, the marginal value theorem** *Theoretical Population Biology* **9**:129–136 [https://doi.org/10.1016/0040-5809\(76\)90040-x](https://doi.org/10.1016/0040-5809(76)90040-x)
15. Haley JA, Chalasani SH (2024) **C. elegans foraging as a model for understanding the neuronal basis of decision-making** *Cellular and Molecular Life Sciences* **81**:252 <https://doi.org/10.1007/s00018-024-05223-1>

16. Brenner S (1974) **The genetics of *Caenorhabditis elegans*** *Genetics* **77**:71–94 <https://doi.org/10.1093/genetics/77.1.71>
17. Faumont S, Lockery SR (2006) **The awake behaving worm: Simultaneous imaging of neuronal activity and behavior in intact animals at millimeter scale** *Journal of Neurophysiology* **95**:1976–1981 <https://doi.org/10.1152/jn.01050.2005>
18. Boulton T, Hobert O (2012) **From genes to function: The *C. elegans* genetic toolbox** *Wiley Interdisciplinary Review Developmental Biology* **1**:114–137 <https://doi.org/10.1002/wdev.1>
19. Samuel BS, Rowedder H, Braendle C, Félix M-A, Ruvkun G (2016) ***Caenorhabditis elegans* responses to bacteria from its natural habitats** *PNAS* **113**:E3941–E3949 <https://doi.org/10.1073/pnas.1607183113>
20. White JG, Southgate E, Thomson JN, Brenner S (1986) **The structure of the nervous system of the nematode *Caenorhabditis elegans*** *Philosophical Transactions of the Royal Society B* **314**:1–340 <https://doi.org/10.1098/rstb.1986.0056>
21. de Bono M, Maricq AV (2005) **Neuronal substrates of complex behaviors in *C. elegans*** *Annual Review of Neuroscience* **28**:451–501 <https://doi.org/10.1146/annurev.neuro.27.070203.144259>
22. Colbert HA, Bargmann CI (1995) **Odorant-specific adaptation pathways generate olfactory plasticity in *C. elegans*** *Neuron* **14**:803–812 [https://doi.org/10.1016/0896-6273\(95\)90224-4](https://doi.org/10.1016/0896-6273(95)90224-4)
23. Zhang Y, Lu H, Bargmann CI (2005) **Pathogenic bacteria induce aversive olfactory learning in *Caenorhabditis elegans*** *Nature* **438**:179–184 <https://doi.org/10.1038/nature04216>
24. Ardiel EL, Rankin CH (2010) **An elegant mind: Learning and memory in *Caenorhabditis elegans*** *Learning & Memory* **17**:191–201 <https://doi.org/10.1101/lm.960510>
25. Calhoun AJ, Tong A, Sharpee TO, Chalasani SH (2015) **Neural mechanisms for evaluating environmental variability in *Caenorhabditis elegans*** *Neuron* **86**:428–441 <https://doi.org/10.1016/j.neuron.2015.03.026>
26. Bendesky A, Tsunozaki M, Rockman MV, Kruglyak L, Bargmann CI (2011) **Catecholamine receptor polymorphisms affect decision-making in *C. elegans*** *Nature* **472**:313–318 <https://doi.org/10.1038/nature09821>
27. Faumont S, Lindsay TH, Lockery SR (2012) **Neuronal microcircuits for decision making in *C. elegans*** *Current Opinion in Neurobiology* **22**:580–591 <https://doi.org/10.1016/j.conb.2012.05.005>
28. Iwanir S, Brown AS, Nagy S, Najjar D, Kazakov A, Lee KS, Zaslaver A, Levine E, Biron D (2016) **Serotonin promotes exploitation in complex environments by accelerating decision-making** *BMC Biology* **14**:9 <https://doi.org/10.1186/s12915-016-0232-y>
29. Tanimoto Y, Yamazoe-Umemoto A, Fujita K, Kawazoe Y, Miyanishi Y, Yamazaki SJ, Fei X, Busch KE, Gengyo-Ando K, Nakai J, Iino Y, Iwasaki Y, Hashimoto K, Kimura KD (2017) **Calcium dynamics regulating the timing of decision-making in *C. elegans*** *eLife* **6**:e21629 <https://doi.org/10.7554/eLife.21629>
30. Ji N, Madan GK, Fabre GI, Dayan A, Baker CM, Kramer TS, Nwabudike I, Flavell SW (2021) **A neural circuit for flexible control of persistent behavioral states** *eLife* **10**:e62889 <https://doi.org/10.7554/eLife.103191.2>

[.org/10.7554/eLife.62889](https://doi.org/10.7554/eLife.62889)

31. Scheer E, Bargmann CI (2023) **Sensory neurons couple arousal and foraging decisions in *C. elegans*** *eLife* **12**:RP88657 <https://doi.org/10.7554/eLife.88657.3>
32. Petersen C, Dirksen P, Schulenburg H (2015) **Why we need more ecology for genetic models such as *C. elegans*** *Trends in Genetics* **31**:120–127 <https://doi.org/10.1016/j.tig.2014.12.001>
33. Volgin AD, Yakovlev OV, Demin KA, de Abreu MS, Rosemberg DB, Meshalkina DA, Alekseeva PA, Friend AJ, Amstislavskaya TG, Kalueff AV (2018) **Understanding the role of environmental enrichment in zebrafish neurobehavioral models** *Zebrafish* **15**:425–432 <https://doi.org/10.1089/zeb.2018.1592>
34. Kempermann G (2019) **Environmental enrichment, new neurons and the neurobiology of individuality** *Nature Reviews Neuroscience* **20**:235–245 <https://doi.org/10.1038/s41583-019-0120-x>
35. Guisnet A, Maitra M, Pradhan S, Hendricks M (2021) **A three-dimensional habitat for *C. elegans* environmental enrichment** *PLoS One* **16**:e0245139 <https://doi.org/10.1371/journal.pone.0245139>
36. Shtonda BB, Avery L (2006) **Dietary choice behavior in *Caenorhabditis elegans*** *Journal of Experimental Biology* **209**:89–102 <https://doi.org/10.1242/jeb.01955>
37. Hills T, Brockie PJ, Maricq AV (2004) **Dopamine and glutamate control area-restricted search behavior in *Caenorhabditis elegans*** *Journal of Neuroscience* **24**:1217–1225 <https://doi.org/10.1523/JNEUROSCI.1569-03.2004>
38. Calhoun AJ, Chalasani SH, Sharpee TO (2014) **Maximally informative foraging by *Caenorhabditis elegans*** *eLife* **3**:e04220 <https://doi.org/10.7554/eLife.04220.001>
39. Gray JM, Hill JJ, Bargmann CI (2005) **A circuit for navigation in *Caenorhabditis elegans*** *PNAS* **102**:3184–3191 <https://doi.org/10.1073/pnas.0409009101>
40. Milward K, Busch KE, Murphy RJ, de Bono M, Olofsson B (2011) **Neuronal and molecular substrates for optimal foraging in *Caenorhabditis elegans*** *PNAS* **108**:20672–20677 <https://doi.org/10.1073/pnas.1106134109>
41. Olofsson B (2014) **The olfactory neuron AWC promotes avoidance of normally palatable food following chronic dietary restriction** *Journal of Experimental Biology* **217**:1790–1798 <https://doi.org/10.1242/jeb.099929>
42. Pradhan S, Quilez S, Homer K, Hendricks M (2019) **Environmental programming of adult foraging behavior in *C. elegans*** *Current Biology* **29**:2867–2879 <https://doi.org/10.1016/j.cub.2019.07.045>
43. Gloria-Soria A, Azevedo RBR (2008) **npr-1 regulates foraging and dispersal strategies in *Caenorhabditis elegans*** *Current Biology* **18**:1694–1699 <https://doi.org/10.1016/j.cub.2008.09.043>
44. Madirolas G, Al-Asmar A, Gaouar L, Marie-Louise L, Garza-Enríquez A, Rodríguez-Rada V, Khona M, Dal Bello M, Ratzke C, Gore J, Pérez-Escudero A (2023) ***Caenorhabditis elegans* foraging patterns follow a simple rule of thumb** *Communications Biology* **6**:841 <https://doi.org/10.1038/s42003-023-05220-3>

45. Fujiwara M, Sengupta P, McIntire SL (2002) **Regulation of body size and behavioral state of *C. elegans* by sensory perception and the EGL-4 cGMP-dependent protein kinase** *Neuron* **36**:1091–1102 [https://doi.org/10.1016/s0896-6273\(02\)01093-0](https://doi.org/10.1016/s0896-6273(02)01093-0)
46. Sawin ER, Ranganathan R, Horvitz HR (2000) ***C. elegans* locomotory rate is modulated by the environment through a dopaminergic pathway and by experience through a serotonergic pathway** *Neuron* **26**:619–631 [https://doi.org/10.1016/S0896-6273\(00\)81199-X](https://doi.org/10.1016/S0896-6273(00)81199-X)
47. Ahmed MO, Walther G (2012) **Investigating the multimodality of multivariate data with principal curves** *Computational Statistics & Data Analysis* **56**:4462–4469 <https://doi.org/10.1016/j.csda.2012.02.020>
48. Silverman BW (1981) **Using kernel density estimates to investigate multimodality** *Journal of the Royal Statistical Society: Series B (Methodological)* **43**:97–99 <https://doi.org/10.1111/j.2517-6161.1981.tb01155.x>
49. Fawcett TW, McNamara JM, Houston AI (2012) **When is it adaptive to be patient? A general framework for evaluating delayed rewards** *Behavioural Processes* **89**:128–136 <https://doi.org/10.1016/j.beproc.2011.08.015>
50. Susini I, Safryghin A, Hillemann F, Wascher CAF (2021) **Delay of gratification in non-human animals: A review of inter-specific variation in performance** *bioRxiv* <https://doi.org/10.1101/2020.05.05.078659>
51. Constantino SM, Daw ND (2015) **Learning the opportunity cost of time in a patch-foraging task. Cognitive, Affective & Behavioral Neuroscience** **15**:837–853 <https://doi.org/10.3758/s13415-015-0350-y>
52. Carter EC, Pedersen EJ, McCullough ME (2015) **Reassessing intertemporal choice: Human decision-making is more optimal in a foraging task than in a self-control task** *Frontiers in Psychology* **6**:95 <https://doi.org/10.3389/fpsyg.2015.00095>
53. Hayden BY (2018) **Why has evolution not selected for perfect self-control?** *Philosophical Transactions of the Royal Society B* **374** <https://doi.org/10.1098/rstb.2018.0139>
54. Stephens DW, Anderson D (2001) **The adaptive value of preference for immediacy: When shortsighted rules have farsighted consequences** *Behavioral Ecology* **12**:330–339 <https://doi.org/10.1093/beheco/12.3.330>
55. Ferguson TS (1967) **Sequential Decision Problems** In: Birnbaum Z.W., Lukacs E., editors. *Mathematical Statistics: A Decision Theoretic Approach* Academic Press
56. Robbins H (1970) **Optimal stopping** *The American Mathematical Monthly* **77**:333–343 <https://doi.org/10.2307/2316139>
57. Mischel W, Ebbesen EB (1970) **Attention in delay of gratification** *Journal of Personality and Social Psychology* **16**:329–337 <https://doi.org/10.1037/h0029815>
58. Rosati AG, Stevens JR, Hare B, Hauser MD (2007) **The evolutionary origins of human patience: Temporal preferences in chimpanzees, bonobos, and human adults** *Current Biology* **17**:1663–1668 <https://doi.org/10.1016/j.cub.2007.08.033>
59. Schnell AK, Boeckle M, Rivera M, Clayton NS, Hanlon RT (2021) **Cuttlefish exert self-control in a delay of gratification task** *Proceedings of the Royal Society B: Biological Sciences*

288:1946 <https://doi.org/10.1098/rspb.2020.3161>

60. Range F, Brucks D, Virányi Z (2020) **Dogs wait longer for better rewards than wolves in a delay of gratification task: But why?** *Animal Cognition* **23**:443–453 <https://doi.org/10.1007/s10071-020-01346-7>
61. Hillemann F, Bugnyar T, Kotrschal K, Wascher CAF (2014) **Waiting for better, not for more: Corvids respond to quality in two delay maintenance tasks** *Animal Behaviour* **90**:1–10 <https://doi.org/10.1016/j.anbehav.2014.01.007>
62. Miller R, Frohnwieser A, Schiestl M, McCoy DE, Gray RD, Taylor AH, Clayton NS (2020) **Delayed gratification in New Caledonian crows and young children: Influence of reward type and visibility** *Animal Cognition* **23**:71–85 <https://doi.org/10.1007/s10071-019-01317-7>
63. Pujol N, Link EM, Liu LX, Kurz CL, Alloing G, Tan M-W, Ray KP, Solari R, Johnson CD, Ewbank JJ (2001) **A reverse genetic analysis of components of the Toll signaling pathway in *Caenorhabditis elegans*** *Current Biology* **11**:809–821 [https://doi.org/10.1016/S0960-9822\(01\)00241-X](https://doi.org/10.1016/S0960-9822(01)00241-X)
64. Melo JA, Ruvkun G (2012) **Inactivation of conserved *C. elegans* genes engages pathogen- and xenobiotic-associated defenses** *Cell* **149**:452–466 <https://doi.org/10.1016/j.cell.2012.02.050>
65. Pradel E, Zhang Y, Pujol N, Matsuyama T, Bargmann CI, Ewbank JJ (2007) **Detection and avoidance of a natural product from the pathogenic bacterium *Serratia marcescens* by *Caenorhabditis elegans*** *PNAS* **104**:2295–2300 <https://doi.org/10.1073/pnas.0610281104>
66. Quach KT, Chalasani SH (2022) **Flexible reprogramming of *Pristionchus pacificus* motivation for attacking *Caenorhabditis elegans* in predator-prey competition** *Current Biology* **32**:1–14 <https://doi.org/10.1016/j.cub.2022.02.033>
67. Pribadi A, Rieger MA, Rosales K, Reddy KC, Chalasani SH (2023) **Dopamine signaling regulates predator-driven changes in *Caenorhabditis elegans*’ egg laying behavior** *eLife* **12**:e83957 <https://doi.org/10.7554/eLife.83957>
68. Busch KE, Olofsson B (2012) **Should I stay or should I go?** *Worm* **1**:182–186 <https://doi.org/10.4161/worm.20464>
69. Hallatschek O, Hersen P, Ramanathan S, Nelson DR (2007) **Genetic drift at expanding frontiers promotes gene segregation** *PNAS* **104**:19926–19930 <https://doi.org/10.1073/pnas.0710150104>
70. Krebs JR, Inman AJ (1992) **Learning and foraging: Individuals, groups, and populations** *The American Naturalist* **140**:S63–S84 <https://doi.org/10.1086/285397>
71. Frézal L, Félix M-A (2015) ***C. elegans* outside the Petri dish** *eLife* **4**:e05849 <https://doi.org/10.7554/eLife.05849>
72. Matty MA, Lau HE, Haley JA, Singh A, Chakraborty A, Kono K, Reddy KC, Hansen M, Chalasani SH (2022) **Intestine-to-neuronal signaling alters risk-taking behaviors in food-deprived *Caenorhabditis elegans*** *PLoS Genetics* **18**:e1010178 <https://doi.org/10.1371/journal.pgen.1010178>

73. Ezcurra M, Tanizawa Y, Swoboda P, Schafer WR (2011) **Food sensitizes *C. elegans* avoidance behaviors through acute dopamine signaling** *EMBO Journal* **30**:1110–1122 <https://doi.org/10.1038/emboj.2011.22>
74. Anderson JL, Albergotti L, Ellebracht B, Huey RB, Phillips PC (2011) **Does thermoregulatory behavior maximize reproductive fitness of natural isolates of *Caenorhabditis elegans*** *BMC Evolutionary Biology* **11**:157 <https://doi.org/10.1186/1471-2148-11-157>
75. Petersen C, Dirksen P, Prah S, Strathmann EA, Schulenburg H (2014) **The prevalence of *Caenorhabditis elegans* across 1.5 years in selected North German locations: The importance of substrate type, abiotic parameters, and *Caenorhabditis* competitors** *BMC Ecology* **6**:4 <https://doi.org/10.1186/1472-6785-14-4>
76. Fraune S, Bosch TCG (2010) **Why bacteria matter in animal development and evolution** *BioEssays* **32**:571–580 <https://doi.org/10.1002/bies.200900192>
77. Dirksen P, Assié A, Zimmerman J, Zhang F, Tietje A-M, Marsh SA, Félix M-A, Shapira M, Kaleta C, Schulenburg H, Samuel BS (2020) **CeMbio - The *Caenorhabditis elegans* microbiome resource** *G3 Genes | Genomes | Genetics* **10**:3025–3039 <https://doi.org/10.1534/g3.120.401309>
78. Ding SS, Muhle LS, Brown AEX, Schumacher LJ, Endres RG (2020) **Comparison of solitary and collective foraging strategies of *Caenorhabditis elegans* in patchy food distributions** *Philosophical Transactions of the Royal Society B* **375**:37520190382 <https://doi.org/10.1098/rstb.2019.0382>
79. Lipton J, Kleemann G, Ghosh R, Lints R, Emmons SW (2004) **Mate searching in *Caenorhabditis elegans*: A genetic model for sex drive in a simple invertebrate** *Journal of Neuroscience* **24**:7427–7434 <https://doi.org/10.1523/JNEUROSCI.1746-04.2004>
80. Barrios A, Nurrish S, Emmons SW (2008) **Sensory regulation of *C. elegans* male mate-searching behavior** *Current Biology* **18**:1865–1871 <https://doi.org/10.1016/j.cub.2008.10.050>
81. Matsuura T, Sato T, Shingai R (2005) **Interactions between *Caenorhabditis elegans* individuals during chemotactic responses** *Zoological Science* **22**:1095–1103 <https://doi.org/10.2108/zsj.22.1095>
82. Gray J, Lissmann HW (1964) **The locomotion of nematodes** *Journal of Experimental Biology* **41**:135–154 <https://doi.org/10.1242/jeb.41.1.135>
83. Croll NA (1975) **Behavioural analysis of nematode movement** *Advances in Parasitology* **13**:71–122 [https://doi.org/10.1016/s0065-308x\(08\)60319-x](https://doi.org/10.1016/s0065-308x(08)60319-x)
84. Croll NA (1975) **Components and patterns in the behavior of the nematode *Caenorhabditis elegans*** *Journal of Zoology* **176**:159–176 <https://doi.org/10.1111/j.1469-7998.1975.tb03191.x>
85. You Y-j, Kim J, Raizen DM, Avery L (2008) **Insulin, cGMP TGF-beta signals regulate food intake and quiescence in *C. elegans*: A model for satiety.** *Cell Metabolism* **7**:249–257 <https://doi.org/10.1016/j.cmet.2008.01.005>
86. Prevedel R, Yoon Y-G, Hoffmann M, Pak N, Wetzstein G, Kato S, Schrödel T, Raskar R, Zimmer M, Boyden ES, Vaziri A (2014) **Simultaneous whole-animal 3D imaging of neuronal activity using light-field microscopy** *Nature Methods* **11**:727–730 <https://doi.org/10.1038/nmeth.2964>

87. Nguyen JP, Shipley FB, Linder AN, Plummer GS, Liu M, Setru SU, Shaevitz JW, Leifer AM (2015) **Whole-brain calcium imaging with cellular resolution in freely behaving *Caenorhabditis elegans*** *PNAS* **113**:E1074–E1081 <https://doi.org/10.1073/pnas.1507110112>
88. Venkatachalam V, Ji N, Wang X, Clark C, Mitchell JK, Klein M, Tabone CJ, Florman J, Ji H, Greenwood J, Chisholm AD, Srinivasan J, Alkema M, Zhen M, Samuel ADT (2015) **Pan-neuronal imaging in roaming *Caenorhabditis elegans*** *PNAS* **113**:E1082–E1088 <https://doi.org/10.1073/pnas.1507109113>
89. Voleti V, Patel KB, Li W, Perez Campos C, Bharadwaj S, Yu H, Ford C, Casper MJ, Yan RW, Liang W, Wen C, Kimura KD, Targoff KL, Hillman EMC (2019) **Real-time volumetric microscopy of in vivo dynamics and large-scale samples with SCAPE 2.0** *Nature Methods* **16**:1054–1062 <https://doi.org/10.1038/s41592-019-0579-4>
90. Hallinen KM, Dempsey R, Scholz M, Yu X, Linder AN, Randi F, Sharma AK, Shaevitz JW, Leifer AM (2021) **Decoding locomotion from population neural activity in moving *C. elegans*** *eLife* **10**:e66135 <https://doi.org/10.7554/eLife.66135>
91. Davidson JD, El Hady A (2019) **Foraging as an evidence accumulation process** *PLoS Computational Biology* **15**:e1007060 <https://doi.org/10.1371/journal.pcbi.1007060>
92. Labrousse A, Chauvet S, Couillaud C, Kurz CL, Ewbank JJ (2000) ***Caenorhabditis elegans* is a model host for *Salmonella typhimurium*** *Current Biology* **10**:1543–1545 [https://doi.org/10.1016/S0960-9822\(00\)00833-2](https://doi.org/10.1016/S0960-9822(00)00833-2)
93. Quach KT, Hughes GA, Chalasani SH (2024) **Interdependence between SEB-3 and NLP-49 peptides shifts across predator-induced defensive behavioral modes in *Caenorhabditis elegans*** *eLife* **13**:RP98262 <https://doi.org/10.7554/eLife.98262.1>
94. Morse JM (1995) **The Significance of saturation** *Qualitative Health Research* **5**:147–149 <https://doi.org/10.1177/104973239500500201>
95. Sulston JE, Brenner S (1974) **The DNA of *Caenorhabditis elegans*** *Genetics* **77**:95–104 <https://doi.org/10.1093/genetics/77.1.95>
96. Silverman BW (1986) **Density Estimation for Statistics and Data Analysis** Chapman & Hall
97. Saravanan V, Berman GJ, Sober SJ (2020) **Application of the hierarchical bootstrap to multi-level data in neuroscience. *Neurons, Behavior*** *Data analysis, and Theory* **3**:5

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Reviewer #1 (Public review):

Summary:

This work uses a novel, ethologically relevant behavioral task to explore decision-making paradigms in *C. elegans* foraging behavior. By rigorously quantifying multiple features of animal behavior as they navigate in a patch food environment, the authors provide strong evidence that worms exhibit one of three qualitatively distinct behavioral responses upon encountering a patch: (1) "search", in which the encountered patch is below the detection threshold; (2) "sample", in which animals detect a patch encounter and reduce their motor speed, but do not stay to exploit the resource and are therefore considered to have "rejected" it; and (3) "exploit", in which animals "accept" the patch and exploit the resource for tens of minutes. Interestingly, the probability of these outcomes varies with the density of the patch as well as the prior experience of the animal. Together, these experiments provide an interesting new framework for understanding the ability of the *C. elegans* nervous system to use sensory information and internal state to implement behavioral state decisions.

Strengths:

- The work uses a novel, neuroethologically-inspired approach to studying foraging behavior
- The studies are carried out with an exceptional level of quantitative rigor and attention to detail

-Powerful quantitative modeling approaches including GLMs are used to study the behavioral states that worms enter upon encountering food, and the parameters that govern the decision about which state to enter

-The work provides strong evidence that *C. elegans* can make 'accept-reject' decisions upon encountering a food resource

-Accept-reject decisions depend on the quality of the food resource encountered as well as on internally represented features that provide measurements of multiple dimensions of internal state, including feeding status and time

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Reviewer #2 (Public review):

This study provides an experimental and computational framework to examine and understand how *C. elegans* make decisions while foraging environments with patches of food. The authors show that *C. elegans* reject or accept food patches depending on a number of internal and external factors.

The key novelty of this paper is the explicit demonstration of behavior analysis and quantitative modeling to elucidate decision-making processes. In particular, the description of the exploring vs. exploiting phases, and sensing vs. non-sensing categories of foraging behavior based on the clustering of behavioral states defined in a multi-dimensional behavior-metrics space, and the implementation of a generalized linear model (GLM) whose parameters can provide quantitative biological interpretations.

The work builds on the literature of *C. elegans* foraging by adding the reject/accept framework.

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Reviewer #3 (Public review):

Summary:

In this study by Haley et al, the authors investigated explore-exploit foraging using *C. elegans* as a model system. Through an elegant set of patchy environment assays, the authors built a GLM based on past experience that predicts whether an animal will decide to stay on a patch to feed and exploit that resource, instead of choosing to leave and explore other patches.

Strengths:

I really enjoyed reading this paper. The experiments are simple and elegant, and address fundamental questions of foraging theory in a well-defined system. The experimental design is thoroughly vetted, and the authors provide a considerable volume of data to prove their points. My only criticisms have to do with the data interpretation, which I think are easily addressable.

Weaknesses:

History-dependence of the GLM

The logistic GLM seems like a logical way to model a binary choice, and I think the parameters you chose are certainly important. However, the framing of them seem odd to me. I do not doubt the animals are assessing the current state of the patch with an assessment of past experience; that makes perfect logical sense. However, it seems odd to reduce past

experience to the categories of recently exploited patch, recently encountered patch, and time since last exploitation. This implies the animals have some way of discriminating these past patch experiences and committing them to memory. Also, it seems logical that the time on these patches, not just their density, should also matter, just as the time without food matters. Time is inherent to memory. This model also imposes a prior categorization in trying to distinguish between sensed vs. not-sensed patches, which I criticized earlier. Only "sensed" patches are used in the model, but it is questionable whether worms genuinely do not "sense" these patches.

It seems more likely that the worm simply has some memory of chemosensation and relative satiety, both of which increase on patches and decrease while off of patches. The magnitudes are likely a function of patch density. That being said, I leave it up to the reader to decide how best to interpret the data.

osm-6

The argument is that osm-6 animals can't sense food very well, so when they sense it, they enter the exploitation state by default. That is what they appear to do, but why? Clearly they are sensing the food in some other way, correct? Are ciliated neurons the only way worms can sense food? Don't they also actively pump on food, and can therefore sense the food entering their pharynx? I think you could provide further insight by commenting on this. Perhaps your decision model is dependent on comparing environmental sensing with pharyngeal sensing? Food intake certainly influences their decision, no? Perhaps food intake triggers exploitation behavior, which can be over-run by chemo/mechanosensory information?

Impact:

I think this work will have a solid impact on the field, as it provides tangible variables to test how animals assess their environment and decide to exploit resources. I think the strength of this research could be strengthened by a reassessment of their model that would both simplify it and provide testable timescales of satiety/starvation memory.

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Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

(1) The authors repeatedly assert that an individual's behavior in the foraging assay depends on its prior history (particularly cultivation conditions). While this seems like a reasonable expectation, it is not fully fleshed out. The work would benefit from studies in which animals are raised on more or less abundant food before the behavioral task.

Cultivation density: While we agree with the reviewer that testing the effects of varying bacterial density during animal development (cultivation) is an interesting experiment, it is not feasible at this time. We previously attempted this experiment but found it nontrivial to maintain stable bacterial density conditions over long timescales as this requires matching the rate of bacterial growth with the rate of bacterial consumption. Despite our best efforts, we have not been able to identify conditions that satisfy these requirements. Thus, we focused our revised manuscript to include only assertions about the effects of recent experiences and added this inquiry as a future direction (lines 618-624).

(2) The authors convincingly show that the probability of particular behavioral outcomes occurring upon patch encounter depends on time-associated parameters (time since last patch encounter, time since last patch exploitation). There are two concerns here. First, it is not clear how these values are initialized - i.e., what values are used for the first occurrence of each behavioral state? More importantly, the authors don't seem to consider the simplest time parameter, the time since the start of the assay (or time since worm transfer). Transferring animals to a new environment can be associated with significant mechanical stimulus, and it seems quite possible that transferring animals causes them to enter a state of arousal. This arousal, which certainly could alter sensory function or decision-making, would likely decay with time. It would be interesting to know how well the model performs using time since assay starts as the only time-dependent parameter.

Parameter Initialization: We thank the reviewer for pointing out an oversight in our methods section regarding the model parameter values used for the first encounter. We clarified the initialization of parameters in the manuscript (lines 1162-1179). In short, for the first patch encounter where $k = 1$:

- ρ_k is the relative density of the first patch.
- τ_s is the duration of time spent off food since the beginning of the recorded experiment. For the first patch, this is equivalent to the total time elapsed.
- ρ_h is the approximated relative density of the bacterial patch on the acclimation plates (see Assay preparation and recording in Methods). Acclimation plates contained one large 200 μ L patch seeded with $OD_{600} = 1$ and grown for a total of ~48 hours. As with all patches, the relative density was estimated from experiments using fluorescent bacteria OP50-GFP as described in Bacterial patch density estimation in Methods.
- ρ_e is equivalent to ρ_h .

Transfer Method: We thank the reviewer for their thoughtful comment on how the stress of transferring animals to a new plate may have resulted in an increased arousal state and thus a greater probability of rejecting patches. We anticipated this possibility and, in order to mitigate the stress of moving, we used an agar plug method where animals were transferred using the flat surface of small cylinders of agar. Importantly, the use of agar as a medium to transfer animals provides minimal disruption to their environment as all physical properties (e.g. temperature, humidity, surface tension) are maintained. Qualitatively, we observed no marked change in behavior from before to after transfer with the agar plug method, especially as compared to the often drastic changes observed when using a metal or eyelash pick. We added these additional methodological details to the methods (lines 791-796).

Time Parameter: However, the reviewer's concern that the simplest time parameter (time since start of the assay) might better predict animal behavior is valid. We thank the reviewer for pointing out the need to specifically test whether the time-dependent change in explore-exploit decision-making corresponds better with satiety (time off patch) or arousal (time since transfer/start of assay) state. To test this hypothesis, we ran our model with varying combinations of the satiety term τ_s and a transfer term τ_t . We found that when both terms were included in the model, the coefficient of the transfer term was non-significant. This result suggests that the relevant time-dependent term is more likely related to satiety than transfer-induced stress (lines 343-358; Figure 4 - supplement 4D).

(3) Similarly, Figures 2L and M clearly show that the probability of a search event occurring upon a patch encounter decreases markedly with time. Because search events are interpreted as a failure to detect a patch, this implies that the detection of (dilute)

patches becomes more efficient with time. It would be useful for the authors to consider this possibility as well as potential explanations, which might be related to the point above.

Time-dependent changes in sensing: We agree with the reviewer that we observe increased responsiveness to dilute patches with time. Although this is interesting, our primary focus was on what decision an animal made given that they clearly sensed the presence of the bacterial patch. Nonetheless, we added this observation to the discussion as an area of future work to investigate the sensory mechanisms behind this effect (lines 563-568).

(4) Based on their results with mec-4 and osm-6 mutants, the authors assert that chemosensation, rather than mechanosensation, likely accounts for animals' ability to measure patch density. This argument is not well-supported: mec-4 is required only for the function of the six non-ciliated light-touch neurons (AVM, PVM, ALML/R, PLML/R). In contrast, osm-6 is expected to disrupt the function of the ciliated dopaminergic mechanosensory neurons CEP, ADE, and PDE, which have previously been shown to detect the presence of bacteria (Sawin et al 2000). Thus, the paper's results are entirely consistent with an important role of mechanosensation in detecting bacterial abundance. Along these lines, it would be useful for the authors to speculate on why osm-6 mutants are more, rather than less, likely to "accept" when encountering a patch.

Sensory mutant behavior: We thank the reviewer for pointing out the error in our interpretation of the behavior of *osm-6* and *mec-4* animals. We further elaborated on our findings and edited the text to better reflect that *osm-6* mutants lack both chemosensory and mechanosensory ciliated sensory neurons (lines 406-448; lines 567-577). Specifically, we provided some commentary on the finding that *osm-6* mutants show an augmented ability to detect the presence of bacterial patches but a reduced ability to assess their bacterial density. While this finding seems contradictory, it suggests that in the absence of the ability to assess bacterial density, animals must prioritize exploiting food resources when available.

(5) While the evidence for the accept-reject framework is strong, it would be useful for the authors to provide a bit more discussion about the null hypothesis and associated expectations. In other words, what would worm behavior in this assay look like if animals were not able to make accept-reject decisions, relying only on exploit-explore decisions that depend on modulation of food-leaving probability?

Accept-reject vs. stay-switch: We thank the reviewer for alerting us to this gap in our discussion. We have revised the text to further extrapolate upon our point of view on this somewhat philosophical distinction and what it predicts about *C. elegans* behavior (lines 507-533).

Reviewer #3 (Public review):

(1) Sensing vs. non-sensing

*The authors claim that when animals encounter dilute food patches, they do not sense them, as evidenced by the shallow deceleration that occurs when animals encounter these patches. This seems ethologically inaccurate. There is a critical difference between not sensing a stimulus, and not reacting to it. Animals sense numerous stimuli from their environment, but often only behaviorally respond to a fraction of them, depending on their attention and arousal state. With regard to *C. elegans*, it is well-established that their amphid chemosensory neurons are capable of detecting very dilute concentrations of odors. In addition, the authors provide evidence that *osm-6* animals have altered exploit behaviors, further supporting the importance of amphid chemosensory neurons in this behavior.*

Interpretation of “non-sensing” encounters: We thank the reviewer for their comment and agree that we do not know for certain whether the animals sensed these patches or were merely non-responsive to them. We are, however, confident that these encounters lack evidence of sensing. Specifically, we note that our analyses used to classify events as sensing or non-sensing examined whether an animal’s slow-down upon patch entry could be distinguished from either that of events where animals exploited or that of encounters with patches lacking bacteria. We found that “non-sensing” encounters are indeed indistinguishable from encounters with bacteria-free patches where there are no bacteria to be sensed (see Figure 2 - Supplement 8A-C and Patch encounter classification as sensing or non-responding in Methods). Regardless, we agree with the reviewer that all that can be asserted about these events is that animals do not appear to respond to the bacterial patch in any way that we measured. Therefore, we have replaced the term “non-sensing” with “non-responding” to better indicate the ethological interpretation of these events and clarified the text to reflect this change (lines 193-200; lines 211-212).

(2) Search vs. sample & sensing vs. non-sensing

In Figures 2H and 2I, the authors claim that there are three behavioral states based on quantifying average velocity, encounter duration, and acceleration, but I only see three. Based on density distributions alone, there really only seem to be 2 distributions, not 3. The authors claim there are three, but to come to this conclusion, they used a QDA, which inherently is based on the authors training the model to detect three states based on prior annotations. Did the authors perform a model test, such as the Bayesian Information Criterion, to confirm whether 2 vs. 3 Gaussians is statistically significant? It seems like the authors are trying to impose two states on a phenomenon with a broad distribution. This seems very similar to the results observed for roaming vs. dwelling experiments, which again, are essentially two behavioral states.

Validation of sensing clusters: We are grateful to the reviewer for pointing out the difficulty in visualizing the clusters and the need for additional clarity in explaining the semi-supervised QDA approach. We added additional visualizations and methods to validate the clusters we have discovered. Specifically, we used Silverman’s test to show that the sensing vs. non-responding data were bi-modal (i.e. a two-cluster classification method fits best) and accompanied this statistical test with heat maps which better illustrate the clusters (lines 171-173; lines 190-191; lines 948-972; lines 1003-1005; Figure 2 - supplement 6A-C; Figure 2 - supplement 7C-F).

Further, it seems that there may be some confusion as to how we arrived at 3 encounter types (i.e. search, sample, exploit). It’s important to note that two methods were used on two different (albeit related) sets of parameters. We first used a two-cluster GMM to classify encounters as explore or exploit. We then used a two-cluster semi-supervised QDA to classify encounters as sensing or non-sensing (now changed to “non-responding”, see above response) using a different set of parameters. We thus separated the explore cluster into two (sensing and non-responding exploratory events) resulting in three total encounter types: exploit, sample (explore/sensing), and search (explore/non-sensing).

(4) History-dependence of the GLM

The logistic GLM seems like a logical way to model a binary choice, and I think the parameters you chose are certainly important. However, the framing of them seems odd to me. I do not doubt the animals are assessing the current state of the patch with an assessment of past experience; that makes perfect logical sense. However, it seems odd to reduce past experience to the categories of recently exploited patch, recently encountered patch, and time since last exploitation. This implies the animals have some

way of discriminating these past patch experiences and committing them to memory. Also, it seems logical that the time on these patches, not just their density, should also matter, just as the time without food matters. Time is inherent to memory. This model also imposes a prior categorization in trying to distinguish between sensed vs. not-sensed patches, which I criticized earlier. Only "sensed" patches are used in the model, but it is questionable whether worms genuinely do not "sense" these patches.

Model design: We thank the reviewer for their thoughtful comments on the model. We completed a number of analyses involving model selection including model selection criteria (AIC, BIC) and optimization with regularization techniques (LASSO and elastic nets) and found that the problem of model selection was compounded by the enormous array of highly-correlated variables we had to choose from. Additionally, we found that both interaction terms and non-linear terms of our task variables could be predictive of accept-reject decisions but that the precise set of terms selected depended sensitively on which model selection technique was used and generally made rather small contributions to prediction. The diverse array of results and combinatorial number of predictors to possibly include failed to add anything of interpretable value. We therefore chose to take a different approach to this problem. Rather than trying to determine what the “best” model was we instead asked whether a minimal model could be used to answer a set of core questions. Indeed, our goal was not *maximal* predictive performance but rather to distinguish between the effects of different influences enough to determine if encounter history had a significant, independent effect on decision making. We thus chose to only include task variables that spanned the most basic components of behavioral mechanisms to ask very specific questions. For example, we selected a time variable that we thought best encapsulated satiety. While we could have included many additional terms, or made different choices about which terms to include, based on our analyses these choices would not have qualitatively changed our results. Further, we sought to validate the parameters we chose with additional studies (i.e. food-deprived and sensory mutant animals). We regard our study as an initial foray into demonstrating accept-reject decision-making in nematodes. The exact mechanisms and, consequently, the best model design are therefore beyond the scope of this study.

Lastly, in regards to the use of only sensed patches in the model; while we acknowledge that we are not certain as to whether the “non-responding” encounters are truly not sensed, we find qualitatively similar results when including all exploratory patches in our analyses. However, we take the position that sensation is necessary for decision-making and thus believe that while our model’s predictive performance may be better using all encounters, the interpretation of our findings is stronger when we only include sensing events. We have added additional commentary about our model to the discussion section (lines 667-695).

(5) *osm-6*

The osm-6 results are interesting. This seems to indicate that the worms are still sensing the food, but are unable to assess quality, therefore the default response is to exploit. How do you think the worms are sensing the food? Clearly, they sense it, but without the amphid sensory neurons, and not mechanosensation. Perhaps feeding is important? Could you speculate on this?

We thank the reviewer for their thoughtful remarks. We have added additional commentary about the result of our sensory mutant experiments as described above in response to Reviewer #1 under Sensory mutant behavior.

(7) *Impact:*

I think this work will have a solid impact on the field, as it provides tangible variables to test how animals assess their environment and decide to exploit resources. I think the

strength of this research could be strengthened by a reassessment of their model that would both simplify it and provide testable timescales of satiety/starvation memory.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

*The authors title the work as an "ethological study" and emphasize the theme of "foraging in naturalistic environments" in contrast to typical laboratory conditions. The only difference in this study relative to typical laboratory conditions is that the food bacteria is distributed in many small patches as compared to one large patch. First, it is not clear to the reviewer that the size of the food patches in these experiments is more relevant to *C. elegans* in its natural context than the standard sizes of food patches. Furthermore, all the other highly unnatural conditions typical of laboratory cultivation still apply: the use of a 2D agar substrate, a single food bacteria that is not a component of a naturalistic diet, and the use of a laboratory-adapted strain of *C. elegans* with behavior quite distinct from that of natural isolates. The reviewer is not suggesting that the authors need to make their experiments more naturalistic, only that the experiments as described here should not be described as naturalistic or ethological as there is no support for such claims.*

Ethological interpretation: We thank the reviewer for their comments about the use of the term ethological to describe this study. We chose to develop a patchy bacterial assay to mimic the naturalistic “boom-or-bust” environment. While we agree with the reviewer that we do not know if the size and distribution of the food patches in these experiments is more relevant to *C. elegans*, we maintain that these experiments were ecologically-inspired and revealed behavior that is difficult to observe in environments with large, densely-seeded bacterial patches. We have updated our text to better reflect that this study was “ecologically-inspired” rather than truly “ethological” in nature (lines 94, 693).

The main finding of the paper is that worms explore and then exploit, i.e. they frequently reject several bacterial patches before accepting one. This result requires additional scrutiny to reject other possible interpretations. In particular, when worms are transferred to a new plate we would expect some period of increased arousal due to the stressful handling process. A high arousal state might cause rejection of food patches. Could the measured accept/reject decisions be influenced by this effect? One approach to addressing this concern would be to allow the animals to acclimate to the new plate on a bare region before encountering the new food patches.

We thank the reviewer for their comment on how the stress of transferring animals to a new plate may have resulted in an increased arousal state and thus a greater probability of rejecting patches. We addressed this above in response to Reviewer #1 under Transfer Method and Time Parameter. In brief, we used a worm picking method that mitigated stress and added additional analyses showing that a transfer-related term was less predictive than a satiety-related term.

Related to the above, in what circumstances exactly are the authors claiming that worms first explore and then exploit? After being briefly deprived of food? After being handled?

Explore-then-exploit: All animals were well-fed and handled gently as described above under Transfer Method (lines 787-795). Our results suggest that the appearance of an explore-then-exploit strategy is a byproduct of being transferred from an environment with high bacterial density to an environment with low bacterial density as described in the manuscript (lines 461-466).

The authors emphasize their analysis of the accept/reject decision as a critical innovation. However, the accept/reject decision does not strike me as substantially different from the previously described stay/switch decision. When a worm encounters a new patch of bacteria, accepting this bacteria is equivalent to staying on it and rejecting (leaving) it is equivalent to switching away from it. The authors should explain how these concepts are significantly distinct.

Accept-reject vs. stay-switch: We thank the reviewer for alerting us to this gap in our discussion. We have revised the text to further extrapolate upon our point of view on this somewhat philosophical distinction and what it predicts about *C. elegans* behavior (lines 507-533).

During patch encounter classification, the authors computed three of the animals' behavioral metrics (Line 801-804) and claimed that the combination of these three metrics reveals two non-Gaussian clusters representing encounters where animals sensed the patch or did not appear to sense the patch. The authors also refer to a video to demonstrate the two clusters by rotating the 3-dimension scatter plot. However, the supposed clusters, if any, are difficult to see in a 3D (Video 5) or in a 2D scatter plot (Figure 3I). The authors need to clearly demonstrate the distinct clustering as claimed in the paper as this feature is fundamental and necessary for the model implementation and interpretation of results.

We are grateful to the reviewer for pointing out the difficulty in visualizing the clusters. We added additional visualizations and methods to validate the clusters we have discovered as described in our above response to Reviewer #3 under Validation of sensing clusters.

When selecting parameters (covariates) for their model, it is critical to avoid overfitting. Therefore, the authors used AIC and BIC (Figure 4- supplement 1) to demonstrate that the full GLM model has a better model performance than the other models which contain only a subset of the full covariates (in a total of 5). However, the authors compare the full set with only 4 other models whereas the total number of models that need to be compared with is $2^5 - 2$. The authors at least need to include the AIC and BIC scores of all possible models in order to draw the conclusion about the performance of the full model.

Model selection criterion: We thank the reviewer for pointing out this gap in our methodology. We have now run the model with all combinations of subsets of model parameters and have confirmed that the model with all 5 covariates outperforms all other models even when using BIC, the strictest criterion for overfitting (Figure 1 - supplement 1A). The only other model that performs well (though not as often as the 5-term model) is the 4-term model lacking ρ_h . This result is not surprising as ρ_h only changes substantially once in an animal's encounter history for the single-density, multi-patch data that this model was fit to. For example, for an animal foraging on patches of density 10, on the first encounter $\rho_h = \sim 200$ (see Parameter initialization above), but on every subsequent encounter $\rho_h = \sim 10$. Resultantly, the effect of ρ_h on the probability of exploiting is somewhat binary on the single-density, multi-patch data set. Nevertheless, we see significantly improved prediction of behavior in the novel multi-density, multi-patch data (Figure 4F) as we observe an effect of the most recently encountered patch. Additionally, we observe a similar impact (i.e., significant coefficient of negative sign) of the ρ_h term when the model is fit to the multi-density, multi-patch data set (Figure 4 - supplement 4D).

In any bacterial patch, the edges have a higher density of bacteria than the patch center. Thus, it is possible that a worm scans the patch edge density, on the basis of which it decides to accept or reject the patch whose average density is smaller. This could

potentially cause an underestimate of the bacteria density used in the model. Furthermore, the potential inhomogeneity of the patch may further complicate the worm's decision-making, and the discrepancy between the reality and the model assumption will reduce the validity of the model. The authors need to estimate the inhomogeneity of the bacterial patches used in their assays and discuss how the edge effects may affect their results and conclusions.

Bacterial patch inhomogeneity: We extensively tested the landscape of the bacterial patches by imaging fluorescently-labeled bacteria OP50-GFP (Bacterial Patch Density in Methods; Figure 2 - supplement 1-3). As the reviewer mentions, we observe significantly greater bacterial density at the patch edge. This within-patch spatial inhomogeneity results from areas of active proliferation of bacteria and likely complicates an animal's ability to accurately assess the quantity of bacteria within a patch and, consequently, our ability to accurately compute a metric related to our assumptions of what the animal is sensing. In our study, we used the relative density of the patch edge where bacterial density is highest as a proxy for an animal's assessment of bacterial patch density (Figure 2 – supplement 1). This decision was based on a previous finding that the time spent on the edge of a bacterial patch affected the dynamics of subsequent area-restricted search. While within-patch spatial inhomogeneity likely affects an animal's ability to assess patch density, we do not believe that this qualitatively affects the results of our study. Both the patch densities tested (Figure 2 – supplement 3A) as well as our observations of time-dependent changes in exploitation (Figure 2E,N-O; Figure 3H-I) maintained a monotonic relationship. Therefore, alternative methods of patch density estimation should yield similar results. We have added additional discussion on this topic to our manuscript (lines 578-593).

The authors claim that their methods (GMM and semi-supervised QDA) are unbiased. This seems unlikely as the QDA involves supervision. The authors need to provide additional explanation on this point.

Semi-supervised QDA labelling: We have removed the term “unbiased” to avoid any misinterpretation of the methodology and clarified our method of labelling used for “supervising” QDA. Specifically, we made two simple assumptions: 1) animals must have sensed the patch if they exploited it and 2) animals must not have sensed the patch if there was no bacteria to sense. Thus, we labeled encounters as sensing if they were found to be exploitative as we assume that sensation is prerequisite to exploitation; and we labeled encounters as non-sensing for events where animals encountered patches lacking bacteria ($OD_{600} = 0$). All other points were non-labeled prior to learning the model. In this way, our labels were based on the experimental design and results of the GMM, an unsupervised method; rather than any expectations we had about what sensing should look like. The semi-supervised QDA method then used these initial labels to iteratively fit a paraboloid that best separated these clusters, by minimizing the posterior variance of classification (lines 1012-1021). See Figure 2 - supplement 8A-B for a visualization showing the labelled data.

Based on the authors' result, worms behaviorally exhibit their preferences toward food abundance (density), which results in a preference scale for a range of densities. Does this scale vary with the worms' initial cultivation states? The author partially verified that by observing starved worms. This hypothesis could be better tested if the authors could analyze the decision-making of the worms that were initially cultivated with different densities of bacterial food.

While we agree with the reviewer that testing the effects of varying bacterial density during animal development (cultivation) is a very interesting experiment, it is not feasible at this time. We focused our revised manuscript to include only assertions about the effects of recent

experiences and added this inquiry as a future direction as described above in our response to Reviewer #1 under Cultivation density.

It would be helpful to elaborate more on how the framework developed in this paper can be applied more broadly to other behaviors and/or organisms and how it may influence our understanding of decision-making across species.

We thank the reviewer for alerting us to this gap in our discussion. We have added additional commentary about our model and its utility to the discussion section (lines 667-695).

Reviewer #3 (Recommendations for the authors):

Sensing vs. non-sensing

Perhaps a more ethologically accurate term to describe this behavior would be "ignoring" rather than "not sensing". If the authors feel strongly about using the term "not sensing", then they should provide experimental evidence supporting this claim. However, I think simply changing the terminology negates these experiments.

We thank the reviewer for their thoughtful comments. While we agree with the reviewer that the term “non-sensing” may not be ethologically accurate (see response to Public Review above under Interpretation of “non-sensing” encounters), we interpret the term “ignoring” to mean that the animal sensed the patches but decided not to react. We have chosen to replace the term “non-sensing” with “non-responding” to best indicate the ethological interpretation of our observation. Nonetheless, we believe that it remains possible that animals are truly not sensing the bacterial patches as our method of classification compared the behavior against encounters with patches lacking bacteria (as described above in response to Reviewer #2 under Semi-supervised QDA labelling).

History-dependence of the GLM

Perhaps a simpler approach would be to say the worm senses everything, and this accumulative memory affects the decision to exploit. For example, the animal essentially experiences two feeding states: feeding on patches, and starvation off of patches.

The level of satiety could be modeled linearly:

$$\text{Satiety}(t_{\text{enter}}:t_{\text{leave}}) = k_{\text{feed}} * \text{patch_density} * \text{delta_t}$$

Where k_{feed} is some model parameter for rate of satiety signal accumulation, t_{enter} is the time the animal entered the patch, t_{leave} is the time the animal left the patch, and delta_t is the difference between the two. Perhaps you could add a saturation limit to this, but given your data, I doubt that is the case.

Starvation could be modeled as simply a decay from the last satiety signal:

$$\text{Starvation}(t_{\text{leave}}:t_{\text{enter}}) = \text{Satiety}(t_{\text{leave}}) * \exp(-k_{\text{starve}} * \text{delta_t}).$$

Where starvation is the rate constant for the decay of the satiety signal.

For the logistic model, the logistic parameter is simply the difference between the current patch density and the current satiety signal.

A nice thing about this approach is that it negates the need to categorize your patches. All patch encounters matter. Brief patch encounters (categorized as non-sensing and not used in the prior GLM) naturally produce a very small satiety signal and contribute very little to the exploit decision. Another nice thing about this approach is that it gives you

memory timescales, that are testable. There is a rate of satiety accumulation and a rate of satiety loss. You should be able to predict behavior with lower patch density, assuming the rate constants hold. (I am not advocating you do more experiments here, just pointing out a nice feature of this approach).

You could possibly apply this to a GLM for velocity on a non-exploited patch as well, though I assume this would be a linear GLM, given the velocity distributions you provided.

We thank the reviewer for their time and thoughtfulness in thinking about our model. The reviewer's proposed model seems entirely reasonable and could aid in elucidating the time component of how prior experience affects decision-making. However, we decided to keep our paper focused on using a minimal model to answer a set of core questions (e.g., Does encounter history or satiety influence decision-making?) (see above under Model design for a more detailed response). Future studies investigating the mechanisms of these foraging decisions should open the door for more mechanistically accurate models. We have expanded our discussion of the model to include this assertion (lines 667-695).

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